INCH-POUND

MIL-M-38510/53F
13 December 2004

SUPERSEDING
MIL-M-38510/53E
7 August 1987

MILITARY SPECIFICATION
MICROCIRCUITS, DIGITAL, CMOS, COMPLIMENTARY PAIR PLUS INVERTER, AND-OR-SELECT EXCLUSIVE OR GATES, EXPANDABLE 8-INPUT GATE MONOLITHIC SILICON

Reactivated after 13 Dec. 2004 and may be used for new and existing designs and acquisitions.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product herein shall consist of this specification sheet and MIL-PRF 38535

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, CMOS, logic microcircuits. Two product assurance classes and a choice of case outlines, lead finishes, and radiation hardness assurance (RHA) are provided and are reflected in the complete Part or Identifying Number (PIN). For this product, the requirements of MIL-M-38510 have been superseded by MIL-PRF-38535 (see 6.3).

1.2 Part or identifying number (PIN). The PIN is in accordance with MIL-PRF-38535 and as specified herein.

1.2.1 Device types. The device types are as follows:

Device type	Circuit
01	Dual complimentary pair plus inverter
02	Quad AND-OR-select gate
03	Quad 2-input exclusive OR gate
04	Multi-function expandable 8-input gate
51	Dual complimentary pair plus inverter
52	Quad AND-OR-select gate
53	Quad 2-input exclusive OR gate
54	Multi-function expandable 8-input gate

1.2.2 Device class. The device class is the product assurance level as defined in MIL-PRF-38535.

1.2.3 Case outlines. The case outlines are as designated in MIL-STD-1835 and as follows:

Outline letter	Descriptive designator	Terminals	Package style
A	GDFP5-F14 or CDFP6-F14	14	Flat pack
C	GDIP1-T14 or CDIP2-T14	14	Dual-in-line
D	GDFP1-F14 or CDFP2-F14	14	Flat pack
E	GDIP1-T16 or CDIP2-T16	16	Dual-in-line
F	GDFP2-F16 or CDFP3-F16	16	Flat pack
N	CDFP4-F16	16	Flat pack
T	CDFP3-F14	14	Flat pack
X 1/ 2/	GDFP5-F14 or CDFP6-F14	14	Flat pack, except A dimension equals 0.100” (2.54 mm) max
Y 1/ 2/	GDFP1-F14 or CDFP2-F14	14	Flat pack, except A dimension equals 0.100” (2.54 mm) max
Z 1/ 2/	GDFP2-F16 or CDFP3-F16	16	Flat pack, except A dimension equals 0.100” (2.54 mm) max

1/ As an exception to the nickel plate or undercoating paragraph of MIL-PRF-38535, appendix A, for case outlines X, Y, and Z only, the leads of bottom brazed ceramic packages (i.e., configuration 2 of case outlines A, D, and F) may have electroless nickel undercoating which is 50 to 200 microinches (1.27 to 5.08 μm) thick provided the lead finish is hot solder dip (i.e., finish letter A) and provided that, after any lead forming, an additional hot solder dip coating is applied which extends from the outer tip of the lead to no more than 0.015 inch (0.38 mm) from the package edge.

2/ For bottom or side brazed packages, case outlines X, Y and Z only, the S_1 dimension may go to .000 inch (.00 mm) minimum.

Comments, suggestions, or questions on this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or email CMOS@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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1.3 Absolute maximum ratings.

Supply voltage range (V_{DD} - V_{SS}):	
Device types 01, 02, 03, 04	-0.5 V dc to +15.5 V dc 1/
Device types 51, 52, 53, 54	-0.5 V dc to +18.0 V dc
Input current (each input)	±10 mA
Input voltage range	(V_{SS} - 0.5 V) ≤ V_I ≤ (V_{DD} + 0.5 V)
Storage temperature range (T_{STG})	-65° to +175°C
Maximum power dissipation (P_D)	200 mW
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction to case (θ_{JC})	See MIL-STD-1835
Junction temperature (T_J)	175°C

1.4 Recommended operating conditions.

Supply voltage range (V_{DD} - V_{SS}):	
Device types 01, 02, 03, 04	4.5 V dc to 12.5 V dc
Device types 51, 52, 53, and 54	4.5 V dc to 15.0 V dc
Input low voltage range (V_{IL}):	
Device types 01, 02, 03, 04	0.0 V to 0.85 V dc @ V_{DD} = 5.0 V dc
	0.0 V to 2.1 V dc @ V_{DD} = 12.5 V dc
Device type 51	V_{OL} = 10% V_{DD}, V_{OH} = 90% V_{DD}
	0.0 V to 1.0 V dc @ V_{DD} = 5.0 V dc
	0.0 V to 2.0 V dc @ V_{DD} = 10.0 V dc
	0.0 V to 3.0 V dc @ V_{DD} = 15.0 V dc
Device types 52, 53, and 54	V_{OL} = 10% V_{DD}, V_{OH} = 90% V_{DD}
	0.0 V to 1.5 V dc @ V_{DD} = 5.0 V dc
	0.0 V to 2.0 V dc @ V_{DD} = 10.0 V dc
	0.0 V to 4.0 V dc @ V_{DD} = 15.0 V dc
Input high voltage range (V_{IH}):	
Device types 01, 02, 03, 04	3.95 V to 5.0 V dc @ V_{DD} = 5.0 V dc
	10.0 V to 12.5 V dc @ V_{DD} = 12.5 V dc
Device type 51	V_{OL} = 10% V_{DD}, V_{OH} = 90% V_{DD}
	4.0 V to 5.0 V dc @ V_{DD} = 5.0 V dc
	8.0 V to 10.0 V dc @ V_{DD} = 10.0 V dc
	12 V to 15.0 V dc @ V_{DD} = 15.0 V dc
Device types 52, 53, and 54	V_{OL} = 10% V_{DD}, V_{OH} = 90% V_{DD}
	3.5 V to 5.0 V dc @ V_{DD} = 5.0 V dc
	8.0 V to 10.0 V dc @ V_{DD} = 10.0 V dc
	11.0 V to 15.0 V dc @ V_{DD} = 15.0 V dc
Load capacitance	50 pF maximum
Case operating temperature range (T_C)	-55°C to +125°C

1/ Applied voltage (for device type 01 only) on pins 2, 4, 9, and 11 is ≥ V_{SS} and ≤ V_{DD}.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and Standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATION

MIL-PRF-38535 - Integrated Circuits (Microcircuits) Manufacturing, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-883 - Test Method Standard Microcircuits.
MIL-STD-1835 - Interface Standard Electronic Component Case Outlines.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Microcircuits furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.3 and 6.4).

3.2 Item requirements. The individual item requirements shall be in accordance with MIL-PRF-38535 and as specified herein or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not affect the form, fit, or function as described herein.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-PRF-38535 and herein. Although eutectic die bonding is preferred, epoxy die bonding may be performed. However the resin used shall be Dupont 5504 Conductive Silver Paste, or equivalent, which is cured at 200°C ±10°C for a minimum of 2 hours. The use of equivalent epoxies or cure cycles shall be approved by the qualifying activity. Equivalency shall be demonstrated in data submitted to the qualifying activity for verification.

3.3.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.3.2 Logic diagrams. The logic diagrams shall be as specified on figure 2.

3.3.3 Truth tables and logic equations. The truth tables and logic equations shall be as specified on figure 3.

3.3.4 Switching time test circuits and waveforms. The switching time test circuits and waveforms shall be as specified on figures 4 to 16.

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3.3.5 Schematic circuits. The schematic circuits shall be maintained by the manufacturer and made available to the qualifying activity or preparing activity upon request.

3.3.6 Case outlines. The case outlines shall be as specified in 1.2.3.

3.4 Lead material and finish. The lead material and finish shall be in accordance with MIL-PRF-38535 (see 6.6).

3.5 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range.

3.6 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.7 Marking. Marking shall be in accordance with MIL-PRF-38535.

3.7.1 Radiation hardness assurance identifier. The radiation hardness assurance identifier shall be in accordance with MIL-PRF-38535 and 4.5.4 herein.

3.8 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 36 (see MIL-PRF-38535, appendix A).

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions 1/ V_{SS} = 0 V -55°C ≤ T_C ≤ +125°C Unless otherwise specified		Device type	Limits		Unit
					Min	Max	
Positive clamping input to V_{DD}	$V_{IC\ (POS)}$	T_C = +25°C, V_{DD} = GND V_{SS} = Open, Output = Open I_{IN} = 1 mA		All		1.5	V dc
Negative clamping input to V_{SS}	$V_{IC\ (NEG)}$	T_C = +25°C, V_{DD} = Open V_{SS} = GND, Output = Open I_{IN} = -1 mA		All		-6.0	V dc
Quiescent supply current	I_{SS}	Any combination of inputs	V_{DD} = 15 V	01		-500	nA
				02, 03		-750	
				04		-3	µA
			V_{DD} = 18 V	51		-500	nA
				52, 53		-750	
				54		-3	µA
High level output voltage	V_{OH1}	V_{DD} = 4.5 V dc I_{OH} = -0.1 mA Any one input = V_{IL}		01, 02, 03	2.50		V
	V_{OH2}	V_{DD} = 5.0 V dc I_{OH} = -0.13 mA Any one input = V_{IL}		01, 02, 03	4.5		V
	V_{OH3}	V_{DD} = 5.0 V dc I_{OH} = 0.0 A Any one input = V_{IL}		01, 02, 03, 04	4.95		V
	V_{OH4}	V_{DD} = 12.5 V dc I_{OH} = 0.0 A Any one input = V_{IL}		01, 02, 03, 04	11.25		V
	V_{OH5}	V_{DD} = 15.0 V dc I_{OH} = 0.0 A		51, 52, 53, 54	14.95		V
Low level output voltage	V_{OL1}	V_{DD} = 5.5 V dc I_{OL} = 0.23 mA Any one input = V_{IH}		01, 02, 03		0.50	V
	V_{OL2}	V_{DD} = 5.0 V dc I_{OL} = 0.28 mA Any one input = V_{IH}		01, 02, 03		0.50	V
	V_{OL3}	V_{DD} = 5.0 V dc I_{OL} = 0.0 A Any one input = V_{IH}		01, 02, 03, 04		0.05	V
	V_{OL4}	V_{DD} = 12.5 V dc I_{OL} = 0.0 A Any one input = V_{IH}		01, 02, 03, 04		1.25	V
	V_{OL5}	V_{DD} = 5.0 V dc Any one input = V_{IH}		03		0.50	V
	V_{OL6}	V_{DD} = 15.0 V dc I_{OL} = 0.0 A		51, 52, 53, 54		0.05	V
Input high voltage	V_{IH1}	V_{DD} = 5 V dc V_O = 0.5 V $\lvert I_O \rvert \leq 1\mu A$		51	4.0		V dc
				52, 53, 54	3.5		
	V_{IH2}	V_{DD} = 10 V dc V_O = 1.0 V $\lvert I_O \rvert \leq 1\mu A$		51	8.0		V dc
				52, 53, 54	7.0		
	V_{IH3}	V_{DD} = 15 V dc V_O = 1.5 V $\lvert I_O \rvert \leq 1\mu A$		51	12.0		V dc
				52, 53, 54	11.0		

See footnotes at end of table.

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TABLE I. Electrical performance characteristics – Continued.

Test	Symbol	Conditions 1/ V_{SS} = 0 V -55°C ≤ T_C ≤ +125°C Unless otherwise specified		Device type	Limits		Unit
					Min	Max	
Input low voltage	V_{IL1}	V_{DD} = 5 V dc V_O = 4.5 V $\|I_O\|$ ≤ 1µA		51		1.0	V dc
				52, 53, 54		1.5	
	V_{IL2}	V_{DD} = 10 V dc V_O = 9.0 V $\|I_O\|$ ≤ 1µA		51		2.0	V dc
				52, 53, 54		2.0	
	V_{IL3}	V_{DD} = 15 V dc V_O = 13.5 V $\|I_O\|$ ≤ 1µA		51		3.0	V dc
				52, 53, 54		4.0	
Output low (sink) current	I_{OL1}	V_{DD} = 5 V dc V_{IN} = 5.0 V dc V_{OL} = 0.4 V dc		51, 52, 53, 54	0.36		mA
	I_{OL2}	V_{DD} = 15 V dc V_{IN} = 15 V dc V_{OL} = 1.5 V dc		51, 52, 53, 54	2.4		mA
Output high (source) current	I_{OH1}	V_{DD} = 5 V dc All inputs = GND V_{OH} = 4.6 V dc		51, 52, 53, 54	-0.36		mA
	I_{OH2}	V_{DD} = 15 V dc All inputs = GND V_{OH} = 13.5 V dc		51, 52, 53, 54	-2.4		mA
Input leakage 2/ current, high	I_{IH}	Measure inputs sequentially	V_{DD} = 15 V	01, 02		100	nA
				03, 04		45	
			V_{DD} = 18 V	51, 54		45	
				52, 53		100	
Input leakage 2/ current, low	I_{IL}	Measure inputs sequentially	V_{DD} = 15 V	01, 02		-100	nA
				03, 04		-45	
			V_{DD} = 18 V	51, 54		-45	
				52, 53		-100	
Three-state output leakage current	I_{OC1}	Measure inputs sequentially V_O = V_{DD}	V_{DD} = 15 V dc	04		90	nA
			V_{DD} = 18 V dc	54			
Three-state output leakage current	I_{OC2}	Measure outputs sequentially V_O = V_{SS}	V_{DD} = 15 V dc	04		-90	nA
			V_{DD} = 18 V dc	54			
Input capacitance	C_I	V_{DD} = 0 V dc, f = 1 MHz T_C = +25°C		All		12	pF
Propagation delay time, high-to-low level	t_{PHL}	V_{DD} = 5.0 V dc, C_L = 50 pF R_L = 200 kΩ (see figure 4)		01	6	255	ns
				51	6	155	
				02	13	525	ns
				52	13	420	
				03	10	270	ns
				53	10	350	

See footnotes at end of table.

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TABLE I. Electrical performance characteristics – Continued.

Test	Symbol	Conditions 1/ V_{SS} = 0 V -55°C ≤ T_C ≤ +125°C unless otherwise specified	Device type	Limits		Unit
				Min	Max	
Propagation delay time, low-to-high level	t_{PLH}	V_{DD} = 5.0 V dc, C_L = 50 pF R_L = 200 kΩ (see figure 4)	01	6	165	ns
			51	6	155	
			02	13	525	ns
			52	13	420	
			03	10	270	ns
			53	10	350	
Transition time, high-to-low level	t_{THL}	V_{DD} = 5.0 V dc, C_L = 50 pF R_L = 200 kΩ (see figure 4)	01	6	360	ns
			51	6	280	
			02	10	525	ns
			52	10	420	
			03	10	380	ns
			53	10	280	
Transition time, low-to-high level	t_{TLH}	V_{DD} = 5.0 V dc, C_L = 50 pF R_L = 200 kΩ (see figure 4)	01	6	240	ns
			51	6	280	
			02	10	525	ns
			52	10	420	
			03	10	530	ns
			53	10	280	
Propagation delay time, high-to-low level (NOR gate)	t_{PHL1}	V_{DD} = 5.0 V dc, C_L = 50 pF R_L = 200 kΩ (see figures 5-15)	04	30	820	ns
			54			
Propagation delay time, high-to-low level (OR gate)	t_{PHL2}		04	28	755	ns
			54			
Propagation delay time, high-to-low level (AND gate)	t_{PHL3}		04	33	905	ns
			54			
Propagation delay time, high-to-low level (NAND gate)	t_{PHL4}		04	29	670	ns
			54			
Propagation delay time, high-to-low level (OR/AND gate)	t_{PHL5}		04	29	775	ns
			54			
Propagation delay time, high-to-low level (OR/NAND gate)	t_{PHL6}		04	27	755	ns
			54			
Propagation delay time, high-to-low level (AND/NOR gate)	t_{PHL7}		04	30	790	ns
			54			
Propagation delay time, high-to-low level (AND/OR gate)	t_{PHL8}		04	31	840	ns
			54			
Propagation delay time, high-to-low level (Ka to output)	t_{PHL9}		04	30	760	ns
			54			
Propagation delay time, high-to-low level (Kb to output)	t_{PHL10}		04	23	615	ns
			54			

See footnotes at end of table.

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Test	Symbol	Conditions 1/ V_{SS} = 0 V -55°C ≤ T_C ≤ +125°C unless otherwise specified	Device type	Limits		Unit
				Min	Max	
Propagation delay time, high-to-low level (Kc to output)	t_{PHL11}	V_{DD} = 5.0 V dc, C_L = 50 pF R_L = 200 kΩ (see figures 5-15)	04 54	18	475	ns
Propagation delay time, low-to-high level (NOR gate)	t_{PLH1}		04 54	24	640	ns
Propagation delay time, low-to-high level (OR gate)	t_{PLH2}		04 54	28	750	ns
Propagation delay time, low-to-high level (AND gate)	t_{PLH3}		04 54	25	655	ns
Propagation delay time, low-to-high level (NAND gate)	t_{PLH4}		04 54	31	830	ns
Propagation delay time, low-to-high level (OR/AND gate)	t_{PLH5}		04 54	24	645	ns
Propagation delay time, low-to-high level (OR/NAND gate)	t_{PLH6}		04 54	26	700	ns
Propagation delay time, low-to-high level (AND/NOR gate)	t_{PLH7}		04 54	27	730	ns
Propagation delay time, low-to-high level (AND/OR gate)	t_{PLH8}		04 54	27	720	ns
Propagation delay time, low-to-high level (Ka to output)	t_{PLH9}		04 54	28	730	ns
Propagation delay time, low-to-high level (Kb to output)	t_{PLH10}		04 54	23	605	ns
Propagation delay time, low-to-high level (Kc to output)	t_{PLH11}		04 54	17	445	ns
Transition time, high-to-low level	t_{THL}	V_{DD} = 5.0 V dc, C_L = 50 pF R_L = 200 kΩ (see figure 5)	04 54	6	160	ns
Transition time, low-to-high level	t_{TLH}		04 54	5	140	ns
Propagation delay time, low to Z, high to Z (Kd input to output)	t_{PLZ}, t_{PHZ}	V_{DD} = 5.0 V dc, C_L = 50 pF R_L = 1.0 kΩ (see figure 16)	04 54	18	490	ns
Propagation delay time, Z to low, Z to high (Kd input to output)	t_{PZL}, t_{PZH}	V_{DD} = 5.0 V dc, C_L = 50 pF R_L = 1.0 kΩ (see figure 16)	04 54	23	630	ns

1/ Complete terminal conditions shall be as specified in table III.
2/ Input current at one input mode.

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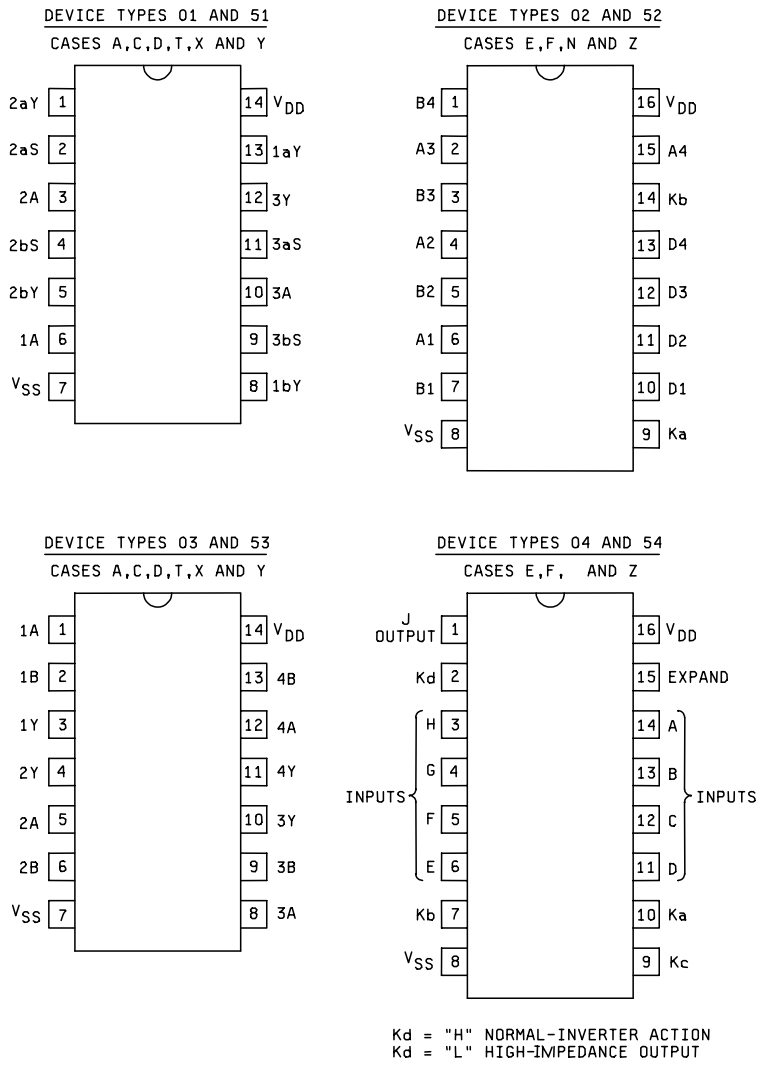


FIGURE 1. Terminal connections.

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Device types 01 and 51

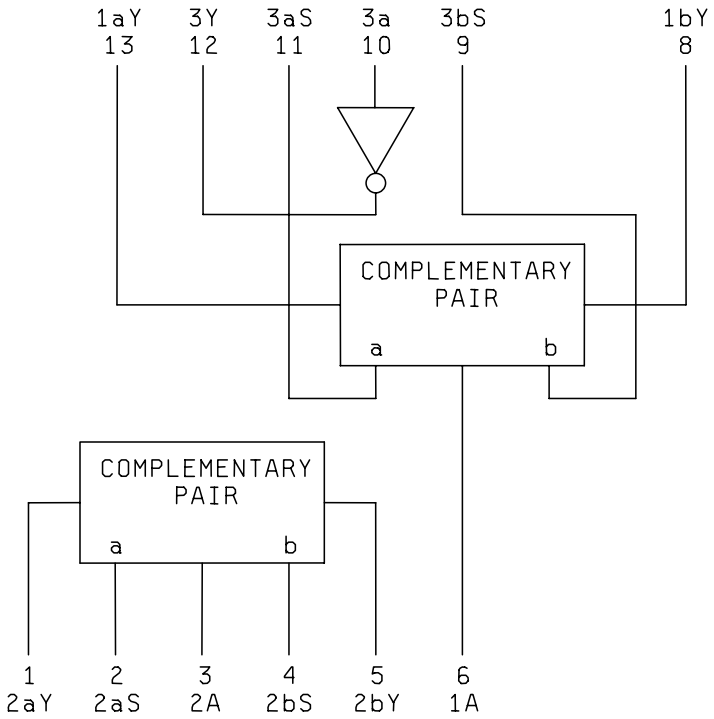
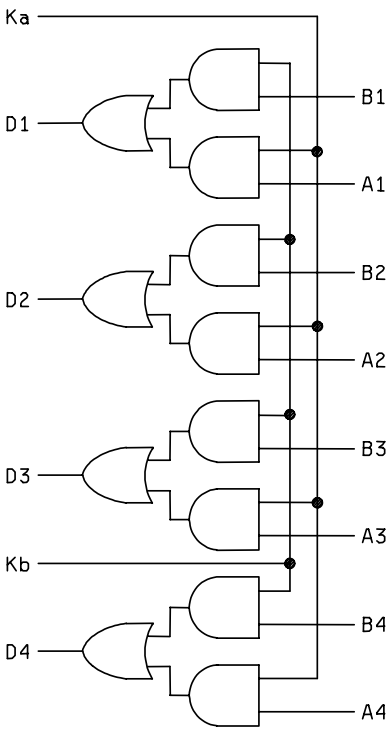


FIGURE 2. Logic diagrams.

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DEVICE TYPES 02 AND 52



DEVICE TYPES 03 AND 53

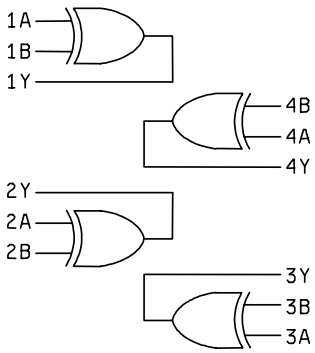


FIGURE 2. Logic diagrams – Continued.

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DEVICE TYPE 04 AND 54

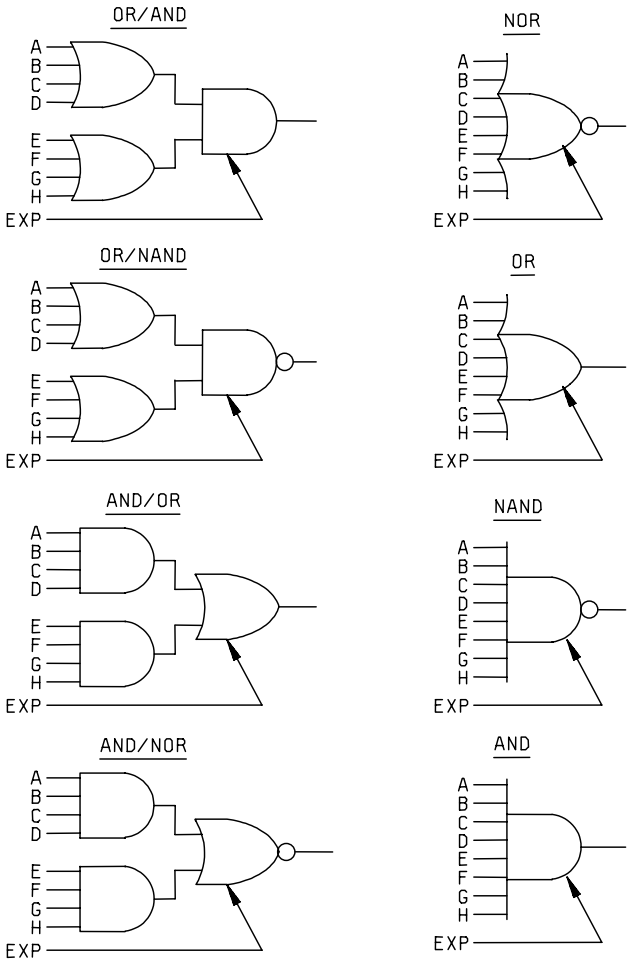


FIGURE 2. Logic diagrams – Continued.

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Device types 01 and 51

Complimentary pair
(not applicable)

INVERTER
Truth table

Input	Output
3A	3Y
H	L
L	H

Positive logic $3Y = \overline{3A}$

Device types 02 and 52

Truth table

Inputs				Ouptut
An	Ka	Bn	Kb	Dn
L	L	L	L	L
H	L	L	L	L
L	H	L	L	L
L	L	H	L	L
L	L	L	H	L
H	H	L	L	H
L	H	H	L	L
L	L	H	H	H
H	L	L	H	L
H	L	H	L	L
L	H	L	H	L
H	H	H	L	H
L	H	H	H	H
H	L	H	H	H
H	H	L	H	H
H	H	H	H	H

Positive logic Dn = (An • Ka) + (Bn • Kb)
n = 1, 2, 3, or 4

Device type 03 and 53

Truth table each gate

Input		Output
A	B	Y
L	L	L
L	H	H
H	L	H
H	H	L

Positive logic Y = A ⊕ B

Device types 04 and 54

Function truth table

Output function	Boolean expression	Ka	Kb	Kc	Unused input
NOR	$J = \overline{A+B+C+D+E+F+G+H}$	0	0	0	V_{SS}
OR	J = A+B+C+D+E+F+G+H	0	0	1	V_{SS}
OR/AND	J = (A+B+C+D) • (E+F+G+H)	0	1	0	V_{SS}
OR/NAND	$J = \overline{(A+B+C+D) \bullet (E+F+G+H)}$	0	1	1	V_{SS}
AND	J = ABCDEFGH	1	0	0	V_{DD}
NAND	$J = \overline{ABCDEFGH}$	1	0	1	V_{DD}
AND/NOR	$J = \overline{ABCD+EFGH}$	1	1	0	V_{DD}
AND/OR	J = ABCD+EFGH	1	1	1	V_{DD}

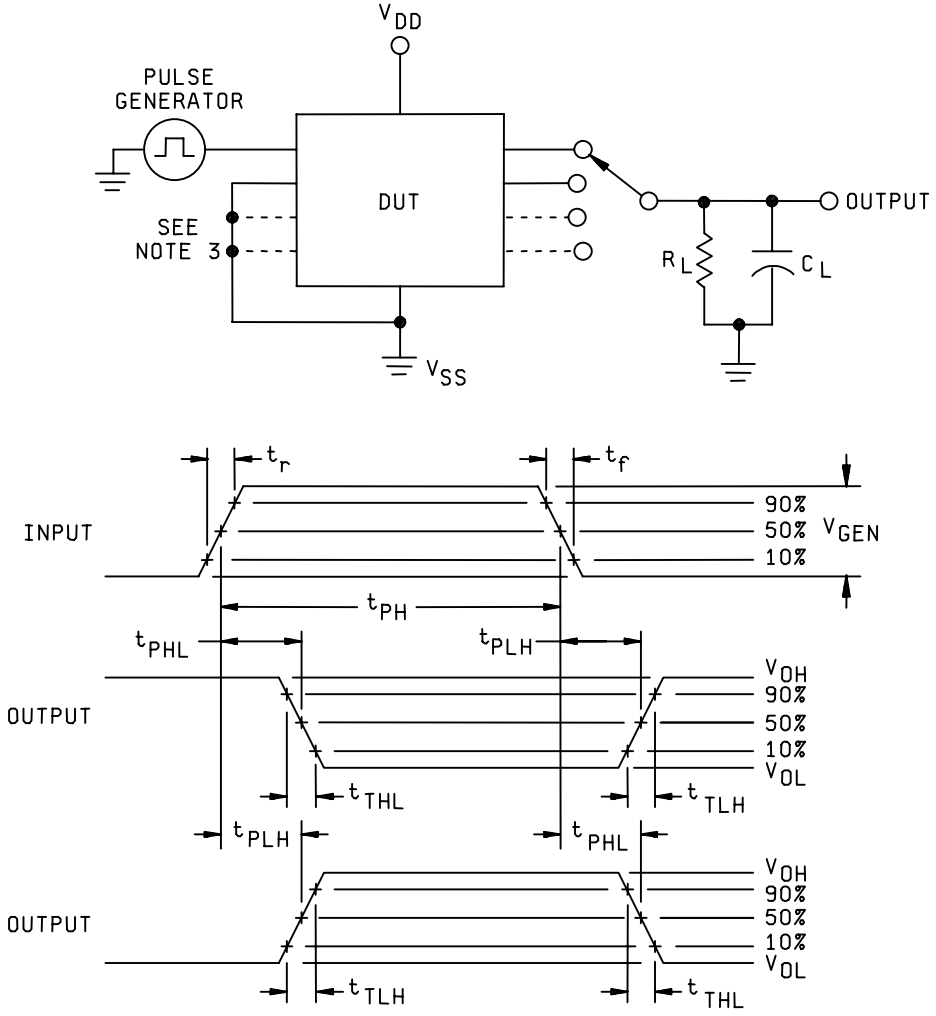
Kd = 1 Normal-inverter action
Kd = 0 High-impedance output

Expand input = 0

H = High level voltage
L = Low level voltage

FIGURE 3. Truth tables and logic equations.

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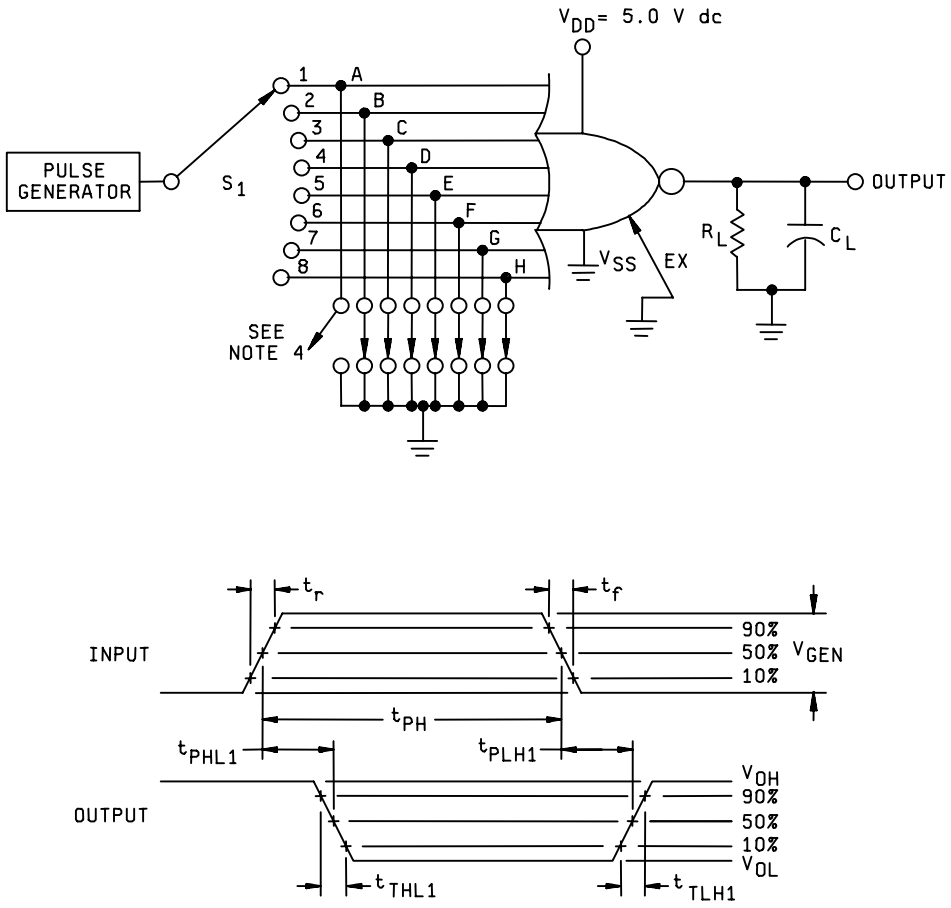


NOTES:

1. Pulse generator has the following characteristics: $V_{GEN} = V_{DD} \pm 1\%$, $t_{PH} = 1.0 \pm 0.1$ µs, t_r and $t_f \leq 10$ ns .
2. See table III for complete terminal conditions.
3. All unused inputs must be tied to V_{SS}.
4. $R_L = 200$ kΩ ±10 % and $C_L = 50$ pF ±5 % including wiring and probe capacitance.

FIGURE 4. Switching time test circuit and waveforms for device types 01, 02, 03, 51, 52, and 53.

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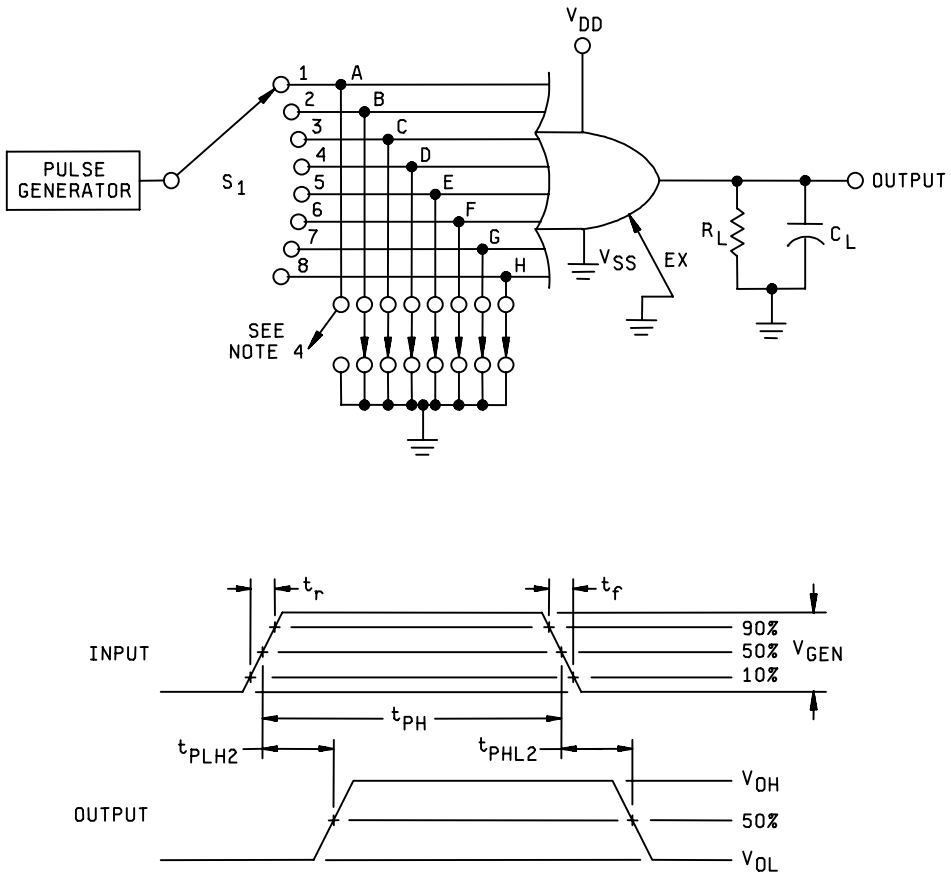


NOTES:

1. Pulse generator has the following characteristics: $V_{GEN} = V_{DD} \pm 1\%$, $t_{PH} = 1.0 \pm 0.1$ µs, t_r and $t_f \leq 10$ ns .
2. $R_L = 200\ k\Omega \pm 10$ % and $C_L = 50$ pF ±5 % including wiring and probe capacitance.
3. See table III for complete terminal conditions.
4. All inputs are tied to V_{SS} except for input connected to the pulse generator.

FIGURE 5. Switching time test circuit and waveforms for device types 04 and 54 (NOR gate).

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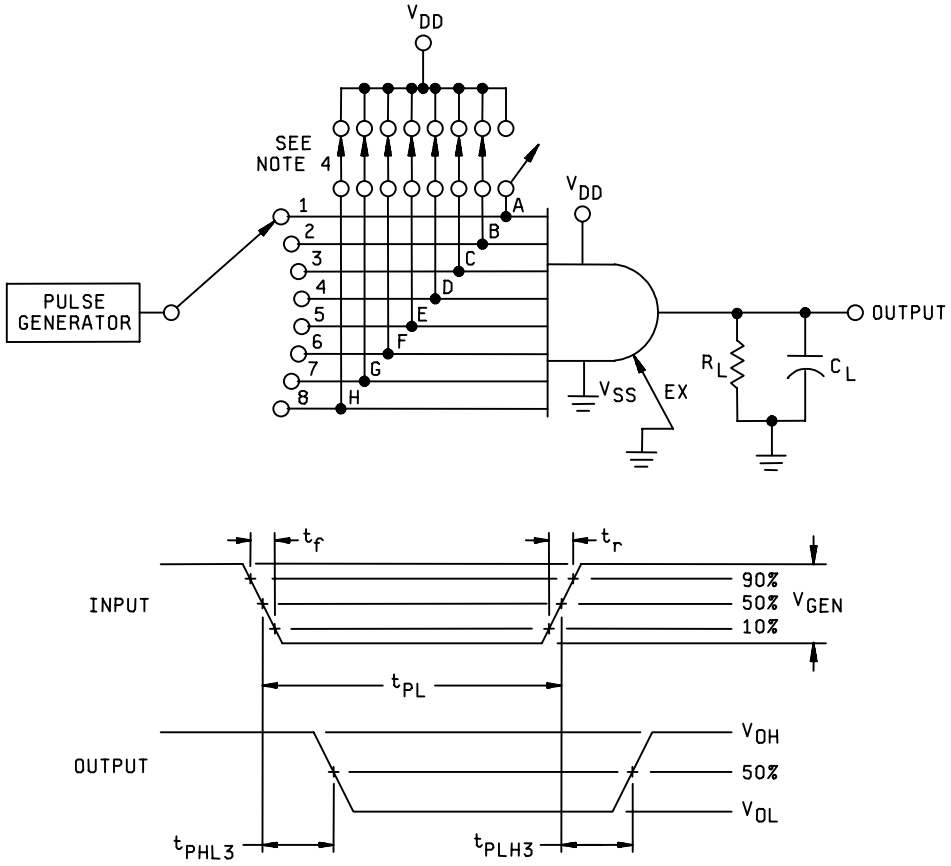


NOTES:

1. Pulse generator has the following characteristics: $V_{GEN} = V_{DD} \pm 1\%$, $t_{PH} = 1.0 \pm 0.1$ µs, t_r and $t_f \leq 10$ ns .
2. $R_L = 200$ kΩ ±10 % and $C_L = 50$ pF ±5 % including wiring and probe capacitance.
3. See table III for complete terminal conditions.
4. All inputs are tied to V_{SS} except for input connected to the pulse generator.

FIGURE 6. Switching time test circuit and waveforms for device types 04 and 54 (OR gate).

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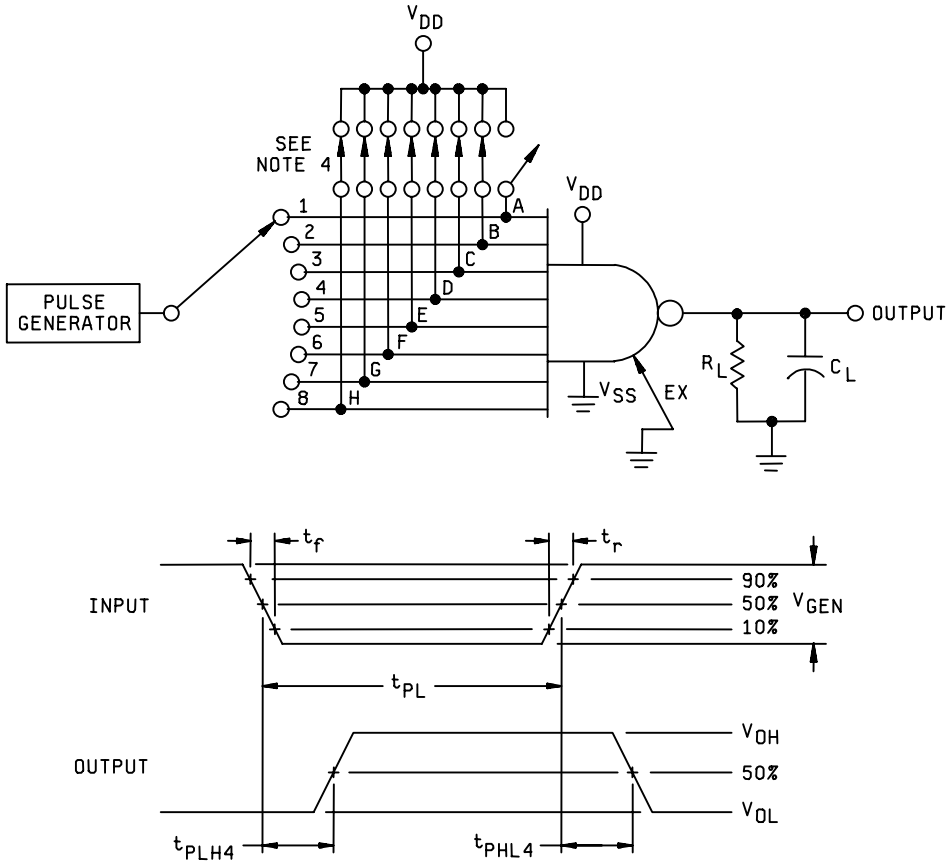


NOTES:

1. Pulse generator has the following characteristics: $V_{GEN} = V_{DD} \pm 1\%$, $t_{PL} = 1.0 \pm 0.1$ µs, t_r and $t_f \leq 10$ ns .
2. $R_L = 200$ kΩ ±10 % and $C_L = 50$ pF ±5 % including wiring and probe capacitance.
3. See table III for complete terminal conditions.
4. All inputs are tied to V_{DD} except for input connected to the pulse generator.

FIGURE 7. Switching time test circuit and waveforms for device types 04 and 54 (AND gate).

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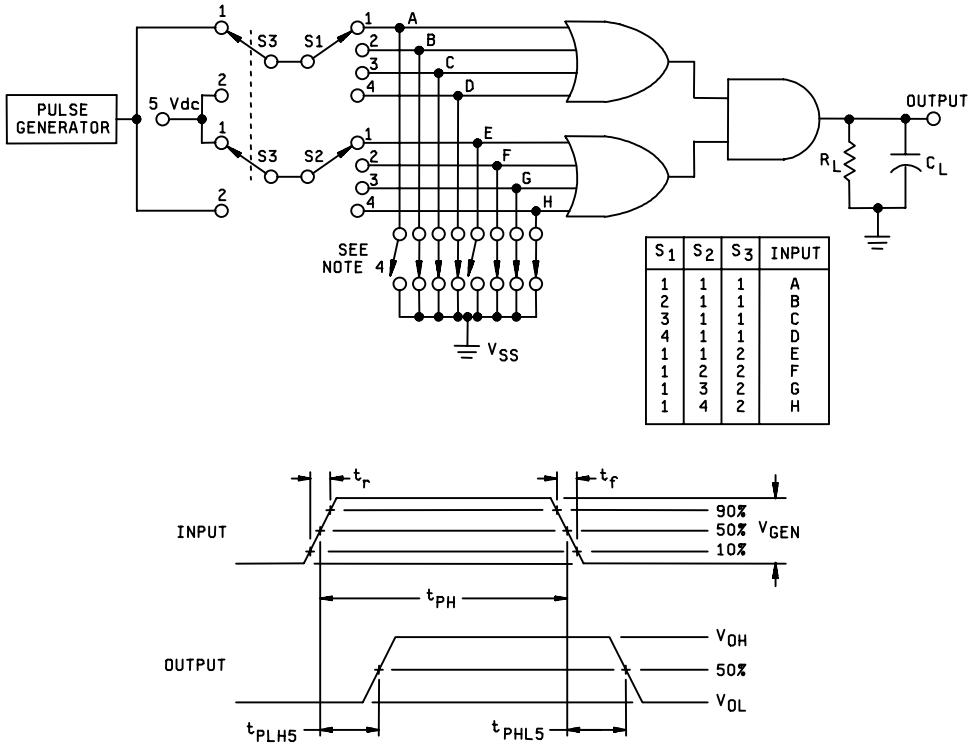


NOTES:

1. Pulse generator has the following characteristics: $V_{GEN} = V_{DD} \pm 1\%$, $t_{PL} = 1.0 \pm 0.1$ µs, t_r and $t_f \leq 10$ ns .
2. $R_L = 200$ kΩ ±10 % and $C_L = 50$ pF ±5 % including wiring and probe capacitance.
3. See table III for complete terminal conditions.
4. All inputs are tied to V_{DD} except for input connected to the pulse generator.

FIGURE 8. Switching time test circuit and waveforms for device types 04 and 54 (NAND gate).

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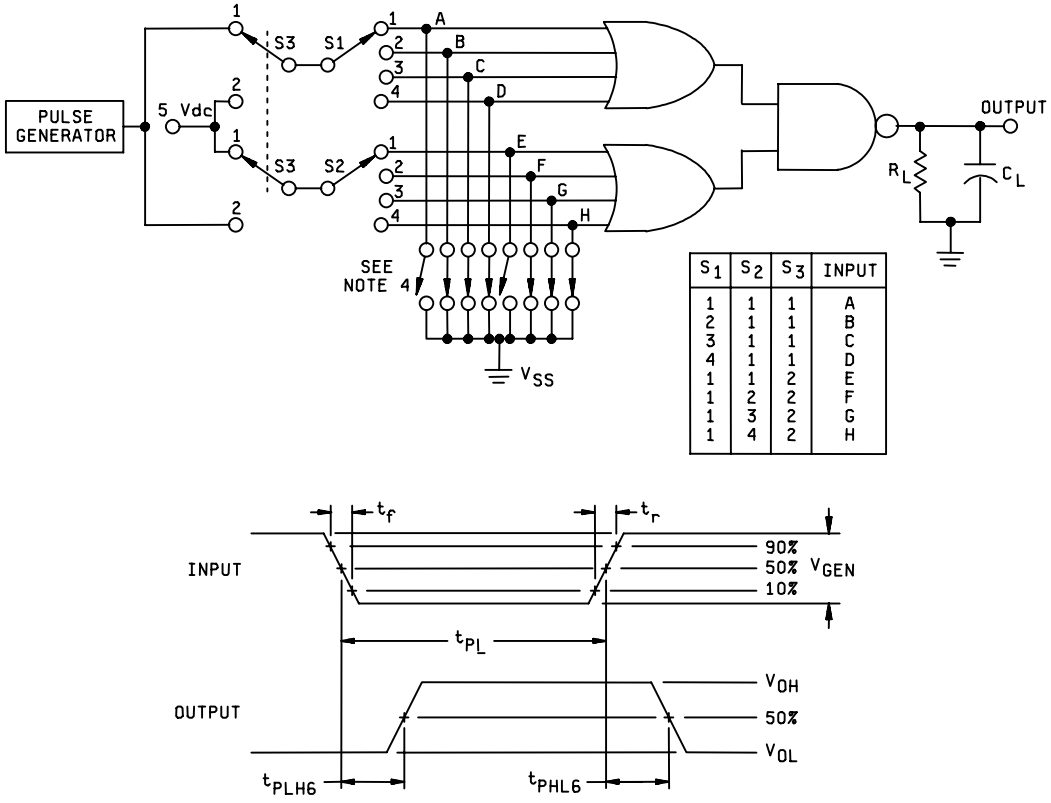


NOTES:

1. Pulse generator has the following characteristics: $V_{GEN} = V_{DD} \pm 1\%$, $t_{PH} = 1.0 \pm 0.1$ µs, t_r and $t_f \leq 10$ ns .
2. $R_L = 200\ k\Omega \pm 10$ % and $C_L = 50$ pF ±5 % including wiring and probe capacitance.
3. See table III for complete terminal conditions.
4. All inputs are tied to V_{SS} except for input connected to S1 and S2.

FIGURE 9. Switching time test circuit and waveforms for device types 04 and 54 (OR/AND gate).

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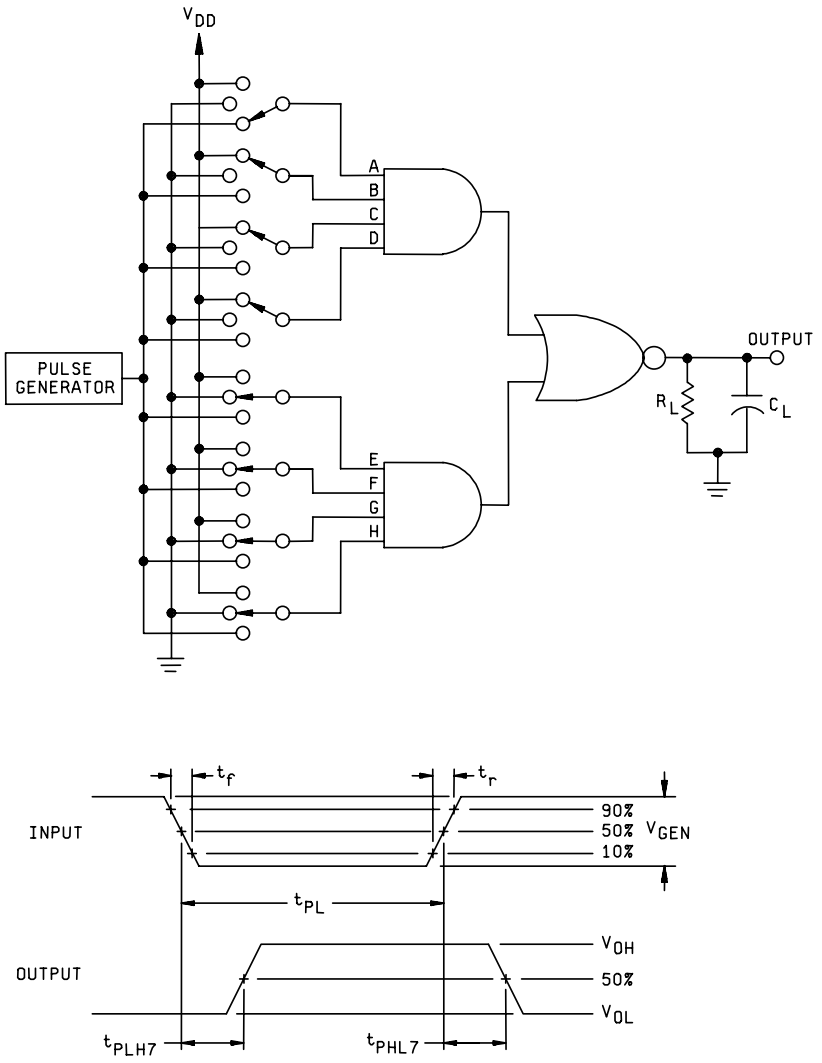
S_1	S_2	S_3	INPUT
1	1	1	A
2	1	1	B
3	1	1	C
4	1	1	D
1	1	2	E
1	2	2	F
1	3	2	G
1	4	2	H

NOTES:

1. Pulse generator has the following characteristics: $V_{GEN} = V_{DD} \pm 1\%$, $t_{PL} = 1.0 \pm 0.1$ µs, t_r and $t_f \leq 10$ ns .
2. $R_L = 200\ k\Omega \pm 10$ % and $C_L = 50$ pF ±5 % including wiring and probe capacitance.
3. See table III for complete terminal conditions.
4. All inputs are tied to V_{SS} except for input connected to S1 and S2.

FIGURE 10. Switching time test circuit and waveforms for device types 04 and 54 (OR/NAND gate).

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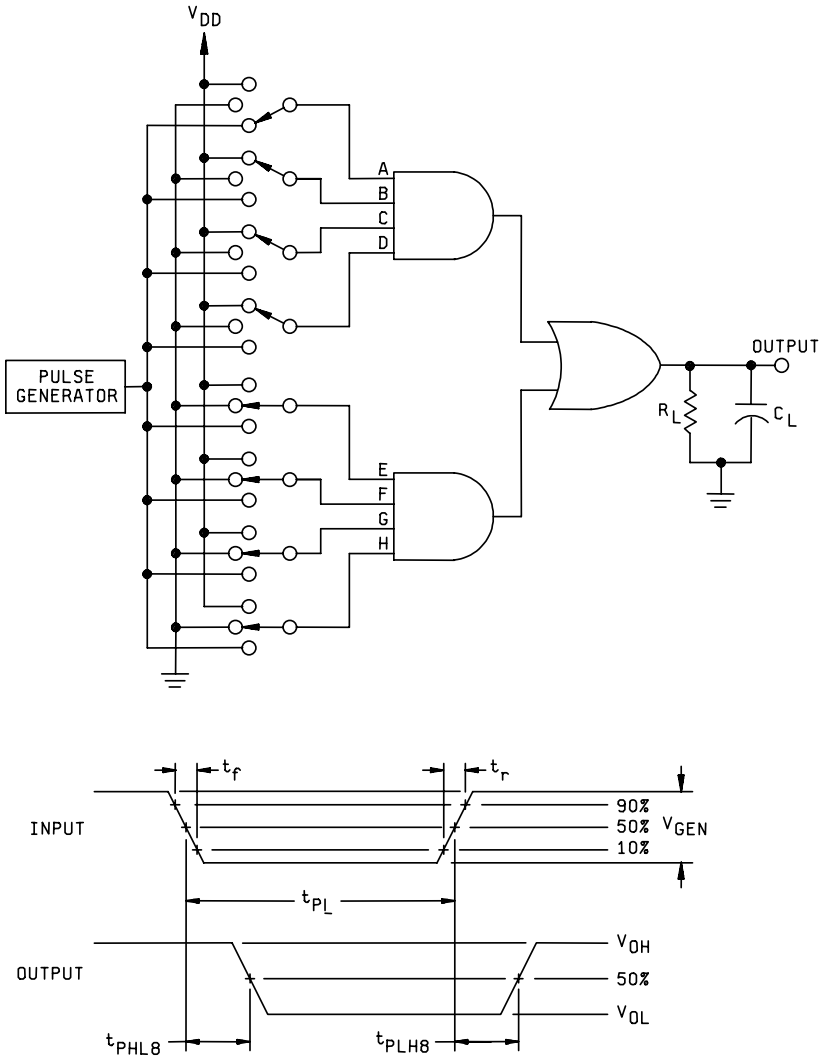


NOTES:

1. Pulse generator has the following characteristics: $V_{GEN} = V_{DD} \pm 1\%$, $t_{PL} = 1.0 \pm 0.1$ µs, t_r and $t_f \leq 10$ ns .
2. $R_L = 200$ kΩ ±10 % and $C_L = 50$ pF ±5 % including wiring and probe capacitance.
3. See table III for complete terminal conditions.

FIGURE 11. Switching time test circuit and waveforms for device types 04 and 54 (AND/NOR gate).

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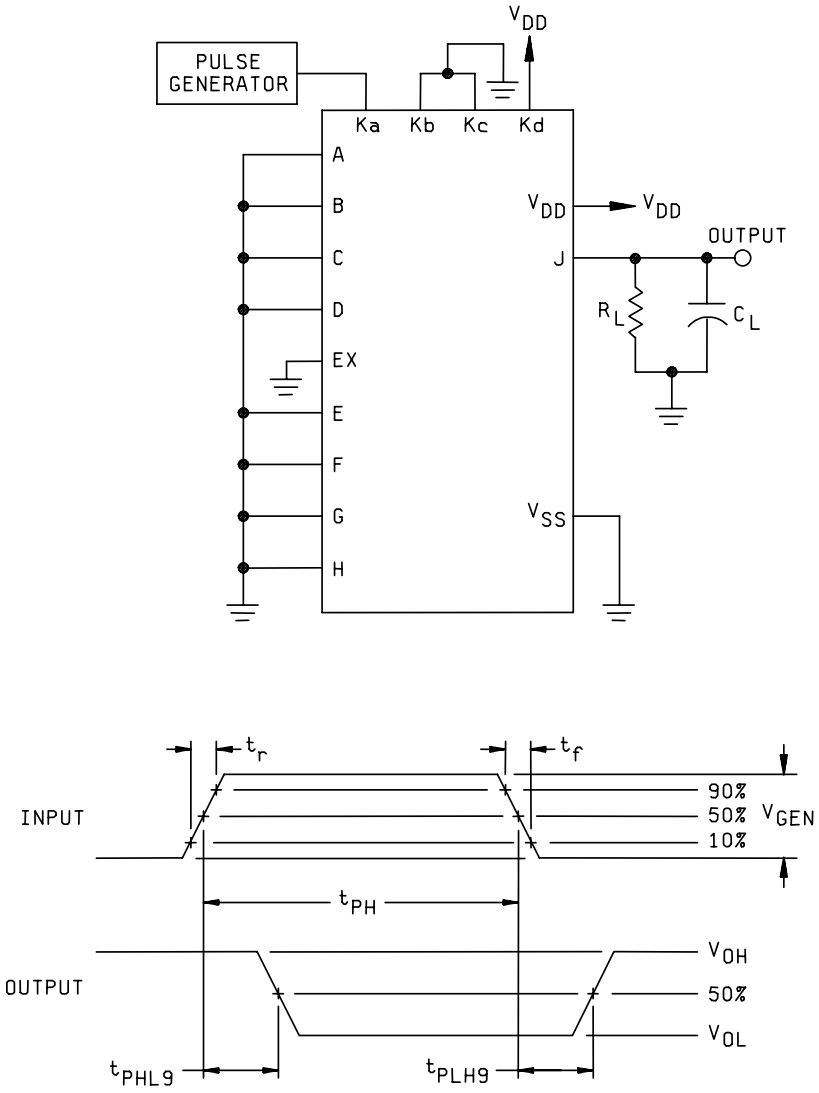


NOTES:

1. Pulse generator has the following characteristics: $V_{GEN} = V_{DD} \pm 1\%$, $t_{PL} = 1.0 \pm 0.1$ µs, t_r and $t_f \leq 10$ ns .
2. $R_L = 200$ kΩ ±10 % and $C_L = 50$ pF ±5 % including wiring and probe capacitance.
3. See table III for complete terminal conditions.

FIGURE 12. Switching time test circuit and waveforms for device types 04 and 54 (AND/OR gate).

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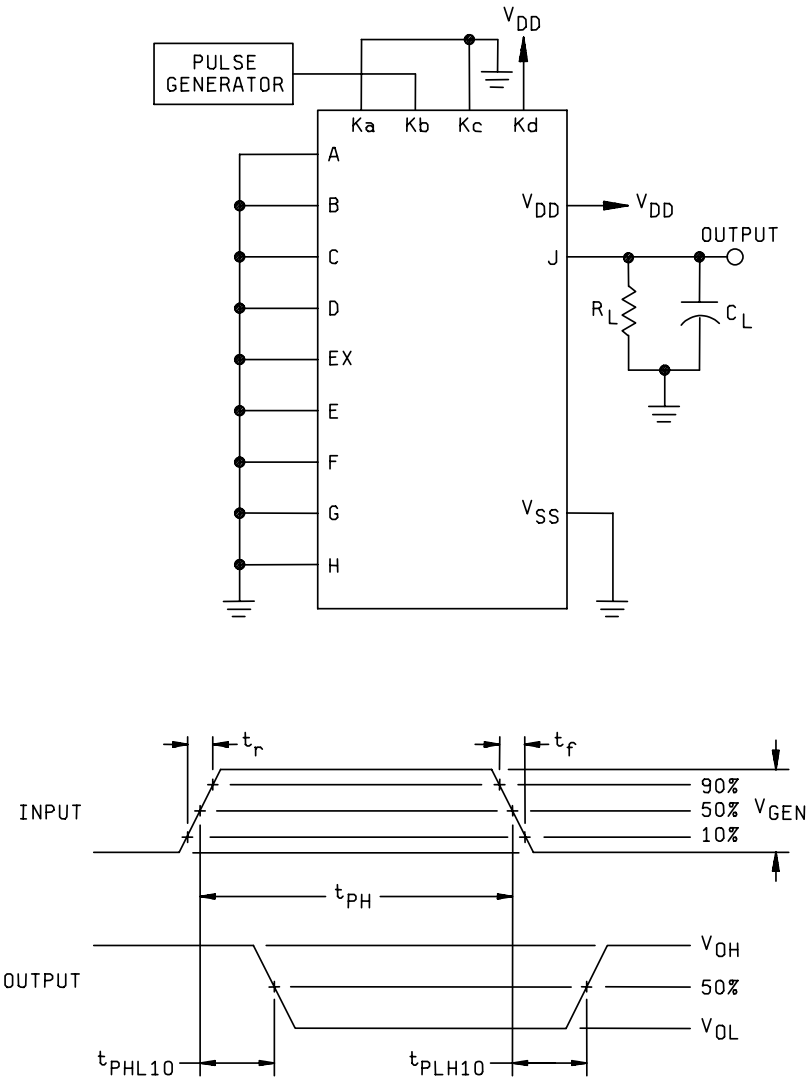


NOTES:

1. Pulse generator has the following characteristics: $V_{GEN} = V_{DD} \pm 1\%$, $t_{PH} = 1.0 \pm 0.1$ µs, t_r and $t_f \leq 10$ ns .
2. $R_L = 200\ k\Omega \pm 10$ % and $C_L = 50$ pF ±5 % including wiring and probe capacitance.
3. See table III for complete terminal conditions.

FIGURE 13. Switching time test circuit and waveforms for device types 04 and 54 (Ka to output).

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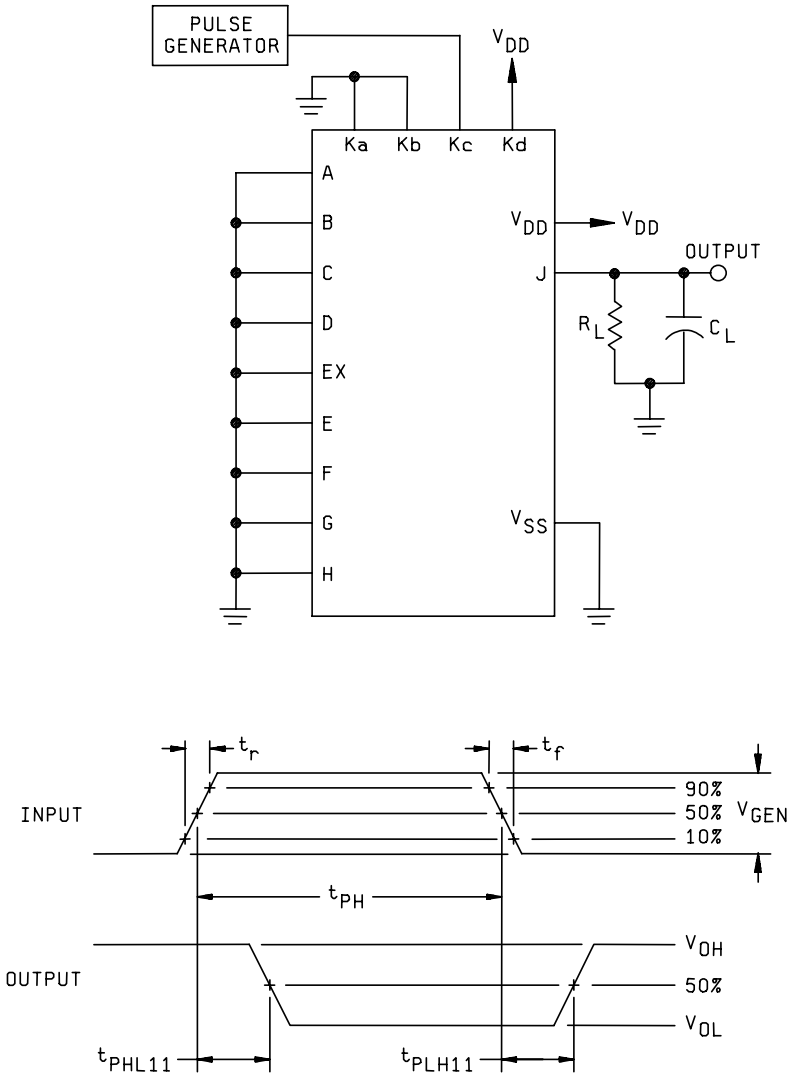


NOTES:

1. Pulse generator has the following characteristics: $V_{GEN} = V_{DD} \pm 1\%$, $t_{PH} = 1.0 \pm 0.1$ µs, t_r and $t_f \leq 10$ ns .
2. $R_L = 200\ k\Omega \pm 10$ % and $C_L = 50$ pF ±5 % including wiring and probe capacitance.
3. See table III for complete terminal conditions.

FIGURE 14. Switching time test circuit and waveforms for device types 04 and 54 (Kb to output).

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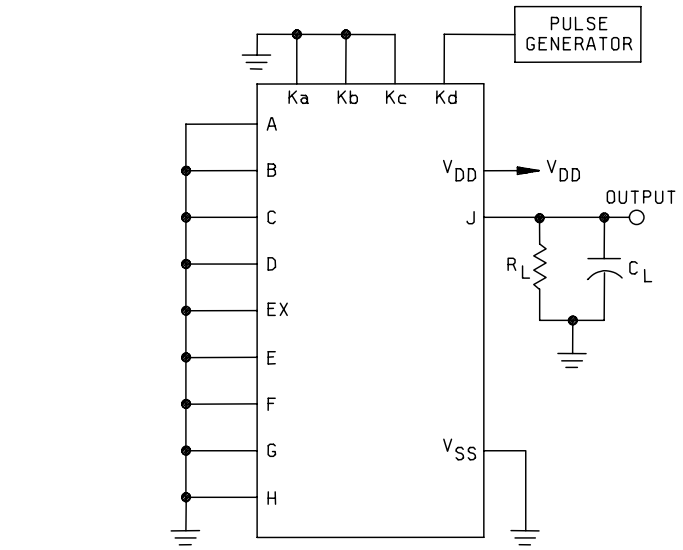


NOTES:

1. Pulse generator has the following characteristics: $V_{GEN} = V_{DD} \pm 1\%$, $t_{PH} = 1.0 \pm 0.1$ µs, t_r and $t_f \leq 10$ ns .
2. $R_L = 200\ k\Omega \pm 10$ % and $C_L = 50$ pF ±5 % including wiring and probe capacitance.
3. See table III for complete terminal conditions.

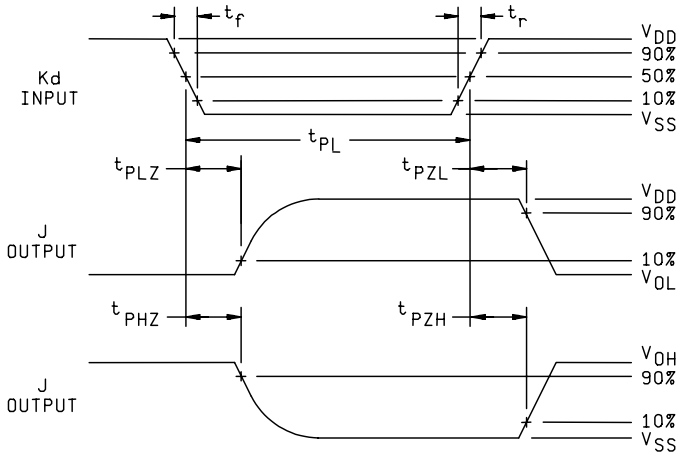
FIGURE 15. Switching time test circuit and waveforms for device types 04 and 54 (Kc to output).

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TEST CIRCUIT

Kd INPUT TO OUTPUT



NOTES:

1. Pulse generator has the following characteristics: $V_{GEN} = V_{DD} \pm 1\%$, $t_{PL} = 1.0 \pm 0.1$ µs, t_r and $t_f \leq 10$ ns .
2. $R_L = 1.0$ kΩ ±5 % and $C_L = 50$ pF ±5 % including wiring and probe capacitance.
3. See table III for complete terminal conditions.
4. $R_L = 1.0$ kΩ ±5 % to V_{DD} for t_{PLZ}.

FIGURE 16. Switching time test circuit and waveforms for device types 04 and 54 (Kd to output).

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4. VERIFICATION

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-PRF-38535 or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not affect the form, fit, or function as described herein.

4.2 Screening. Screening shall be in accordance with MIL-PRF-38535 and shall be conducted on all devices prior to qualification and conformance inspection. The following additional criteria shall apply:

a. The burn-in test duration, test condition, and test temperature, or approved alternatives shall be as specified in the device manufacturer's QM plan in accordance with MIL-PRF-38535. The burn-in test circuit shall be maintained under document control by the device manufacturer's Technology Review Board (TRB) in accordance with MIL-PRF-38535 and shall be made available to the acquiring or preparing activity upon request. The test circuit shall specify the inputs, outputs, biases, and power dissipation, as applicable, in accordance with the intent specified in test method 1015 of MIL-STD-883.

b. Delete the sequence specified as interim (pre-burn-in) electrical parameters through interim (post-burn-in) electrical parameters of table IA of MIL-PRF-38535 and substitute lines 1 through 7 of table II herein.

c. Burn-in (method 1015 of MIL-STD-883).

(1) Unless otherwise specified in the manufacturers QM plan for static tests (test condition A), case temperature (T_C) shall be +125°C minimum. Test duration for each static test shall be 24 hours minimum for class S devices and in accordance with table I of method 1015 for class B devices.

i. For static burn-in I, all inputs shall be connected to 0.0 V.

ii. For static burn-in II, all inputs shall be connected to V_{DD}.

iii. Except for V_{DD} and V_{SS}, the terminal shall be connected through resistors whose value is 2 kΩ to 47 kΩ. The actual measured value of the resistor selected shall not exceed ±20% of its branded value due to use, heat or age.

iv. Output may be open or connected to $V_{DD}/2$.

v. For device types 01 and 51:
(a) Connect pins 2, 11, and 14 together.
(b) Connect pins 4, 7, and 9 together.
(c) Connect pins 8 and 13 together.
(d) Connect pins 1 and 5 together.

vi. V_{DD} = 12.5 V minimum and 15 V maximum for device types 01, 02, 03, and 04.
V_{DD} = 15 V minimum and 18 V maximum for device types 51, 52, 53, and 54.
$V_{DD}/2 = V_{DD}/2 \pm 1.0$ V for all devices. V_{SS} = 0.0 V.

(2) Unless otherwise specified in the manufacturers QM plan for dynamic test (test condition D), case temperature shall be +125°C minimum. Test duration shall be in accordance with table I of method 1015.

i. Except for V_{DD} and V_{SS}, the terminals shall be connected through resistors whose value is 2 kΩ to 47 kΩ. The actual measured value of the resistor selected shall not exceed ±20% of its branded value due to use, heat or age.

ii. Input signal requirements: Square wave, 50% duty cycle; 25 kHz < PRR < 1 MHz; t_{TLH} and t_{THL} < 1 µs. Voltage level: Minimum = V_{SS} – 0.5 V, +10% V_{DD}; maximum = V_{DD} + 0.5 V, -10% V_{DD}.

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iii For device types 01 and 51:
(a) Connect pins 2, 11, and 14 together.
(b) Connect pins 4, 7, and 9 together.
(c) Connect pins 8 and 13 together.
(d) Connect pins 1 and 5 together.

iv V_{DD} = 12.5 V minimum and 15 V maximum for device types 01, 02, 03, and 04.
V_{DD} = 15 V minimum and 18 V maximum for device types 51, 52, 53, and 54.
$V_{DD}/2 = V_{DD}/2 \pm 1.0$ V. V_{SS} = 0.0 V.

d. Interim and final electrical test parameters shall be as specified in table II.

e. For class S devices, post dynamic burn-in, or class B devices, post static burn-in, electrical parameter measurements may, at the manufacturer's option, be performed separately or included in the final electrical parameter requirements.

f. When device types 01 through 04 are qualified by extension (see 4.3.1), they shall be screened in accordance with the requirements for corresponding device types 51 through 54, respectively.

TABLE II. Electrical test requirements.

Line no.	MIL-PRF-38535 test requirements	Class S device 1/			Class B device 1/		
		Ref. par.	Table III Subgroups 2/	Table IV delta limits 3/	Ref. par.	Table III subgroups 2/	Table IV delta limits 3/
1	Interim electrical parameters		1			1	
2	Static burn-in I (method 1015)	4.2c 4.5.2					
3	Same as line 1		1	Δ			
4	Static burn-in II (method 1015)	4.2c 4.5.2			4.2c 4.5.2	4/	
5	Same as line 1	4.2e	1*	Δ	4.2e	1*	Δ
6	Dynamic burn-in (method 1015)	4.2c 4.5.2					
7	Same as line 1	4.2e	1*	Δ			
8	Final electrical parameters (method 5004)		1*, 2, 3, 9			1*, 2, 3, 9	
9	Group A test requirements (method 5005)	4.4.1	1, 2, 3, 4, 9, 10, 11		4.4.1	1, 2, 3, 4, 9, 10, 11	
10	Group B test when using method 5005 QCI option	4.4.2	1, 2, 3, 9, 10, 11	Δ			
11	Group C end-point electrical parameters (method 5005)				4.4.3	1, 2, 3	Δ
12	Group D end-point electrical parameters (method 5005)	4.4.4	1, 2, 3		4.4.4	1, 2, 3	

1/ Blank spaces indicate tests are not applicable.
2/ * indicates PDA applies to subgroup 1 (see 4.2.1).
3/ Δ indicates delta limits shall be required only on table III subgroup 1, where specified, and the delta values shall be computed with reference to the previous interim electrical parameters.
4/ The device manufacturer may at his option either perform delta measurements or within 24 hours after burn-in (or removal of bias) perform the final electrical parameter measurements.

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4.2.1 Percent defective allowable (PDA).

a. The PDA for class S devices shall be 5 percent for static burn-in and 5 percent for dynamic burn-in, based on the exact number of devices submitted to each separate burn-in.

b. Static burn-in I and II failure shall be cumulative for determining the PDA.

c. The PDA for class B devices shall be in accordance with MIL-PRF-38535 for static burn-in. Dynamic burn-in is not required.

d. Those devices whose measured characteristics, after burn-in, exceed the specified delta (Δ) limits or electrical parameter limits specified in table III, subgroup 1, are defective and shall be removed from the lot. The verified failures divided by the total number of devices in the lot initially submitted to burn-in shall be used to determine the percent defective for the lot and the lot shall be accepted or rejected based on the specified PDA.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-38535.

4.3.1 Qualification extension. When authorized by the qualifying activity, for qualification inspection, if a manufacturer qualifies to a 51-54 device type which is manufactured identically to a 01-04 device type on this specification, then the 01-04 device type may be part I qualified by conducting only worst case group A electrical tests and any electrical tests specified as additional group C subgroups and submitting data in accordance with MIL-PRF-38535.

4.4 Technology Conformance inspection (TCI). Technology conformance inspection shall be in accordance with MIL-PRF-38535 and herein for groups A, B, C, D, and E inspections (see 4.4.1 through 4.4.5).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table III of MIL-PRF-38535 and as follows:

a. Tests shall be performed in accordance with table II herein.

b. Subgroups 5, 6, 7, and 8 shall be omitted.

c. Subgroup 4 (C_I measurement) shall be measured only for initial qualification and after process or design changes that may affect input capacitance. Capacitance shall be measured between the designated terminal and V_{SS} at a frequency of 1 MHz.

d. Subgroups 9 and 11 shall be measured only for initial qualification and after process or design changes which may affect dynamic performance.

e. When device types 01 through 04 are qualified by extension (see 4.3.1), these device types will be inspected (QCI) according to the requirements for device types 51 through 54, respectively.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of MIL-PRF-38535.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table IV of MIL-PRF-38535 and as follows:

a. End-point electrical parameters shall be as specified in table II herein. Delta limits shall apply only to subgroup 1 of group C inspection and shall consist of tests specified in table IV herein.

b. The steady-state life test duration, test condition, and test temperature, or approved alternatives shall be as specified in the device manufacturer's QM plan in accordance with MIL-PRF-38535. The burn-in test circuit shall be maintained under document control by the device manufacturer's Technology Review Board (TRB) in accordance with MIL-PRF-38535 and shall be made available to the acquiring or preparing activity upon request. The test circuit shall specify the inputs, outputs, biases, and power dissipation, as applicable, in accordance with the intent specified in test method 1005 of MIL-STD-883.

c. When device types 01 through 04 are qualified by extension (see 4.3.1), these device types will be inspected (QCI) according to the requirements for device types 51 through 54, respectively.

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TABLE III. Group A inspection for device type 01.

Symbol	MIL-STD-883 method	Cases A,C,D,T,X,Y	Terminal conditions 1/									
		Symbol	2aY	2aS	2A	2bS	2bY	1A	V_{SS}	1bY	3bS	3A
		Test no.	1	2	3	4	5	6	7	8	9	10
$V_{IC(POS)}$		1	(5) 2/	GND			(1) 2/	1 mA		(13) 2/		
		2	"	"	1 mA		"			"		
		3	"	"			"			"		1 mA
$V_{IC(NEG)}$		4	"			GND	"	-1 mA	GND	"	GND	
		5	"		-1 mA	"	"		"	"	"	
		6	"			"	"		"	"	"	-1 mA
I_{SS}	3005	7	"	15.0V	GND	"	"	GND	"	"	"	GND
		8	"	15.0V	15.0V	"	"	15.0V	"	"	"	15.0V
V_{OH1}	3006	9	" 3/	4.5V	GND	"	"	V_{IL1}	"	"	"	GND
		10	" I_{OH1}	"	V_{IL1}	"	"	GND	"	"	"	GND
		11	"	"	GND	"	"	GND	"	"	"	V_{IL1}
V_{OH2}		12	" 4/	5.0V	GND	"	"	V_{IL1}	"	"	"	GND
		13	" I_{OH2}	"	V_{IL1}	"	"	GND	"	"	"	GND
		14	"	"	GND	"	"	GND	"	"	"	V_{IL1}
V_{OH3}		15	"	"	GND	"	"	V_{IL1}	"	"	"	GND
		16	"	"	V_{IL1} 5/	"	"	GND	"	"	"	GND
		17	"	"	GND	"	"	GND	"	"	"	V_{IL1}
V_{OH4}		18	"	12.5V	GND	"	"	V_{IL2}	"	"	"	GND
		19	"	"	V_{IL2} 5/	"	"	GND	"	"	"	GND
		20	"	"	GND	"	"	GND	"	"	"	V_{IL2}
V_{OL1}	3007	21	" 7/	5.5V	GND	"	"	V_{IH1}	"	"	"	GND
		22	" I_{OL1}	"	V_{IH1} 8/	"	"	GND	"	"	"	GND
		23	"	"	GND	"	"	GND	"	"	"	V_{IH1}
V_{OL2}		24	" 9/	5.0V	GND	"	"	V_{IH1}	"	"	"	GND
		25	" I_{OL2}	"	V_{IH1}	"	"	GND	"	"	"	GND
		26	"	"	GND	"	"	GND	"	"	"	V_{IH1}
V_{OL3}		27	"	"	GND	"	"	V_{IH1}	"	"	"	GND
		28	"	"	V_{IH1}	"	"	GND	"	"	"	GND
		29	"	"	GND	"	"	GND	"	"	"	V_{IH1}
V_{OL4}		30	"	12.5V	GND	"	"	V_{IH2}	"	"	"	GND
		31	"	"	V_{IH2} 10/	"	"	GND	"	"	"	GND
		32	"	"	GND	"	"	GND	"	"	"	V_{IH2}
I_{IH1} 11/	3010	33	"	15.0V	15.0V	"	"	15.0V	"	"	"	15.0V
I_{IH2}		34	"	"	GND	"	"	15.0V	"	"	"	GND
		35	"	"	15.0V	"	"	GND	"	"	"	GND
		36	"	"	GND	"	"	"	"	"	"	15.0V
I_{IL1} 11/	3009	37	"	"	GND	"	"	"	"	"	"	GND
I_{IL2}		38	"	"	15.0V	"	"	"	"	"	"	15.0V
		39	"	"	GND	"	"	15.0V	"	"	"	15.0V
		40	"	"	15.0V	"	"	15.0V	"	"	"	GND

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 – Continued.

Symbol	MIL-STD-883 method	Cases A,C,D,T,X,Y	Terminal conditions 1/				Measured terminal	Test limits						Unit
		Symbol	3aS	3Y	1aY	V_{DD}		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Test no.	11	12	13	14		Min	Max	Min	Max	Min	Max	
$V_{IC(POS)}$		1	GND		(8) 2/	GND	1A		1.5					V
		2	"		"	"	2A		"					"
		3	"		"	"	3A		"					"
$V_{IC(NEG)}$		4			"		1A		-6					"
		5			"		2A		"					"
		6			"		3A		"					"
I_{SS}	3005	7	15.0V		"	15.0V	V_{SS}		-50		-500			nA
		8	15.0V		"	15.0V	V_{SS}		-50		-500			nA
V_{OH1}	3006	9	4.5V		" I_{OH1}	4.5V	1aY	2.5		2.5		2.5		V
		10	"		"	"	2aY	"		"		"		"
		11	"	I_{OH1}	"	"	3Y	"		"		"		"
V_{OH2}		12	5.0V		" I_{OH2}	5.0V	1aY	4.5		4.5		4.5		"
		13	"		"	"	2aY	"		"		"		"
		14	"	I_{OH2}	"	"	3Y	"		"		"		"
V_{OH3}		15	"		"	"	1aY	4.95		4.95		4.95		"
		16	"		"	"	2aY	"		"		"		"
		17	"		"	"	3Y	"		"		"		"
V_{OH4}		18	12.5V		"	12.5V	1aY	11.25		11.25		11.25		"
		19	"		"	"	2aY	"		"		"		"
		20	"		"	"	3Y	"		"		"		"
V_{OL1}	3007	21	5.5V		" I_{OL1}	5.5V	1aY		0.4		0.5		0.4	"
		22	"		"	"	2aY		"		"		"	"
		23	"	I_{OL1}	"	"	3Y		"		"		"	"
V_{OL2}		24	5.0V		" I_{OL2}	5.0V	1aY		0.5		"		0.5	"
		25	"		"	"	2aY		"		"		"	"
		26	"	I_{OL2}	"	"	3Y		"		"		"	"
V_{OL3}		27	"		"	"	1aY		50		50		50	mV dc
		28	"		"	"	2aY		"		"		"	"
		29	"		"	"	3Y		"		"		"	"
V_{OL4}		30	12.5V		"	12.5V	1aY		1.25		1.25		1.25	V
		31	"		"	"	2aY		"		"		"	"
		32	"		"	"	3Y		"		"		"	"
I_{IH1} 11/	3010	33	15.0V		"	15.0V	All inputs together		300.0					nA
I_{IH2}		34	"		"	"	1A		100.0		100.0			"
		35	"		"	"	2A		"		"			"
		36	"		"	"	3A		"		"			"
I_{IL1} 11/	3009	37	"		"	"	All inputs together		-300.0					"
I_{IL2}		38	"		"	"	1A		-100.0		-100.0			"
		39	"		"	"	2A		"		"			"
		40	"		"	"	3A		"		"			"

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TABLE III. Group A inspection for device type 01 – Continued.

Symbol	MIL-STD-883 method	Cases A,C,D,T,X,Y	Terminal conditions 1/									
		Symbol	2aY	2aS	2A	2bS	2bY	1A	V_{SS}	1bY	3bS	3A
		Test no.	1	2	3	4	5	6	7	8	9	10
C_I	3012	41	(5) 2/	GND	12/	GND		12/	GND		GND	12/
		42	"	"		"			"		"	
		43	"	"		"			"		"	
t_{PHL}	3003 (Fig. 4)	44	(5)	5.0V	GND	GND	(1)	IN	GND	(13)	GND	GND
		45	"	"	IN	"	"	GND	"	"	"	GND
		46	"	"	GND	"	"	GND	"	"	"	IN
t_{PLH}		47	"	"	GND	"	"	IN	"	"	"	GND
		48	"	"	IN	"	"	GND	"	"	"	GND
		49	"	"	GND	"	"	GND	"	"	"	IN
t_{THL}	3004 (Fig. 4)	50	"	"	GND	"	"	IN	"	"	"	GND
		51	"	"	IN	"	"	GND	"	"	"	GND
		52	"	"	GND	"	"	GND	"	"	"	IN
t_{TLH}		53	"	"	GND	"	'	IN	"	"	"	GND
		54	"	"	IN	"	"	GND	"	"	"	GND
		55	"	"	GND	"	"	GND	"	"	"	IN

1/ Pins not designated may be "high" level logic, "low" level logic, or open. Exceptions are as follows: For $V_{IC(POS)}$ tests, the V_{SS} terminal shall be open; for $V_{IC(NEG)}$ tests, the V_{DD} terminal shall be open; for I_{SS} tests, the outputs shall be open.

2/ Terminals in parentheses are connected together as indicated by the included number.

3/ I_{OH1} = -0.1 mA at +25°C; -0.1 mA at +125°C; -0.1 mA at -55°C.

4/ I_{OH2} = -0.30 mA at +25°C; -0.21 mA at +125°C; -0.375 mA at -55°C.

5/ V_{IL1} = 0.9 V at +25°C; 0.65 V at +125°C; 0.95 V at -55°C.

6/ V_{IL2} = 2.25 V at +25°C; 1.95 V at +125°C; 2.40 V at -55°C.

7/ I_{OL1} = 0.23 mA at +25°C; 0.23 mA at +125°C; 0.23 mA at -55°C.

8/ V_{IH1} = 3.95 V at +25°C; 3.85 V at +125°C; 4.05 V at -55°C.

9/ I_{OL2} = 600 μA at +25°C; 400 μA at +125°C; 750 μA at -55°C.

10/ V_{IH2} = 10.25 V at +25°C; 10.0 V at +125°C; 10.5 V at -55°C.

11/ The device manufacturer may, at his option, measure I_{IL} and I_{IH} at +25°C for each individual input or measure all inputs together.

12/ See 4.4.1c.

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TABLE III. Group A inspection for device type 01 – Continued.

Symbol	MIL-STD-883 method	Cases A C,D,T,X,Y	Terminal conditions 1/				Measured terminal	Test limits						Unit
		Symbol	3aS	3Y	1aY	V_{DD}		Subgroup 4 T_A = 25°C						
		Test no.	11	12	13	14		Min	Max					
C_I	3012	41 42 43	GND “ “			GND “ “	1A 2A 3A		12 “ “					pF “ “
								Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
								Min	Max	Min	Max	Min	Max	
t_{PHL}	3003 (Fig. 4)	44 45 46	5.0V “ “		(8) “ “	5.0V “ “	1A to 1aY 2A to 2aY 3A to 3Y	6 “ “	170 “ “	9 “ “	255 “ “	6 “ “	170 “ “	ns “ “
t_{PLH}		47 48 49	“ “ “		“ “ “	“ “ “	1A to 1aY 2A to 2aY 3A to 3Y	“ “ “	110 “ “	“ “ “	165 “ “	“ “ “	110 “ “	“ “ “
t_{THL}	3004 (Fig. 4)	50 51 52	“ “ “		“ “ “	“ “ “	1aY 2aY 3Y	“ “ “	240 “ “	“ “ “	360 “ “	“ “ “	240 “ “	“ “ “
t_{TLH}		53 54 55	“ “ “		“ “ “	“ “ “	1aY 2aY 3Y	“ “ “	160 “ “	“ “ “	240 “ “	“ “ “	160 “ “	“ “ “

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TABLE III. Group A inspection for device type 02.

Symbol	MIL-STD-883 method	Cases E,F,N,Z	Terminal conditions 1/									
		Symbol	B4	A3	B3	A2	B2	A1	B1	V_{SS}	Ka	D1
		Test no.	1	2	3	4	5	6	7	8	9	10
$V_{IC(POS)}$		1 2 3 4 5 6 7 8 9 10	1 mA	1 mA	1 mA	1 mA	1 mA	1 mA	1 mA		1 mA	
$V_{IC(NEG)}$		11 12 13 14 15 16 17 18 19 20	-1 mA	-1 mA	-1 mA	-1 mA	-1 mA	-1 mA	-1 mA	GND " " " " " " " " "	-1 mA	
I_{SS}	3005	21 22 23 24	GND GND 15.0V 15.0V	15.0V GND GND 15.0V	GND GND 15.0V 15.0V	15.0V GND GND 15.0V	GND GND 15.0V 15.0V	15.0V GND GND 15.0V	GND GND 15.0V 15.0V	" " " "	15.0V " " GND	
V_{OH1}	3006	25 26 27 28 29 30 31 32	V_{IH1} 2/ GND " " " " " "	GND V_{IH1} GND " " " " "	GND GND V_{IH1} GND " " " "	GND " " V_{IH1} GND " " "	GND " " " V_{IH1} GND " "	GND " " " " V_{IH1} GND "	GND " " " " " V_{IH1} GND	" " " " " " " "	4.5V " " " " " " "	I_{OH1} 3/ I_{OH1}
V_{OH2}	3006	33 34 35 36 37 38 39 40	V_{IH1} 2/ GND " " " " " "	" V_{IH1} GND " " " " "	" " V_{IH1} GND " " " "	" " " V_{IH1} GND " " "	" " " " V_{IH1} GND " "	" " " " " V_{IH1} GND "	" " " " " " V_{IH1} GND	" " " " " " " "	5.0V " " " " " " "	I_{OH2} 4/ I_{OH2}
V_{OH3}	3006	41 42 43 44 45 46 47 48	V_{IH1} 2/ GND " " " " " "	" V_{IH1} GND " " " " "	" " V_{IH1} GND " " " "	" " " V_{IH1} GND " " "	" " " " V_{IH1} GND " "	" " " " " V_{IH1} GND "	" " " " " " V_{IH1} GND	" " " " " " " "		
V_{OH4} 5/	3006	49 50 51 52 53 54 55 56	V_{IH2} 5/ " " " GND " " "	" " " " V_{IH2} 5/ " " "	V_{IH2} 5/ " " " GND " " "	" " " " V_{IH2} 5/ " " "	V_{IH2} 5/ " " " GND " " "	" " " " V_{IH2} 5/ " " "	V_{IH2} 5/ " " " GND " " "	" " " " " " " "		

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 – Continued.

Symbol	MIL-STD-883 method	Cases E,F,N,Z	Terminal conditions 1/						Measured terminal	Test limits						Unit
		Symbol	D2	D3	D4	Kb	A4	V_{DD}		Subgroup 1 T_A = +25°C		Subgroup 2 T_A = +125°C		Subgroup 3 T_A = -55°C		
		Test no.	11	12	13	14	15	16		Min	Max	Min	Max	Min	Max	
$V_{IC(POS)}$		1						GND	B1		1.5					V
		2						"	A1		"					"
		3						"	B2		"					"
		4						"	A2		"					"
		5						"	B3		"					"
		6						"	A3		"					"
		7						"	B4		"					"
		8					1 mA	"	A4		"					"
		9						"	Ka		"					"
		10				1 mA		"	Kb		"					"
$V_{IC(NEG)}$		11							B1		-6.0					"
		12							A1		"					"
		13							B2		"					"
		14							A2		"					"
		15							B3		"					"
		16							A3		"					"
		17							B4		"					"
		18					-1 mA		A4		"					"
		19							Ka		"					"
		20				-1 mA			Kb		"					"
I_{SS}	3005	21				15.0V	15.0V	15.0V	V_{SS}		-75		-750			nA
		22				"	GND	"	"		"		"			"
		23				"	GND	"	"		"		"			"
		24				GND	15.0V	"	"		"		"			"
V_{OH1}	3006	25			I_{OH1}	4.5V	GND	4.5V	D4	2.5		2.5		2.5		V
		26		I_{OH1}		"	"	"	D3	"		"		"		"
		27		I_{OH1}		"	"	"	D3	"		"		"		"
		28	I_{OH1}			"	"	"	D2	"		"		"		"
		29	I_{OH1}			"	"	"	D2	"		"		"		"
		30				"	"	"	D1	"		"		"		"
		31				"	"	"	D1	"		"		"		"
		32			I_{OH1}	"	V_{IH1}	"	D4	"		"		"		"
V_{OH2}	3006	33			I_{OH2}	5.0V	GND	5.0V	D4	4.5		4.5		4.5		"
		34		I_{OH2}		"	"	"	D3	"		"		"		"
		35		I_{OH2}		"	"	"	D3	"		"		"		"
		36	I_{OH2}			"	"	"	D2	"		"		"		"
		37	I_{OH2}			"	"	"	D2	"		"		"		"
		38				"	"	"	D1	"		"		"		"
		39				"	"	"	D1	"		"		"		"
		40			I_{OH2}	"	V_{IH1}	"	D4	"		"		"		"
V_{OH3}	3006	41				"	GND	"	D4	4.95		4.95		4.95		"
		42				"	"	"	D3	"		"		"		"
		43				"	"	"	D3	"		"		"		"
		44				"	"	"	D2	"		"		"		"
		45				"	"	"	D2	"		"		"		"
		46				"	"	"	D1	"		"		"		"
		47				"	"	"	D1	"		"		"		"
		48				"	V_{IH1}	"	D4	"		"		"		"
V_{OH4} 5/	3006	49				12.5V	GND	12.5V	D4	11.25		11.25		11.25		"
		50				"	"	"	D3	"		"		"		"
		51				"	"	"	D2	"		"		"		"
		52				"	"	"	D1	"		"		"		"
		53				"	V_{IH2} 5/	"	D4	"		"		"		"
		54				"	"	"	D3	"		"		"		"
		55				"	"	"	D2	"		"		"		"
		56				"	"	"	D4	"		"		"		"

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TABLE III. Group A inspection for device type 02 – Continued.

Symbol	MIL-STD-883 method	Cases E,F,N,Z	Terminal conditions 1/									
		Symbol	B4	A3	B3	A2	B2	A1	B1	V_{SS}	Ka	D1
		Test no.	1	2	3	4	5	6	7	8	9	10
V_{OL1}	3007	57	5.5V	5.5V	5.5V	5.5V	5.5V	5.5V	5.5V	GND	V_{IL1} 6/	
		58	"	"	"	"	"	"	"	"	"	
		59	"	"	"	"	"	"	"	"	"	
		60	"	"	"	"	"	"	"	"	"	I_{OL1}
		61	V_{IL1}	V_{IL1}	V_{IL1}	V_{IL1}	V_{IL1}	V_{IL1}	V_{IL1}	"	5.5V	
		62	"	"	"	"	"	"	"	"	"	
		63	"	"	"	"	"	"	"	"	"	
		64	"	"	"	"	"	"	"	"	"	I_{OL1}
V_{OL2}		65	5.0V	5.0V	5.0V	5.0V	5.0V	5.0V	5.0V	"	V_{IL1}	
		66	"	"	"	"	"	"	"	"	"	
		67	"	"	"	"	"	"	"	"	"	
		68	"	"	"	"	"	"	"	"	"	I_{OL2}
		69	V_{IL1}	V_{IL1}	V_{IL1}	V_{IL1}	V_{IL1}	V_{IL1}	V_{IL1}	"	5.0V	
		70	"	"	"	"	"	"	"	"	"	
		71	"	"	"	"	"	"	"	"	"	
		72	"	"	"	"	"	"	"	"	"	I_{OL2}
V_{OL3}		73	5.0V	5.0V	5.0V	5.0V	5.0V	5.0V	5.0V	"	V_{IL1}	
		74	"	"	"	"	"	"	"	"	"	
		75	"	"	"	"	"	"	"	"	"	
		76	"	"	"	"	"	"	"	"	"	
		77	V_{IL1}	V_{IL1}	V_{IL1}	V_{IL1}	V_{IL1}	V_{IL1}	V_{IL1}	"	5.0V	
		78	"	"	"	"	"	"	"	"	"	
		79	"	"	"	"	"	"	"	"	"	
		80	"	"	"	"	"	"	"	"	"	
V_{OL4}		81	12.5V	12.5V	12.5V	12.5V	12.5V	12.5V	12.5V	"	V_{IL2} 9/	
		82	"	"	"	"	"	"	"	"	"	
		83	"	"	"	"	"	"	"	"	"	
		84	"	"	"	"	"	"	"	"	"	
		85	V_{IL2}	V_{IL2}	V_{IL2}	V_{IL2}	V_{IL2}	V_{IL2}	V_{IL2}	"	12.5V	
		86	"	"	"	"	"	"	"	"	"	
		87	"	"	"	"	"	"	"	"	"	
		88	"	"	"	"	"	"	"	"	"	
I_{IH1} 10/	3010	89	15.0V	15.0V	15.0V	15.0V	15.0V	15.0V	15.0V	"	15.0V	
I_{IH2}		90	15.0V	GND	GND	GND	GND	GND	GND	"	GND	
		91	GND	15.0V	GND	"	"	"	"	"	"	
		92	"	GND	15.0V	"	"	"	"	"	"	
		93	"	"	GND	15.0V	"	"	"	"	"	
		94	"	"	"	GND	15.0V	"	"	"	"	
		95	"	"	"	"	GND	15.0V	"	"	"	
		96	"	"	"	"	"	GND	15.0V	"	"	
		97	"	"	"	"	"	"	GND	"	15.0V	
		98	"	"	"	"	"	"	"	"	GND	
		99	"	"	"	"	"	"	"	"	"	
I_{IL1} 10/	3009	100	"	"	"	"	"	"	"	"	"	
I_{IL2}		101	"	15.0V	15.0V	15.0V	15.0V	15.0V	15.0V	"	15.0V	
		102	15.0V	GND	15.0V	"	"	"	"	"	"	
		103	"	15.0V	GND	"	"	"	"	"	"	
		104	"	"	15.0V	GND	"	"	"	"	"	
		105	"	"	"	15.0V	GND	"	"	"	"	
		106	"	"	"	"	15.0V	GND	"	"	"	
		107	"	"	"	"	"	15.0V	GND	"	"	
		108	"	"	"	"	"	"	15.0V	"	GND	
		109	"	"	"	"	"	"	"	"	15.0V	
		110	"	"	"	"	"	"	"	"	15.0V	

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 – Continued.

Symbol	MIL-STD-883 method	Cases E,F,N,Z	Terminal conditions 1/						Measured terminal	Test limits						Unit
		Symbol	D2	D3	D4	Kb	A4	V_{DD}		Subgroup 1 T_A = +25°C		Subgroup 2 T_A = +125°C		Subgroup 3 T_A = -55°C		
		Test no.	11	12	13	14	15	16		Min	Max	Min	Max	Min	Max	
V_{OL1}	3007	57			I_{OL1} 7/	V_{IL1}	5.5V	5.5V	D4		0.4		0.5		0.4	V
		58		I_{OL1}		"	"	"	D3		"		"		"	"
		59	I_{OL1}			"	"	"	D2		"		"		"	"
		60				"	"	"	D1		"		"		"	"
		61			I_{OL1}	5.5V	V_{IL1}	"	D4		"		"		"	"
		62		I_{OL1}		"	"	"	D3		"		"		"	"
		63	I_{OL1}			"	"	"	D2		"		"		"	"
		64				"	"	"	D1		"		"		"	"
V_{OL2}		65			I_{OL2} 8/	V_{IL1}	5.0V	5.0V	D4		0.5		"		0.5	"
		66		I_{OL2}		"	"	"	D3		"		"		"	"
		67	I_{OL2}			"	"	"	D2		"		"		"	"
		68				"	"	"	D1		"		"		"	"
		69			I_{OL2}	5.0V	V_{IL1}	"	D4		"		"		"	"
		70		I_{OL2}		"	"	"	D3		"		"		"	"
		71	I_{OL2}			"	"	"	D2		"		"		"	"
		72				"	"	"	D1		"		"		"	"
V_{OL3}		73				V_{IL1}	5.0V	"	D4		50		50		50	mV dc
		74				"	"	"	D3		"		"		"	"
		75				"	"	"	D2		"		"		"	"
		76				"	"	"	D1		"		"		"	"
		77				5.0V	V_{IL1}	"	D4		"		"		"	"
		78				"	"	"	D3		"		"		"	"
		79				"	"	"	D2		"		"		"	"
		80				"	"	"	D1		"		"		"	"
V_{OL4}		81				V_{IL2}	12.5V	12.5V	D4		1.25		1.25		1.25	V
		82				"	"	"	D3		"		"		"	"
		83				"	"	"	D2		"		"		"	"
		84				"	"	"	D1		"		"		"	"
		85				12.5V	V_{IL2}	"	D4		"		"		"	"
		86				"	"	"	D3		"		"		"	"
		87				"	"	"	D2		"		"		"	"
		88				"	"	"	D1		"		"		"	"
I_{IH1} 10/	3010	89				15.0V	15.0V	15.0V	All inputs together		1000.0					nA
I_{IH2}		90				GND	GND	"	B4		100.0		100.0			"
		91				"	"	"	A3		"		"			"
		92				"	"	"	B3		"		"			"
		93				"	"	"	A2		"		"			"
		94				"	"	"	B2		"		"			"
		95				"	"	"	A1		"		"			"
		96				"	"	"	B1		"		"			"
		97				"	"	"	Ka		"		"			"
		98				15.0V	"	"	Kb		"		"			"
		99				GND	15.0V	"	A4		"		"			"
I_{IL1} 10/	3009	100				GND	GND	"	All inputs together		-1000.0					"
I_{IL2}		101				15.0V	15.0V	"	B4		-100.0		-100.0			"
		102				"	"	"	A3		"		"			"
		103				"	"	"	B3		"		"			"
		104				"	"	"	A2		"		"			"
		105				"	"	"	B2		"		"			"
		106				"	"	"	A1		"		"			"
		107				"	"	"	B1		"		"			"
		108				"	"	"	Ka		"		"			"
		109				GND	"	"	Kb		"		"			"
		110				15.0V	GND	"	A4		"		"			"

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TABLE III. Group A inspection for device type 02 – Continued.

Symbol	MIL-STD-883 method	Cases E,F,N,Z	Terminal conditions 1/									
		Symbol	B4	A3	B3	A2	B2	A1	B1	V_{SS}	Ka	D1
		Test no.	1	2	3	4	5	6	7	8	9	10
C_I	3012	111	11/							GND		
		112		11/						"		
		113			11/					"		
		114				11/				"		
		115					11/			"		
		116						11/		"		
		117							11/	"		
		118								"		
		119								"	11/	
		120								"		
t_{PHL}	3003 (Fig. 4)	121	IN	GND	GND	GND	GND	GND	GND	GND	5.0V	
		122	GND	"	IN	"	GND	"	"	"	"	
		123	"	"	GND	"	IN	"	"	"	"	
		124	"	"	"	"	GND	"	IN	"	"	OUT
		125	"	"	"	"	"	"	GND	"	"	
		126	"	IN	"	"	"	"	"	"	"	
		127	"	GND	"	IN	"	"	"	"	"	
		128	"	"	"	GND	"	IN	"	"	"	OUT
		129	"	"	"	"	"	GND	"	"	IN	
		130	"	5.0V	"	"	"	"	"	"	"	
		131	"	GND	"	5.0V	"	"	"	"	"	
		132	"	"	"	GND	"	5.0V	"	"	"	OUT
		133	5.0V	"	"	"	"	GND	"	"	5.0V	
		134	GND	"	5.0V	"	"	"	"	"	"	
		135	"	"	GND	"	5.0V	"	"	"	"	
		136	"	"	"	"	GND	"	5.0V	"	"	OUT
t_{PLH}		137	IN	"	"	"	"	"	GND	"	"	
		138	GND	"	IN	"	"	"	"	"	"	
		139	"	"	GND	"	IN	"	"	"	"	
		140	"	"	"	"	GND	"	IN	"	"	OUT
		141	"	"	"	"	"	"	GND	"	"	
		142	"	IN	"	"	"	"	"	"	"	
		143	"	GND	"	IN	"	"	"	"	"	
		144	"	"	"	GND	"	IN	"	"	"	OUT
		145	"	"	"	"	"	GND	"	"	IN	
		146	"	5.0V	"	"	"	"	"	"	"	
		147	"	GND	"	5.0V	"	"	"	"	"	
		148	"	"	"	GND	"	5.0V	"	"	"	OUT
		149	5.0V	"	"	"	"	GND	"	"	5.0V	
		150	GND	"	5.0V	"	"	"	"	"	"	
		151	"	"	GND	"	5.0V	"	"	"	"	
		152	"	"	"	"	GND	"	5.0V	"	"	OUT
t_{THL}	3004 (Fig. 4)	153	IN	"	"	"	"	"	GND	"	"	
		154	GND	"	IN	"	"	"	"	"	"	
		155	"	"	GND	"	IN	"	"	"	"	
		156	"	"	"	"	GND	"	IN	"	"	OUT
		157	"	"	"	"	"	"	GND	"	"	
		158	"	IN	"	"	"	"	"	"	"	
		159	"	GND	"	IN	"	"	"	"	"	
		160	"	"	"	GND	"	IN	"	"	"	OUT
		161	"	"	"	"	"	GND	"	"	IN	
		162	"	5.0V	"	"	"	"	"	"	"	
		163	"	GND	"	5.0V	"	"	"	"	"	
		164	"	"	"	GND	"	5.0V	"	"	"	OUT
		165	5.0V	"	"	"	"	GND	"	"	5.0V	
		166	GND	"	5.0V	"	"	"	"	"	"	
		167	"	"	GND	"	5.0V	"	"	"	"	
		168	"	"	GND	"	GND	"	5.0V	"	"	OUT

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 – Continued.

Symbol	MIL-STD-883 method	Cases E,F,N,Z	Terminal conditions 1/						Measured terminal	Test limits						Unit
		Symbol	D2	D3	D4	Kb	A4	V_{DD}		Subgroup 4 T_A = +25°C						
		Test no.	11	12	13	14	15	16		Min	Max					
C_I	3012	111						GND	B4		12					pF
		112						"	A3		"					"
		113						"	B3		"					"
		114						"	A2		"					"
		115						"	B2		"					"
		116						"	A1		"					"
		117						"	B1		"					"
		118					11/	"	A4		"					"
		119						"	Ka		"					"
		120				11/		"	Kb		"					"
										Subgroup 9 T_A = +25°C		Subgroup 10 T_A = +125°C		Subgroup 11 T_A = -55°C		
										Min	Max	Min	Max	Min	Max	
t_{PHL}	3003 (Fig. 4)	121			OUT	5.0V	GND	5.0V	B4 to D4	13	350	18	525	13	350	ns
		122		OUT		"	"	"	B3 to D3	"	"	"	"	"	"	"
		123	OUT			"	"	"	B2 to D2	"	"	"	"	"	"	"
		124				"	"	"	B1 to D1	"	"	"	"	"	"	"
		125			OUT	"	IN	"	A4 to D4	"	"	"	"	"	"	"
		126		OUT		"	GND	"	A3 to D3	"	"	"	"	"	"	"
		127	OUT			"	"	"	A2 to D2	"	"	"	"	"	"	"
		128				"	"	"	A1 to D1	"	"	"	"	"	"	"
		129			OUT	"	5.0V	"	Ka to D4	"	"	"	"	"	"	"
		130		OUT		"	GND	"	Ka to D3	"	"	"	"	"	"	"
		131	OUT			"	"	"	Ka to D2	"	"	"	"	"	"	"
		132				"	"	"	Ka to D1	"	"	"	"	"	"	"
		133			OUT	IN	"	"	Kb to D4	"	"	"	"	"	"	"
		134		OUT		"	"	"	Kb to D3	"	"	"	"	"	"	"
		135	OUT			"	"	"	Kb to D2	"	"	"	"	"	"	"
		136				"	"	"	Kb to D1	"	"	"	"	"	"	"
t_{PLH}		137			OUT	5.0V	"	"	B4 to D4	"	"	"	"	"	"	"
		138		OUT		"	"	"	B3 to D3	"	"	"	"	"	"	"
		139	OUT			"	"	"	B2 to D2	"	"	"	"	"	"	"
		140				"	"	"	B1 to D1	"	"	"	"	"	"	"
		141			OUT	"	IN	"	A4 to D4	"	"	"	"	"	"	"
		142		OUT		"	GND	"	A3 to D3	"	"	"	"	"	"	"
		143	OUT			"	"	"	A2 to D2	"	"	"	"	"	"	"
		144				"	"	"	A1 to D1	"	"	"	"	"	"	"
		145			OUT	"	5.0V	"	Ka to D4	"	"	"	"	"	"	"
		146		OUT		"	GND	"	Ka to D3	"	"	"	"	"	"	"
		147	OUT			"	"	"	Ka to D2	"	"	"	"	"	"	"
		148				"	"	"	Ka to D1	"	"	"	"	"	"	"
		149			OUT	IN	"	"	Kb to D4	"	"	"	"	"	"	"
		150		OUT		"	"	"	Kb to D3	"	"	"	"	"	"	"
		151	OUT			"	"	"	Kb to D2	"	"	"	"	"	"	"
		152				"	"	"	Kb to D1	"	"	"	"	"	"	"
t_{THL}	3004 (Fig. 4)	153			OUT	5.0V	"	"	D4	10	"	14	"	10	"	"
		154		OUT		"	"	"	D3	"	"	"	"	"	"	"
		155	OUT			"	"	"	D2	"	"	"	"	"	"	"
		156				"	"	"	D1	"	"	"	"	"	"	"
		157			OUT	"	IN	"	D4	"	"	"	"	"	"	"
		158		OUT		"	GND	"	D3	"	"	"	"	"	"	"
		159	OUT			"	"	"	D2	"	"	"	"	"	"	"
		160				"	"	"	D1	"	"	"	"	"	"	"
		161			OUT	"	5.0V	"	D4	"	"	"	"	"	"	"
		162		OUT		"	GND	"	D3	"	"	"	"	"	"	"
		163	OUT			"	"	"	D2	"	"	"	"	"	"	"
		164				"	"	"	D1	"	"	"	"	"	"	"
		165			OUT	IN	"	"	D4	"	"	"	"	"	"	"
		166		OUT		"	"	"	D3	"	"	"	"	"	"	"
		167	OUT			"	"	"	D2	"	"	"	"	"	"	"
		168				"	"	"	D1	"	"	"	"	"	"	"

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TABLE III. Group A inspection for device type 02 – Continued.

Symbol	MIL-STD-883 method	Cases E,F,N,Z	Terminal conditions 1/									
		Symbol	B4	A3	B3	A2	B2	A1	B1	V_{SS}	Ka	D1
		Test no.	1	2	3	4	5	6	7	8	9	10
t_{TLH}	3004 (Fig. 4)	169	IN	GND	GND	GND	GND	GND	GND	GND	5.0V	
		170	GND	"	IN	"	GND	"	"	"	"	
		171	"	"	GND	"	IN	"	"	"	"	
		172	"	"	"	"	GND	"	IN	"	"	OUT
		173	"	"	"	"	"	"	GND	"	"	
		174	"	IN	"	"	"	"	"	"	"	
		175	"	GND	"	IN	"	"	"	"	"	
		176	"	"	"	GND	"	IN	"	"	"	OUT
		177	"	"	"	"	"	GND	"	"	IN	
		178	"	5.0V	"	"	"	"	"	"	"	
		179	"	GND	"	5.0V	"	"	"	"	"	
		180	"	"	"	GND	"	5.0V	"	"	"	OUT
		181	5.0V	"	"	"	"	GND	"	"	5.0V	
		182	GND	"	5.0V	"	"	"	"	"	"	
		183	"	"	GND	"	5.0V	"	"	"	"	
		184	"	"	GND	"	GND	"	5.0V	"	"	OUT

1/ Pins not designated may be "high" level logic, "low" level logic, or open. Exceptions are as follows: For $V_{IC(POS)}$ tests, the V_{SS} terminal shall be open; for $V_{IC(NEG)}$ tests, the V_{DD} terminal shall be open; for I_{SS} tests, the outputs shall be open.

2/ V_{IH1} = 3.8 V at +25°C, 3.5 V at +125°C; 3.95 V at -55°C.

3/ I_{OH1} = -0.1 mA at +25°C; -0.1 mA at +125°C; -0.1 mA at -55°C.

4/ I_{OH2} = -0.25 mA at +25°C; -0.175 mA at +125°C; -0.31 mA at -55°C.

5/ V_{IH2} = 9.5 V at +25°C; 9.25 V at +125°C; 9.75 V at -55°C.

6/ V_{IL1} = 1.1 V at +25°C; 0.85 V at +125°C; 1.35 V at -55°C.

7/ I_{OL1} = 0.23 mA at +25°C; 0.23 mA at +125°C; 0.23 mA at -55°C.

8/ I_{OL2} = 0.5 mA at +25°C; 0.4 mA at +125°C; 0.55 mA at -55°C.

9/ V_{IL2} = 2.8 V at +25°C; 2.55 V at +125°C; 3.05 V at -55°C.

10/ The device manufacturer may, at his option, measure I_{IL} and I_{IH} at +25°C for each individual input or measure all inputs together.

11/ See 4.4.1c.

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TABLE III. Group A inspection for device type 02 – Continued.

Symbol	MIL-STD-883 method	Cases E,F,N,Z	Terminal conditions 1/						Measured terminal	Test limits						Unit
		Symbol	D2	D3	D4	Kb	A4	V_{DD}		Subgroup 9 T_A = +25°C		Subgroup 10 T_A = +125°C		Subgroup 11 T_A = -55°C		
		Test no.	11	12	13	14	15	16		Min	Max	Min	Max	Min	Max	
t_{TLH}	3004 (Fig. 4)	169			OUT	5.0V	GND	5.0V	D4	10	350	14	525	10	350	ns
		170		OUT		"	"	"	D3	"	"	"	"	"	"	"
		171	OUT			"	"	"	D2	"	"	"	"	"	"	"
		172				"	"	"	D1	"	"	"	"	"	"	"
		173			OUT	"	IN	"	D4	"	"	"	"	"	"	"
		174		OUT		"	GND	"	D3	"	"	"	"	"	"	"
		175	OUT			"	"	"	D2	"	"	"	"	"	"	"
		176				"	"	"	D1	"	"	"	"	"	"	"
		177			OUT	"	5.0V	"	D4	"	"	"	"	"	"	"
		178		OUT		"	GND	"	D3	"	"	"	"	"	"	"
		179	OUT			"	"	"	D2	"	"	"	"	"	"	"
		180				"	"	"	D1	"	"	"	"	"	"	"
		181			OUT	IN	"	"	D4	"	"	"	"	"	"	"
		182		OUT		"	"	"	D3	"	"	"	"	"	"	"
		183	OUT			"	"	"	D2	"	"	"	"	"	"	"
		184				"	"	"	D1	"	"	"	"	"	"	"

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TABLE III. Group A inspection for device type 03.

Symbol	MIL-STD-883 method	Cases A,C,D,T,X,Y	Terminal conditions 1/										
		Symbol	1A	1B	1Y	2Y	2A	2B	V_{SS}	3A	3B	3Y	4Y
		Test no.	1	2	3	4	5	6	7	8	9	10	11
$V_{IC(POS)}$		1 2 3 4 5 6 7 8	1 mA	1 mA			1 mA	1 mA		1 mA	1 mA		
$V_{IC(NEG)}$		9 10 11 12 13 14 15 16	-1 mA	-1 mA			-1 mA	-1 mA	GND “ “ “ “ “ “	-1 mA	-1 mA		
I_{SS} 2/	3005	17 18 19	15.0V GND GND	GND 15.0V GND			GND 15.0V GND	15.0V GND GND	“ “ “	15.0V GND GND	GND 15.0V GND		
V_{OH1}	3006	20 21 22 23	V_{IL1} 3/ 4.5V “ “	4.5V “ “ “	I_{OH1} 4/	I_{OH1}	4.5V V_{IL1} 4.5V 4.5V	4.5V “ “ “	“ “ “ “	4.5V 4.5V V_{IL1} 4.5V	4.5V “ “ “	I_{OH1}	I_{OH1}
V_{OH2}		24 25 26 27	V_{IL1} 5.0V “ “	5.0V “ “ “	I_{OH2} 5/	I_{OH2}	5.0V V_{IL1} 5.0V “	5.0V “ “ “	“ “ “ “	5.0V 5.0V V_{IL1} 5.0V	5.0V “ “ “	I_{OH2}	I_{OH2}
V_{OH3}		28 29 30 31	V_{IL1} 5.0V “ “	“ “ “ “			“ V_{IL1} 5.0V 5.0V	“ “ “ “	“ “ “ “	5.0V 5.0V V_{IL1} 5.0V	“ “ “ “		
V_{OH4}		32 33 34 35	V_{IL2} 6/ 12.5V “ “	12.5V “ “ “			12.5V V_{IL2} 12.5V 12.5V	12.5V “ “ “	“ “ “ “	12.5V 12.5V V_{IL2} 12.5V	12.5V “ “ “		
V_{OL1}	3007	36 37 38 39 40 41 42 43	V_{IH1} 7/ GND “ “ “ “ “ “	V_{IH1} GND “ “ “ “ “ “	I_{OL1} 8/ I_{OL1}	I_{OL1} I_{OL1}	GND GND V_{IH1} GND “ “ “ “	GND GND V_{IH1} GND “ “ “ “	“ “ “ “ “ “ “ “	GND “ “ “ V_{IH1} GND “ “	GND “ “ “ V_{IH1} GND “ “	I_{OL1} I_{OL1}	I_{OL1} I_{OL1}
V_{OL2}		44 45 46 47 48 49 50 51	V_{IH1} GND “ “ “ “ “ “	V_{IH1} GND “ “ “ “ “ “	I_{OL2} 9/ I_{OL2}	I_{OL2} I_{OL2}	“ “ V_{IH1} GND “ “ “ “	“ “ V_{IH1} GND “ “ “ “	“ “ “ “ “ “ “ “	“ “ “ “ V_{IH1} GND “ “	“ “ “ “ V_{IH1} GND “ “	I_{OL2} I_{OL2}	I_{OL2} I_{OL2}

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 03 – Continued.

Symbol	MIL-STD-883 method	Cases A, C,D,T,X,Y	Terminal conditions 1/			Measured terminal	Test limits						Unit
		Symbol	4A	4B	V_{DD}		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	12	13	14		Min	Max	Min	Max	Min	Max	
$V_{IC(POS)}$		1			GND	1A		1.5					V
		2			"	1B		"					"
		3			"	2A		"					"
		4			"	2B		"					"
		5			"	3A		"					"
		6			"	3B		"					"
		7	1 mA		"	4A		"					"
		8		1 mA	"	4B		"					"
$V_{IC(NEG)}$		9				1A		-6					"
		10				1B		"					"
		11				2A		"					"
		12				2B		"					"
		13				3A		"					"
		14				3B		"					"
		15	-1 mA			4A		"					"
		16		-1 mA		4B		"					"
I_{SS} 2/	3005	17	GND	15.0V	15.0V	V_{SS}		-25		-750			nA
		18	15.0V	GND	"	"		"		"			"
		19	GND	GND	"	"		"		"			"
V_{OH1}	3006	20	4.5V	4.5V	4.5V	1Y	2.5		2.5		2.5		V
		21	"	"	"	2Y	"		"		"		"
		22	"	"	"	3Y	"		"		"		"
		23	V_{IL1}	"	"	4Y	"		"		"		"
V_{OH2}		24	5.0V	5.0V	5.0V	1Y	4.5		4.5		4.5		"
		25	"	"	"	2Y	"		"		"		"
		26	"	"	"	3Y	"		"		"		"
		27	V_{IL1}	"	"	4Y	"		"		"		"
V_{OH3}		28	5.0V	"	"	1Y	4.95		4.95		4.95		"
		29	"	"	"	2Y	"		"		"		"
		30	"	"	"	3Y	"		"		"		"
		31	V_{IL1}	"	"	4Y	"		"		"		"
V_{OH4}		32	12.5V	12.5V	12.5V	1Y	11.25		11.25		11.25		"
		33	"	"	"	2Y	"		"		"		"
		34	"	"	"	3Y	"		"		"		"
		35	V_{IL2}	"	"	4Y	"		"		"		"
V_{OL1}	3007	36	GND	GND	5.5V	1Y		0.4		0.5		0.4	"
		37	"	"	"	1Y		"		"		"	"
		38	"	"	"	2Y		"		"		"	"
		39	"	"	"	2Y		"		"		"	"
		40	"	"	"	3Y		"		"		"	"
		41	"	"	"	3Y		"		"		"	"
		42	V_{IH1}	V_{IH1}	"	4Y		"		"		"	"
		43	GND	GND	"	4Y		"		"		"	"
V_{OL2}		44	"	"	5.0V	1Y		0.5		"		0.5	"
		45	"	"	"	1Y		"		"		"	"
		46	"	"	"	2Y		"		"		"	"
		47	"	"	"	2Y		"		"		"	"
		48	"	"	"	3Y		"		"		"	"
		49	"	"	"	3Y		"		"		"	"
		50	V_{IH1}	V_{IH1}	"	4Y		"		"		"	"
		51	GND	GND	"	4Y		"		"		"	"

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TABLE III. Group A inspection for device type 03 – Continued.

Symbol	MIL-STD-883 method	Cases A, C,D,T,X,Y	Terminal conditions 1/										
		Symbol	1A	1B	1Y	2Y	2A	2B	V_{SS}	3A	3B	3Y	4Y
		Test no.	1	2	3	4	5	6	7	8	9	10	11
V_{OL3}	3007	52	V_{IH1}	V_{IH1}			GND	GND	GND	GND	GND		
		53	GND	GND			GND	GND	"	"	"		
		54	"	"			V_{IH1}	V_{IH1}	"	"	"		
		55	"	"			GND	GND	"	"	"		
		56	"	"			"	"	"	V_{IH1}	V_{IH1}		
		57	"	"			"	"	"	GND	GND		
		58	"	"			"	"	"	"	"		
		59	"	"			"	"	"	"	"		
V_{OL4}		60	V_{IH2} 10/	V_{IH2}			"	"	"	"	"		
		61	GND	GND			"	"	"	"	"		
		62	"	"			V_{IH2}	V_{IH2}	"	"	"		
		63	"	"			GND	GND	"	"	"		
		64	"	"			"	"	"	V_{IH2}	V_{IH2}		
		65	"	"			"	"	"	GND	GND		
		66	"	"			"	"	"	"	"		
		67	"	"			"	"	"	"	"		
V_{OL5}		68	V_{IH1}	V_{IH1}	I_{OL3} 11/		"	"	"	"	"		
		69	GND	GND	I_{OL3}		"	"	"	"	"		
		70	"	"		I_{OL3}	V_{IH1}	V_{IH1}	"	"	"		
		71	"	"			GND	GND	"	"	"		
		72	"	"			"	"	"	V_{IH1}	V_{IH1}	I_{OL3}	
		73	"	"			"	"	"	GND	GND	I_{OL3}	
		74	"	"			"	"	"	"	"		I_{OL3}
		75	"	"			"	"	"	"	"		
I_{IH1} 12/	3010	76	15.0V	15.0V			15.0V	15.0V	"	15.0V	15.0V		
I_{IH2}		77	15.0V	GND			GND	GND	"	GND	GND		
		78	GND	15.0V			GND	"	"	"	"		
		79	"	GND			15.0V	"	"	"	"		
		80	"	"			GND	15.0V	"	"	"		
		81	"	"			"	GND	"	15.0V	"		
		82	"	"			"	"	"	GND	15.0V		
		83	"	"			"	"	"	"	GND		
		84	"	"			"	"	"	"	"		
I_{IL1} 12/	3009	85	"	"			"	"	"	"	"		
I_{IL2}		86	"	"			"	"	"	"	"		
		87	"	"			"	"	"	"	"		
		88	"	"			"	"	"	"	"		
		89	"	"			"	"	"	"	"		
		90	"	"			"	"	"	"	"		
		91	"	"			"	"	"	"	"		
		92	"	"			"	"	"	"	"		
		93	"	"			"	"	"	"	"		
C_I	3012	94	13/						GND				
		95		13/					"				
		96					13/		"				
		97						13/	"				
		98							"	13/			
		99							"		13/		
		100							"				
		101							"				

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 03 – Continued.

Symbol	MIL-STD-883 method	Cases A, C,D,T,X,Y	Terminal conditions 1/			Measured terminal	Test limits						Unit
		Symbol	4A	4B	V_{DD}		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	12	13	14		Min	Max	Min	Max	Min	Max	
V_{OL3}	3007	52	GND	GND	5.0V	1Y		50		50		50	mV dc
		53	"	"	"	1Y		"		"		"	"
		54	"	"	"	2Y		"		"		"	"
		55	"	"	"	2Y		"		"		"	"
		56	"	"	"	3Y		"		"		"	"
		57	"	"	"	3Y		"		"		"	"
		58	V_{IH1}	V_{IH1}	"	4Y		"		"		"	"
		59	GND	GND	"	4Y		"		"		"	"
V_{OL4}		60	"	"	12.5V	1Y		1.25		1.25		1.25	V
		61	"	"	"	1Y		"		"		"	"
		62	"	"	"	2Y		"		"		"	"
		63	"	"	"	2Y		"		"		"	"
		64	"	"	"	3Y		"		"		"	"
		65	"	"	"	3Y		"		"		"	"
		66	V_{IH2}	V_{IH2}	"	4Y		"		"		"	"
		67	GND	GND	"	4Y		"		"		"	"
V_{OL5}		68	"	"	5.0V	1Y		0.5		0.5		0.5	"
		69	"	"	"	1Y		"		"		"	"
		70	"	"	"	2Y		"		"		"	"
		71	"	"	"	2Y		"		"		"	"
		72	"	"	"	3Y		"		"		"	"
		73	"	"	"	3Y		"		"		"	"
		74	V_{IH1}	V_{IH1}	"	4Y		"		"		"	"
		75	GND	GND	"	4Y		"		"		"	"
I_{IH1} 12/	3010	76	15.0V	15.0V	15.0V	All inputs together		8.0					nA
I_{IH2}		77	GND	GND	"	1A		1.0		45			"
		78	"	"	"	1B		"		"			"
		79	"	"	"	2A		"		"			"
		80	"	"	"	2B		"		"			"
		81	"	"	"	3A		"		"			"
		82	"	"	"	3B		"		"			"
		83	15.0V	"	"	4A		"		"			"
		84	GND	15.0V	"	4B		"		"			"
I_{IL1} 12/	3009	85	"	GND	"	All inputs together		-8.0					"
I_{IL2}		86	"	"	"	1A		-1.0		-45			"
		87	"	"	"	1B		"		"			"
		88	"	"	"	2A		"		"			"
		89	"	"	"	2B		"		"			"
		90	"	"	"	3A		"		"			"
		91	"	"	"	3B		"		"			"
		92	"	"	"	4A		"		"			"
		93	"	"	"	4B		"		"			"
							Subgroup 4 T_C = +25°C						
							Min	Max					
C_I	3012	94			GND	1A		12					pF
		95			"	1B		"					"
		96			"	2A		"					"
		97			"	2B		"					"
		98			"	3A		"					"
		99			"	3B		"					"
		100	13/		"	4A		"					"
		101		13/	"	4B		"					"

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TABLE III. Group A inspection for device type 03 – Continued.

Symbol	MIL-STD-883 method	Cases A, C,D,T,X,Y	Terminal conditions 1/										
		Symbol	1A	1B	1Y	2Y	2A	2B	V_{SS}	3A	3B	3Y	4Y
		Test no.	1	2	3	4	5	6	7	8	9	10	11
t_{PHL}	3003 (Fig. 4)	102	IN	GND	OUT		GND	GND	GND	GND	GND		
		103	GND	IN	OUT		GND	"	"	"	"		
		104	"	GND		OUT	IN	"	"	"	"		
		105	"	"		OUT	GND	IN	"	"	"		
		106	"	"			"	GND	"	IN	"	OUT	
		107	"	"			"	"	"	GND	IN	OUT	
		108	"	"			"	"	"	"	GND		OUT
		109	"	"			"	"	"	"	"		OUT
t_{PLH}		110	IN	"	OUT		"	"	"	"	"		
		111	GND	IN	OUT		"	"	"	"	"		
		112	"	GND		OUT	IN	"	"	"	"		
		113	"	"		OUT	GND	IN	"	"	"		
		114	"	"			"	GND	"	IN	"	OUT	
		115	"	"			"	"	"	GND	IN	OUT	
		116	"	"			"	"	"	"	GND		OUT
		117	"	"			"	"	"	"	"		OUT
t_{THL}	3004 (Fig. 4)	118	IN	"	OUT		"	"	"	"	"		
		119	GND	IN	OUT		"	"	"	"	"		
		120	"	GND		OUT	IN	"	"	"	"		
		121	"	"		OUT	GND	IN	"	"	"		
		122	"	"			"	GND	"	IN	"	OUT	
		123	"	"			"	"	"	GND	IN	OUT	
		124	"	"			"	"	"	"	GND		OUT
		125	"	"			"	"	"	"	"		OUT
t_{TLH}		126	IN	"	OUT		"	"	"	"	"		
		127	GND	IN	OUT		"	"	"	"	"		
		128	"	GND		OUT	IN	"	"	"	"		
		129	"	"		OUT	GND	IN	"	"	"		
		130	"	"			"	GND	"	IN	"	OUT	
		131	"	"			"	"	"	GND	IN	OUT	
		132	"	"			"	"	"	"	GND		OUT
		133	"	"			"	"	"	"	"		OUT

1/ Pins not designated may be "high" level logic, "low" level logic, or open. Exceptions are as follows: For $V_{IC(POS)}$ tests, the V_{SS} terminal shall be open; for $V_{IC(NEG)}$ tests, the V_{DD} terminal shall be open; for I_{SS} tests, the outputs shall be open.

2/ I_{SS} measurements shall be performed in sequence.

3/ V_{IL1} = 1.1 V at +25°C; 0.35 V at +125°C; 1.35 V at -55°C.

4/ I_{OH1} = -0.1 mA at +25°C; -0.1 mA at +125°C; -0.1 mA at -55°C.

5/ I_{OH2} = -0.2 mA at +25°C; -0.13 mA at +125°C; -0.25 mA at -55°C.

6/ V_{IL2} = 2.8 V at +25°C; 2.55 V at +125°C; 3.05 V at -55°C.

7/ V_{IH1} = 3.8 V at +25°C; 3.5 V at +125°C; 3.95 V at -55°C.

8/ I_{OL1} = 0.23 mA at +25°C; 0.23 mA at +125°C; 0.23 mA at -55°C.

9/ I_{OL2} = 0.40 mA at +25°C; 0.28 mA at +125°C; 0.5 mA at -55°C.

10/ V_{IH2} = 9.5 V at +25°C; 9.25 V at +125°C; 9.75 V at -55°C.

11/ I_{OL3} = 0.60 mA at +25°C; 0.56 mA at +125°C; 1.0 mA at -55°C.

12/ The device manufacturer may, at his option, measure I_{IL} and I_{IH} at +25°C for each individual input or measure all inputs together.

13/ See 4.4.1c.

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TABLE III. Group A inspection for device type 03 – Continued.

Symbol	MIL-STD-883 method	Cases A, C,D,T,X,Y	Terminal conditions 1/			Measured terminal	Test limits						Unit
		Symbol	4A	4B	V_{DD}		Subgroup 9 T_C = +25°C		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C		
		Test no.	12	13	14		Min	Max	Min	Max	Min	Max	
t_{PHL}	3003 (Fig. 4)	102	GND	GND	5.0 V	1A to 1Y	10	210	14	270	10	155	ns
		103	"	"	"	1B to 1Y	"	"	"	"	"	"	"
		104	"	"	"	2A to 2Y	"	"	"	"	"	"	"
		105	"	"	"	2B to 2Y	"	"	"	"	"	"	"
		106	"	"	"	3A to 3Y	"	"	"	"	"	"	"
		107	"	"	"	3B to 3Y	"	"	"	"	"	"	"
		108	IN	"	"	4A to 4Y	"	"	"	"	"	"	"
		109	GND	IN	"	4B to 4Y	"	"	"	"	"	"	"
t_{PLH}		110	"	GND	"	1A to 1Y	"	"	"	"	"	"	"
		111	"	"	"	1B to 1Y	"	"	"	"	"	"	"
		112	"	"	"	2A to 2Y	"	"	"	"	"	"	"
		113	"	"	"	2B to 2Y	"	"	"	"	"	"	"
		114	"	"	"	3A to 3Y	"	"	"	"	"	"	"
		115	"	"	"	3B to 3Y	"	"	"	"	"	"	"
		116	IN	"	"	4A to 4Y	"	"	"	"	"	"	"
		117	GND	IN	"	4B to 4Y	"	"	"	"	"	"	"
t_{THL}	3004 (Fig. 4)	118	"	GND	"	1Y	"	300	"	380	"	220	"
		119	"	"	"	1Y	"	"	"	"	"	"	"
		120	"	"	"	2Y	"	"	"	"	"	"	"
		121	"	"	"	2Y	"	"	"	"	"	"	"
		122	"	"	"	3Y	"	"	"	"	"	"	"
		123	"	"	"	3Y	"	"	"	"	"	"	"
		124	IN	"	"	4Y	"	"	"	"	"	"	"
		125	GND	IN	"	4Y	"	"	"	"	"	"	"
t_{TLH}		126	"	GND	"	1Y	"	410	"	530	"	305	"
		127	"	"	"	1Y	"	"	"	"	"	"	"
		128	"	"	"	2Y	"	"	"	"	"	"	"
		129	"	"	"	2Y	"	"	"	"	"	"	"
		130	"	"	"	3Y	"	"	"	"	"	"	"
		131	"	"	"	3Y	"	"	"	"	"	"	"
		132	IN	"	"	4Y	"	"	"	"	"	"	"
		133	GND	IN	"	4Y	"	"	"	"	"	"	"

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TABLE III. Group A inspection for device type 04.

Symbol	MIL-STD-883 method	Cases E,F,Z	Terminal conditions 1/																Measured terminal	Test limits						Unit
		Symbol	J	Kd	H	G	F	E	Kb	V_{SS}	Kc	Ka	D	C	B	A	EX	V_{DD}		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Min	Max	Min	Max	Min	Max	
V_{IC} (pos) 2/		1		1mA														GND	Kd		1.5					V
		2			1mA													"	H		"					"
		3				1mA												"	G		"					"
		4					1mA											"	F		"					"
		5						1mA										"	E		"					"
		6							1mA									"	Kb		"					"
		7									1mA							"	Kc		"					"
		8										1mA						"	Ka		"					"
		9											1mA					"	D		"					"
		10												1mA				"	C		"					"
		11													1mA			"	B		"					"
		12														1mA		"	A		"					"
		13															1mA	"	EX		"					"
V_{IC} (NEG) 2/		14		-1mA						GND									Kd		-6.0					"
		15			-1mA					"									H		"					"
		16				-1mA				"									G		"					"
		17					-1mA			"									F		"					"
		18						-1mA		"									E		"					"
		19							-1mA	"									Kb		"					"
		20								"	-1mA								Kc		"					"
		21								"		-1mA							Ka		"					"
		22								"			-1mA						D		"					"
		23								"				-1mA					C		"					"
		24								"					-1mA				B		"					"
		25								"						-1mA			A		"					"
		26								"							-1mA		EX		"					"
I_{SS} 3/	3005	27		GND	GND	GND	GND	GND	GND	"	GND	GND	GND	GND	GND	GND	GND	15.0V	V_{SS}		-1.9		-3			µA
		28		15.0V	GND	GND	GND	GND	15.0V	"	15.0V	GND	GND	GND	GND	GND	GND	"	"		"		"			"
		29		GND	15.0V	15.0V	15.0V	15.0V	GND	"	GND	GND	15.0V	15.0V	15.0V	15.0V	15.0V	"	"		"		"			"
		30		15.0V	15.0V	15.0V	15.0V	15.0V	15.0V	"	15.0V	15.0V	15.0V	15.0V	15.0V	15.0V	15.0V	"	"		"		"			"
V_{OH3}	3006	31		5.0V	V_{IL1} 4/	V_{IL1}	V_{IL1}	V_{IL1}	GND	"	GND	GND	V_{IL1}	V_{IL1}	V_{IL1}	V_{IL1}	GND	5.0V	J	4.95		4.95		4.95		V
		32		"	V_{IL1}	V_{IL1}	V_{IL1}	V_{IL1}	"	"	5.0V	5.0V	V_{IL1}	V_{IL1}	V_{IL1}	V_{IL1}	"	5.0V	"	4.95		4.95		4.95		"
V_{OH4}		33		"	V_{IL2} 5/	V_{IL2}	V_{IL2}	V_{IL2}	"	"	GND	GND	V_{IL2}	V_{IL2}	V_{IL2}	V_{IL2}	"	12.5V	"	11.25		11.25		11.25		"
		34		"	V_{IL2}	V_{IL2}	V_{IL2}	V_{IL2}	"	"	5.0V	5.0V	V_{IL2}	V_{IL2}	V_{IL2}	V_{IL2}	"	12.5V	"	11.25		11.25		11.25		"
V_{OL3}	3007	35		"	V_{IH1} 6/	V_{IH1}	V_{IH1}	V_{IH1}	"	"	GND	GND	V_{IH1}	V_{IH1}	V_{IH1}	V_{IH1}	"	5.0V	"		0.05		0.05		0.05	"
		36		"	V_{IH1}	V_{IH1}	V_{IH1}	V_{IH1}	"	"	5.0V	5.0V	V_{IH1}	V_{IH1}	V_{IH1}	V_{IH1}	"	5.0V	"		0.05		0.05		0.05	"
V_{OL4}		37		"	V_{IH2} 7/	V_{IH2}	V_{IH2}	V_{IH2}	"	"	GND	GND	V_{IH2}	V_{IH2}	V_{IH2}	V_{IH2}	"	12.5V	"		1.25		1.25		1.25	"
		38		"	V_{IH2}	V_{IH2}	V_{IH2}	V_{IH2}	"	"	5.0V	5.0V	V_{IH2}	V_{IH2}	V_{IH2}	V_{IH2}	"	12.5V	"		1.25		1.25		1.25	"
I_{IH1} 8/	3010	39		15.0V	15.0V	15.0V	15.0V	15.0V	15.0V	"	15.0V	15.0V	15.0V	15.0V	15.0V	15.0V	15.0V	15.0V	All inputs together		13					nA
I_{IH2}		40		15.0V	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	"	Kd		1.0		45			"
		41		GND	15.0V	GND	"	"	"	"	"	"	"	"	"	"	"	"	H		"		"			"
		42		"	GND	15.0V	15.0V	"	"	"	"	"	"	"	"	"	"	"	G		"		"			"
		43		"	"	GND	GND	15.0V	"	"	"	"	"	"	"	"	"	"	F		"		"			"
		44		"	"	"	"	GND	15.0V	"	"	"	"	"	"	"	"	"	E		"		"			"
		45		"	"	"	"	"	GND	"	"	"	"	"	"	"	"	"	Kb		"		"			"
		46		"	"	"	"	"	"	"	15.0V	"	"	"	"	"	"	"	Kc		"		"			"
		47		"	"	"	"	"	"	"	GND	15.0V	"	"	"	"	"	"	Ka		"		"			"
		48		"	"	"	"	"	"	"	"	GND	15.0V	"	"	"	"	"	D		"		"			"
		49		"	"	"	"	"	"	"	"	"	GND	15.0V	"	"	"	"	C		"		"			"
		50		"	"	"	"	"	"	"	"	"	"	GND	15.0V	"	"	"	B		"		"			"
		51		"	"	"	"	"	"	"	"	"	"	"	GND	15.0V	"	"	A		"		"			"
		52		"	"	"	"	"	"	"	"	"	"	"	"	GND	15.0V	"	EX		"		"			"

See footnotes at end of device type 04.

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TABLE III. Group A inspection for device type 04 – Continued.

Symbol	MIL-STD-883 method	Cases E,F,Z Symbol	Terminal conditions 1/																Measured terminal	Test limits						Unit
			J	Kd	H	G	F	E	Kb	V_{SS}	Kc	Ka	D	C	B	A	EX	V_{DD}		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Min	Max	Min	Max	Min	Max	
I_{IL1} 8/	3009	53		GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	15.0V	All inputs together		-13					nA
I_{IL2}		54		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	Kd		-1.0		-45			"
		55		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	H		"		"			"
		56		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	G		"		"			"
		57		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	F		"		"			"
		58		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	E		"		"			"
		59		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	Kb		"		"			"
		60		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	Kc		"		"			"
		61		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	Ka		"		"			"
		62		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	D		"		"			"
		63		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	C		"		"			"
		64		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B		"		"			"
		65		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A		"		"			"
		66		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	EX		"		"			"
I_{OC1}		67	15.0V	"	15.0V	15.0V	15.0V	15.0V	15.0V	"	15.0V	15.0V	15.0V	15.0V	15.0V	15.0V	"	15.0V	J		25		90		25	"
I_{OC2}		68	GND	"	15.0V	15.0V	15.0V	15.0V	15.0V	"	15.0V	15.0V	15.0V	15.0V	15.0V	15.0V	"	15.0V	J		-25		-90		-25	"
																				Subgroup 4 T_C = +25°C						
																				Min	Max					
C_I	3012	69		9/						GND								GND	Kd		12					pF
		70			9/					"								"	H		"					"
		71				9/				"								"	G		"					"
		72					9/			"								"	F		"					"
		73						9/		"								"	E		"					"
		74							9/	"								"	Kb		"					"
		75								"	9/							"	Kc		"					"
		76								"		9/						"	Ka		"					"
		77								"			9/					"	D		"					"
		78								"				9/				"	C		"					"
		79								"					9/			"	B		"					"
		80								"						9/		"	A		"					"
		81								"							9/	"	EX		"					"
																				Subgroup 9 T_C = +25°C		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C		
																				Min	Max	Min	Max	Min	Max	
t_{PHL1}	3003 (Fig. 5)	82	OUT	5.0V	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	IN	GND	5.0V	A to J	34	685	41	820	30	600	ns
		83	"	"	"	"	"	"	"	"	"	"	"	"	IN	GND	"	"	B to J	"	"	"	"	"	"	"
		84	"	"	"	"	"	"	"	"	"	"	"	IN	GND	"	"	"	C to J	"	"	"	"	"	"	"
		85	"	"	"	"	"	"	"	"	"	"	IN	GND	"	"	"	"	D to J	"	"	"	"	"	"	"
		86	"	"	"	"	"	IN	"	"	"	"	GND	"	"	"	"	"	E to J	"	"	"	"	"	"	"
		87	"	"	"	"	IN	GND	"	"	"	"	"	"	"	"	"	"	F to J	"	"	"	"	"	"	"
		88	"	"	"	IN	GND	"	"	"	"	"	"	"	"	"	"	"	G to J	"	"	"	"	"	"	"
		89	"	"	IN	GND	"	"	"	"	"	"	"	"	"	"	"	"	H to J	"	"	"	"	"	"	"
t_{PLH1}		90	"	"	GND	"	"	"	"	"	"	"	"	"	"	IN	GND	"	A to J	26	535	32	640	24	475	"
		91	"	"	"	"	"	"	"	"	"	"	"	"	IN	GND	"	"	B to J	"	"	"	"	"	"	"
		92	"	"	"	"	"	"	"	"	"	"	"	IN	GND	"	"	"	C to J	"	"	"	"	"	"	"
		93	"	"	"	"	"	"	"	"	"	"	IN	GND	"	"	"	"	D to J	"	"	"	"	"	"	"
		94	"	"	"	"	"	IN	"	"	"	"	GND	"	"	"	"	"	E to J	"	"	"	"	"	"	"
		95	"	"	"	"	IN	GND	"	"	"	"	"	"	"	"	"	"	F to J	"	"	"	"	"	"	"
		96	"	"	"	IN	GND	"	"	"	"	"	"	"	"	"	"	"	G to J	"	"	"	"	"	"	"
		97	"	"	IN	GND	"	"	"	"	"	"	"	"	"	"	"	"	H to J	"	"	"	"	"	"	"

See footnotes on end of device type 04.

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TABLE III. Group A inspection for device type 04 – Continued.

Symbol	MIL-STD-883 method	Cases E,F,Z Symbol	Terminal conditions 1/ J	Kd	H	G	F	E	Kb	V_{SS}	Kc	Ka	D	C	B	A	EX	V_{DD}	Measured terminal	Test limits Subgroup 9 T_C = +25°C Min	Max	Subgroup 10 T_C = +125°C Min	Max	Subgroup 11 T_C = -55°C Min	Max	Unit
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
t_{PHL2}	3003 (Fig. 6)	98	OUT	5.0V	GND	GND	GND	GND	GND	GND	5.0 V	GND	GND	GND	GND	IN	GND	5.0V	A to J	32	635	38	755	28	555	ns
		99	"	"	"	"	"	"	"	"	"	"	"	GND	IN	GND	"	"	B to J	"	"	"	"	"	"	"
		100	"	"	"	"	"	"	"	"	"	"	"	IN	GND	"	"	"	C to J	"	"	"	"	"	"	"
		101	"	"	"	"	"	IN	"	"	"	"	IN	GND	"	"	"	"	D to J	"	"	"	"	"	"	"
		102	"	"	"	"	IN	GND	"	"	"	"	GND	"	"	"	"	"	E to J	"	"	"	"	"	"	"
		103	"	"	"	IN	GND	"	"	"	"	"	"	"	"	"	"	"	F to J	"	"	"	"	"	"	"
		104	"	"	"	GND	"	"	"	"	"	"	"	"	"	"	"	"	G to J	"	"	"	"	"	"	"
		105	"	"	IN	"	"	"	"	"	"	"	"	"	"	"	"	"	H to J	"	"	"	"	"	"	"
t_{PLH2}		106	"	"	GND	"	"	"	"	"	"	"	"	"	"	IN	"	"	A to J	31	620	"	750	"	550	"
		107	"	"	"	"	"	"	"	"	"	"	"	"	IN	GND	"	"	B to J	"	"	"	"	"	"	"
		108	"	"	"	"	"	"	"	"	"	"	"	IN	GND	"	"	"	C to J	"	"	"	"	"	"	"
		109	"	"	"	"	"	"	"	"	"	"	IN	GND	"	"	"	"	D to J	"	"	"	"	"	"	"
		110	"	"	"	"	"	IN	"	"	"	"	GND	"	"	"	"	"	E to J	"	"	"	"	"	"	"
		111	"	"	"	"	IN	GND	"	"	"	"	"	"	"	"	"	"	F to J	"	"	"	"	"	"	"
		112	"	"	"	IN	GND	"	"	"	"	"	"	"	"	"	"	"	G to J	"	"	"	"	"	"	"
		113	"	"	IN	GND	GND	"	"	"	"	"	"	"	"	"	"	"	H to J	"	"	"	"	"	"	"
t_{PHL3}	3003 (Fig. 7)	114	"	"	5.0V	5.0V	5.0V	5.0V	"	"	GND	5.0V	5.0V	5.0V	5.0V	IN	"	"	A to J	38	760	45	905	33	665	"
		115	"	"	"	"	"	"	"	"	"	"	"	5.0V	IN	5.0V	"	"	B to J	"	"	"	"	"	"	"
		116	"	"	"	"	"	"	"	"	"	"	"	IN	5.0V	"	"	"	C to J	"	"	"	"	"	"	"
		117	"	"	"	"	"	"	"	"	"	"	IN	5.0V	"	"	"	"	D to J	"	"	"	"	"	"	"
		118	"	"	"	"	"	IN	"	"	"	"	5.0V	"	"	"	"	"	E to J	"	"	"	"	"	"	"
		119	"	"	"	"	IN	5.0V	"	"	"	"	"	"	"	"	"	"	F to J	"	"	"	"	"	"	"
		120	"	"	"	IN	5.0V	"	"	"	"	"	"	"	"	"	"	"	G to J	"	"	"	"	"	"	"
		121	"	"	IN	5.0V	"	"	"	"	"	"	"	"	"	"	"	"	H to J	"	"	"	"	"	"	"
t_{PLH3}		122	"	"	5.0V	"	"	"	"	"	"	"	"	"	"	IN	"	"	A to J	33	650	32	655	25	490	"
		123	"	"	"	"	"	"	"	"	"	"	"	"	IN	5.0V	"	"	B to J	"	"	"	"	"	"	"
		124	"	"	"	"	"	"	"	"	"	"	"	IN	5.0V	"	"	"	C to J	"	"	"	"	"	"	"
		125	"	"	"	"	"	"	"	"	"	"	IN	5.0V	"	"	"	"	D to J	"	"	"	"	"	"	"
		126	"	"	"	"	"	IN	"	"	"	"	5.0V	"	"	"	"	"	E to J	"	"	"	"	"	"	"
		127	"	"	"	"	IN	5.0V	"	"	"	"	"	"	"	"	"	"	F to J	"	"	"	"	"	"	"
		128	"	"	"	IN	5.0V	"	"	"	"	"	"	"	"	"	"	"	G to J	"	"	"	"	"	"	"
		129	"	"	IN	5.0V	"	"	"	"	"	"	"	"	"	"	"	"	H to J	"	"	"	"	"	"	"
t_{PHL4}	3003 (Fig. 8)	130	"	"	5.0V	"	"	"	"	"	5.0V	"	"	"	"	IN	"	"	A to J	32	"	33	670	29	570	"
		131	"	"	"	"	"	"	"	"	"	"	"	"	IN	5.0V	"	"	B to J	"	"	"	"	"	"	"
		132	"	"	"	"	"	"	"	"	"	"	"	IN	5.0V	"	"	"	C to J	"	"	"	"	"	"	"
		133	"	"	"	"	"	"	"	"	"	"	IN	5.0V	"	"	"	"	D to J	"	"	"	"	"	"	"
		134	"	"	"	"	"	IN	"	"	"	"	5.0V	"	"	"	"	"	E to J	"	"	"	"	"	"	"
		135	"	"	"	"	IN	5.0V	"	"	"	"	"	"	"	"	"	"	F to J	"	"	"	"	"	"	"
		136	"	"	"	IN	5.0V	"	"	"	"	"	"	"	"	"	"	"	G to J	"	"	"	"	"	"	"
		137	"	"	IN	5.0V	"	"	"	"	"	"	"	"	"	"	"	"	H to J	"	"	"	"	"	"	"
t_{PLH4}		138	"	"	5.0V	"	"	"	"	"	"	"	"	"	"	IN	"	"	A to J	35	695	42	830	31	610	"
		139	"	"	"	"	"	"	"	"	"	"	"	"	IN	5.0V	"	"	B to J	"	"	"	"	"	"	"
		140	"	"	"	"	"	"	"	"	"	"	"	IN	5.0V	"	"	"	C to J	"	"	"	"	"	"	"
		141	"	"	"	"	"	"	"	"	"	"	IN	5.0V	"	"	"	"	D to J	"	"	"	"	"	"	"
		142	"	"	"	"	"	IN	"	"	"	"	5.0V	"	"	"	"	"	E to J	"	"	"	"	"	"	"
		143	"	"	"	"	IN	5.0V	"	"	"	"	"	"	"	"	"	"	F to J	"	"	"	"	"	"	"
		144	"	"	"	IN	5.0V	"	"	"	"	"	"	"	"	"	"	"	G to J	"	"	"	"	"	"	"
		145	"	"	IN	5.0V	"	"	"	"	"	"	"	"	"	"	"	"	H to J	"	"	"	"	"	"	"
t_{PHL5}	3003 (Fig. 9)	146	"	"	GND	GND	GND	5.0V	5.0V	"	GND	GND	GND	GND	GND	IN	"	"	A to J	33	655	39	775	29	575	"
		147	"	"	"	"	5.0V	GND	"	"	"	"	"	GND	IN	GND	"	"	B to J	"	"	"	"	"	"	"
		148	"	"	"	5.0V	GND	"	"	"	"	"	"	IN	GND	"	"	"	C to J	"	"	"	"	"	"	"
		149	"	"	5.0V	GND	"	"	"	"	"	"	IN	GND	"	"	"	"	D to J	"	"	"	"	"	"	"
		150	"	"	GND	"	"	IN	"	"	"	"	GND	"	"	"	"	"	E to J	"	"	"	"	"	"	"
		151	"	"	"	"	IN	GND	"	"	"	"	"	"	"	5.0V	"	"	F to J	"	"	"	"	"	"	"
		152	"	"	"	IN	GND	"	"	"	"	"	"	"	5.0V	GND	"	"	G to J	"	"	"	"	"	"	"
		153	"	"	IN	GND	"	"	"	"	"	"	5.0V	GND	GND	"	"	"	H to J	"	"	"	"	"	"	"

See footnotes at end of device type 04.

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TABLE III. Group A inspection for device type 04 – Continued.

Symbol	MIL-STD-883 method	Cases E,F,Z	Terminal conditions 1/																Measured terminal	Test limits						Unit
		Symbol	J	Kd	H	G	F	E	Kb	V_{SS}	Kc	Ka	D	C	B	A	EX	V_{DD}		Subgroup 9 T_C = +25°C		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C		
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Min	Max	Min	Max	Min	Max	
t_{PLH5}	3003 (Fig. 9)	154	OUT	5.0V	GND	GND	GND	5.0V	5.0V	GND	GND	GND	GND	GND	GND	IN	GND	5.0V	A to J	27	540	32	645	24	485	ns
		155	"	"	"	GND	5.0V	GND	"	"	"	"	"	GND	IN	GND	"	"	B to J	"	"	"	"	"	"	"
		156	"	"	"	5.0V	GND	"	"	"	"	"	"	IN	GND	"	"	"	C to J	"	"	"	"	"	"	"
		157	"	"	5.0V	GND	"	"	"	"	"	"	IN	GND	"	"	"	"	D to J	"	"	"	"	"	"	"
		158	"	"	GND	"	"	IN	"	"	"	"	GND	"	"	5.0V	"	"	E to J	"	"	"	"	"	"	"
		159	"	"	"	"	IN	GND	"	"	"	"	"	"	5.0V	GND	"	"	F to J	"	"	"	"	"	"	"
		160	"	"	"	IN	GND	"	"	"	"	"	"	5.0V	GND	"	"	"	G to J	"	"	"	"	"	"	"
		161	"	"	IN	GND	"	"	"	"	"	"	5.0V	GND	"	"	"	"	H to J	"	"	"	"	"	"	"
t_{PHL6}	3003 (Fig. 10)	162	"	"	GND	"	"	5.0V	"	"	5.0V	"	GND	"	"	IN	"	"	A to J	32	640	38	755	27	540	"
		163	"	"	"	"	"	"	"	"	"	"	"	"	IN	GND	"	"	B to J	"	"	"	"	"	"	"
		164	"	"	"	"	"	"	"	"	"	"	"	IN	GND	"	"	"	C to J	"	"	"	"	"	"	"
		165	"	"	"	"	"	"	"	"	"	"	IN	GND	"	"	"	"	D to J	"	"	"	"	"	"	"
		166	"	"	"	"	"	IN	"	"	"	"	GND	"	"	5.0V	"	"	E to J	"	"	"	"	"	"	"
		167	"	"	"	"	IN	GND	"	"	"	"	"	"	"	"	"	"	F to J	"	"	"	"	"	"	"
		168	"	"	"	IN	GND	"	"	"	"	"	"	"	"	"	"	"	G to J	"	"	"	"	"	"	"
		169	"	"	IN	GND	"	"	"	"	"	"	"	"	"	"	"	"	H to J	"	"	"	"	"	"	"
t_{PLH6}		170	"	"	GND	"	"	5.0V	"	"	"	"	"	"	"	IN	"	"	A to J	30	585	35	700	26	520	"
		171	"	"	"	"	"	"	"	"	"	"	"	"	IN	GND	"	"	B to J	"	"	"	"	"	"	"
		172	"	"	"	"	"	"	"	"	"	"	"	IN	GND	"	"	"	C to J	"	"	"	"	"	"	"
		173	"	"	"	"	"	"	"	"	"	"	IN	GND	"	"	"	"	D to J	"	"	"	"	"	"	"
		174	"	"	"	"	"	IN	"	"	"	"	GND	"	"	5.0V	"	"	E to J	"	"	"	"	"	"	"
		175	"	"	"	"	IN	GND	"	"	"	"	"	"	"	"	"	"	F to J	"	"	"	"	"	"	"
		176	"	"	"	IN	GND	"	"	"	"	"	"	"	"	"	"	"	G to J	"	"	"	"	"	"	"
		177	"	"	IN	GND	"	"	"	"	"	"	"	"	"	"	"	"	H to J	"	"	"	"	"	"	"
t_{PHL7}	3003 (Fig. 11)	178	"	"	GND	"	"	"	"	"	GND	5.0 V	5.0 V	5.0 V	5.0 V	IN	5.0 V	"	A to J	34	675	40	790	30	590	"
		179	"	"	"	"	"	"	"	"	"	"	"	5.0 V	IN	5.0 V	"	"	B to J	"	"	"	"	"	"	"
		180	"	"	"	"	"	"	"	"	"	"	"	IN	5.0 V	"	"	"	C to J	"	"	"	"	"	"	"
		181	"	"	"	"	"	"	"	"	"	"	IN	5.0 V	5.0 V	"	"	"	D to J	"	"	"	"	"	"	"
		182	"	"	5.0 V	5.0 V	5.0 V	IN	"	"	"	"	GND	GND	GND	GND	"	"	E to J	"	"	"	"	"	"	"
		183	"	"	"	5.0 V	IN	5.0 V	"	"	"	"	"	"	"	"	"	"	F to J	"	"	"	"	"	"	"
		184	"	"	"	IN	5.0 V	"	"	"	"	"	"	"	"	"	"	"	G to J	"	"	"	"	"	"	"
		185	"	"	IN	5.0 V	5.0 V	"	"	"	"	"	"	"	"	"	"	"	H to J	"	"	"	"	"	"	"
t_{PLH7}		186	"	"	GND	GND	GND	GND	"	"	"	"	5.0 V	5.0 V	5.0 V	IN	"	"	A to J	31	615	36	730	27	550	"
		187	"	"	"	"	"	"	"	"	"	"	"	5.0 V	IN	5.0 V	"	"	B to J	"	"	"	"	"	"	"
		188	"	"	"	"	"	"	"	"	"	"	"	IN	5.0 V	"	"	"	C to J	"	"	"	"	"	"	"
		189	"	"	"	"	"	"	"	"	"	"	IN	5.0 V	5.0 V	"	"	"	D to J	"	"	"	"	"	"	"
		190	"	"	5.0 V	5.0 V	5.0 V	IN	"	"	"	"	GND	GND	GND	GND	"	"	E to J	"	"	"	"	"	"	"
		191	"	"	"	5.0 V	IN	5.0 V	"	"	"	"	"	"	"	"	"	"	F to J	"	"	"	"	"	"	"
		192	"	"	"	IN	5.0 V	"	"	"	"	"	"	"	"	"	"	"	G to J	"	"	"	"	"	"	"
		193	"	"	IN	5.0 V	5.0 V	"	"	"	"	"	"	"	"	"	"	"	H to J	"	"	"	"	"	"	"
t_{PHL8}	3003 (Fig. 12)	194	"	"	GND	GND	GND	GND	"	"	5.0 V	"	5.0 V	5.0 V	5.0 V	IN	"	"	A to J	36	715	42	840	31	630	"
		195	"	"	"	"	"	"	"	"	"	"	"	5.0 V	IN	5.0 V	"	"	B to J	"	"	"	"	"	"	"
		196	"	"	"	"	"	"	"	"	"	"	"	IN	5.0 V	"	"	"	C to J	"	"	"	"	"	"	"
		197	"	"	"	"	"	"	"	"	"	"	IN	5.0 V	5.0 V	"	"	"	D to J	"	"	"	"	"	"	"
		198	"	"	5.0 V	5.0 V	5.0 V	IN	"	"	"	"	GND	GND	GND	GND	"	"	E to J	"	"	"	"	"	"	"
		199	"	"	"	5.0 V	IN	5.0 V	"	"	"	"	"	"	"	"	"	"	F to J	"	"	"	"	"	"	"
		200	"	"	"	IN	5.0 V	"	"	"	"	"	"	"	"	"	"	"	G to J	"	"	"	"	"	"	"
		201	"	"	IN	5.0 V	5.0 V	"	"	"	"	"	"	"	"	"	"	"	H to J	"	"	"	"	"	"	"
t_{PLH8}		202	"	"	GND	GND	GND	GND	"	"	"	"	5.0 V	5.0 V	5.0 V	IN	"	"	A to J	30	600	36	720	27	535	"
		203	"	"	"	"	"	"	"	"	"	"	"	5.0 V	IN	5.0 V	"	"	B to J	"	"	"	"	"	"	"
		204	"	"	"	"	"	"	"	"	"	"	"	IN	5.0 V	"	"	"	C to J	"	"	"	"	"	"	"
		205	"	"	"	"	"	"	"	"	"	"	IN	5.0 V	5.0 V	"	"	"	D to J	"	"	"	"	"	"	"
		206	"	"	5.0 V	5.0 V	5.0 V	IN	"	"	"	"	GND	GND	GND	GND	"	"	E to J	"	"	"	"	"	"	"
		207	"	"	"	5.0 V	IN	5.0 V	"	"	"	"	"	"	"	"	"	"	F to J	"	"	"	"	"	"	"
		208	"	"	"	IN	5.0 V	"	"	"	"	"	"	"	"	"	"	"	G to J	"	"	"	"	"	"	"
		209	"	"	IN	5.0 V	5.0 V	"	"	"	"	"	"	"	"	"	"	"	H to J	"	"	"	"	"	"	"

See footnotes at end of device type 04.

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TABLE III. Group A inspection for device type 04 – Continued.

Symbol	MIL-STD-883 method	Cases E,F,Z	Terminal conditions 1/																Measured terminal	Test limits						Unit
		Symbol	J	Kd	H	G	F	E	Kb	V_{SS}	Kc	Ka	D	C	B	A	EX	V_{DD}		Subgroup 9 T_C = +25°C		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C		
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Min	Max	Min	Max	Min	Max	
t_{PHL9}	3003 (Fig. 13)	210	OUT	5.0 V	GND	GND	GND	GND	GND	GND	GND	IN	GND	GND	GND	GND	GND	5.0 V	Ka to J	34	690	38	760	30	595	ns
t_{PLH9}		211	"	"	"	"	"	"	GND	"	"	IN	"	"	"	"	"	"	Ka to J	31	625	37	730	28	555	"
t_{PHL10}	3003 (Fig. 14)	212	"	"	"	"	"	"	IN	"	"	GND	"	"	"	"	"	"	Kb to J	27	535	31	615	23	460	"
t_{PLH10}		213	"	"	"	"	"	"	IN	"	"	"	"	"	"	"	"	"	Kb to J	26	515	30	605	23	465	"
t_{PHL11}	3003 (Fig. 15)	214	"	"	"	"	"	"	GND	"	IN	"	"	"	"	"	"	"	Kc to J	21	420	24	475	18	360	"
t_{PLH11}		215	"	"	"	"	"	"	"	"	IN	"	"	"	"	"	"	"	Kc to J	19	380	22	445	17	340	"
t_{THL}	3004 (Fig. 5)	216	"	"	"	"	"	"	"	"	GND	"	"	"	"	IN	"	"	J	7	135	8	160	6	115	"
		217	"	"	"	"	"	"	"	"	"	"	"	"	IN	GND	"	"	"	"	"	"	"	"	"	"	"
		218	"	"	"	"	"	"	"	"	"	"	"	IN	GND	"	"	"	"	"	"	"	"	"	"	"	"
		219	"	"	"	"	"	"	"	"	"	"	IN	GND	"	"	"	"	"	"	"	"	"	"	"	"	"
		220	"	"	"	"	"	IN	"	"	"	"	GND	"	"	"	"	"	"	"	"	"	"	"	"	"	"
		221	"	"	"	"	IN	GND	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
		222	"	"	"	IN	GND	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
		223	"	"	IN	GND	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
t_{TLH}		224	"	"	GND	"	"	"	"	"	"	"	"	"	"	IN	"	"	"	6	115	7	140	5	100	"
		225	"	"	"	"	"	"	"	"	"	"	"	"	IN	GND	"	"	"	"	"	"	"	"	"	"	"
		226	"	"	"	"	"	"	"	"	"	"	"	IN	GND	"	"	"	"	"	"	"	"	"	"	"	"
		227	"	"	"	"	"	"	"	"	"	"	IN	GND	"	"	"	"	"	"	"	"	"	"	"	"	"
		228	"	"	"	"	"	IN	"	"	"	"	GND	"	"	"	"	"	"	"	"	"	"	"	"	"	"
		229	"	"	"	"	IN	GND	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
		230	"	"	"	IN	GND	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
		231	"	"	IN	GND	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
t_{PLZ}	3003 (Fig. 16)	232	"	IN	GND	"	"	"	"	"	"	"	"	"	"	5.0 V	"	"	Kd to J	18	350	25	490	18	350	"
t_{PHZ}		233	"	"	"	"	"	"	"	"	"	"	"	"	"	GND	"	"	"	18	350	25	490	18	350	"
t_{PZL}		234	"	"	"	"	"	"	"	"	"	"	"	"	"	5.0 V	"	"	"	23	450	32	630	23	450	"
t_{PZH}		235	"	"	"	"	"	"	"	"	"	"	"	"	"	GND	"	"	"	23	450	32	630	23	450	"

1/ Pins not designated shall be "high" level logic, "low" level logic, or open. Exceptions are as follows: For $V_{IC(POS)}$ tests, the V_{SS} terminal shall be open; for $V_{IC(NEG)}$ tests, the V_{DD} terminal shall be open; for I_{SS} tests, the output shall be open.

2/ During $V_{IC(POS)}$ and $V_{IC(NEG)}$ tests, the V_{SS} terminals respectively shall be open.

3/ I_{SS} measurements shall be performed in sequence.

4/ V_{IL1} = 1.1 V at +25°C; 0.8 V at +125°C; 1.35 V at -55°C.

5/ V_{IL2} = 2.8 V at +25°C; 2.55 V at +125°C; 1.35 V at -55°C.

6/ V_{IH1} = 3.8 V at +25°C; 3.6 V at +125°C; 3.95 V at -55°C.

7/ V_{IH2} = 9.5 V at +25°C; 9.25 V at +125°C; 9.75 V at -55°C.

8/ The device manufacturer may, at his option, measure I_{IL} and I_{IH} at +25°C for each individual input or measure all inputs together.

9/ See 4.4.1c.

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TABLE III. Group A inspection for device type 51.

Symbol	MIL-STD-883 method	Cases A, C,D,T,X,Y Symbol	Terminal conditions 1/														Measured terminal	Test limits						Unit
			2aY	2aS	2A	2bS	2bY	1A	V_{SS}	1bY	3bS	3A	3aS	3Y	1aY	V_{DD}		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14		Min	Max	Min	Max	Min	Max	
$V_{IC(POS)}$		1 2 3	(5) 2/ " "	GND " "	1mA		(1) 2/ " "	1mA		(13) 2/ " "		1mA	GND " "		(8) 2/ " "	GND " "	1A 2A 3A		1.5 " "					V " "
$V_{IC(NEG)}$		4 5 6	" " "		-1mA	GND " "	" " "	-1mA	GND " "	" " "	GND " "				" " "		1A 2A 3A		-6 " "					" " "
I_{SS} 3/	3005	7 8	" "	18.0V 18.0V	GND 18.0V	" "	" "	GND 18.0V	" "	" "	" "	GND 18.0V	18.0V 18.0V		" "	18.0V 18.0V	V_{SS} V_{SS}		-50 -50		-500 -500			nA nA
V_{OH5}	3006	9 10 11	" " "	15.0V " "	GND " "	" " "	" " "	GND " "	" " "	" " "	" " "	GND " "	15.0V " "		" " "	15.0V " "	1aY 2aY 3Y	14.95 " "		14.95 " "		14.95 " "		V " "
V_{OL6}	3007	12 13 14	" " "	" " "	" 15.0V GND	" " "	" " "	15.0V GND GND	" " "	" " "	" " "	" " 15.0V	" " "		" " "	" " "	1aY 2aY 3Y		0.05 " "		0.05 " "		0.05 " "	" " "
V_{IH1} 4/		15 16 17	" " "	5.0V " "	GND 4.0V GND	" " "	" " "	4.0V GND GND	" " "	" " "	" " "	GND GND 4.0V	5.0V " "		" " "	5.0V " "	1aY 2aY 3Y		0.5 " "		0.5 " "		0.5 " "	" " "
V_{IH2} 4/		18 19 20	" " "	10.0V " "	GND 8.0V GND	" " "	" " "	8.0V GND GND	" " "	" " "	" " "	GND GND 8.0V	10.0V " "		" " "	10.0V " "	1aY 2aY 3Y		1.0 " "		1.0 " "		1.0 " "	" " "
V_{IH3} 4/		21 22 23	" " "	15.0V " "	GND 12.0V GND	" " "	" " "	12.0V GND GND	" " "	" " "	" " "	GND GND 12.0V	15.0V " "		" " "	15.0V " "	1aY 2aY 3Y		1.5 " "		1.5 " "		1.5 " "	" " "
V_{IL1} 4/		24 25 26	" " "	5.0V " "	GND 1.0V GND	" " "	" " "	1.0V GND GND	" " "	" " "	" " "	GND GND 1.0V	5.0V " "		" " "	5.0V " "	1aY 2aY 3Y	4.5 " "		4.5 " "		4.5 " "		" " "
V_{IL2} 4/		27 28 29	" " "	10.0V " "	GND 2.0V GND	" " "	" " "	2.0V GND GND	" " "	" " "	" " "	GND GND 2.0V	10.0V " "		" " "	10.0V " "	1aY 2aY 3Y	9.0 " "		9.0 " "		9.0 " "		" " "
V_{IL3} 4/		30 31 32	" " "	15.0V " "	GND 3.0V GND	" " "	" " "	3.0V GND GND	" " "	" " "	" " "	GND GND 3.0V	15.0V " "		" " "	15.0V " "	1aY 2aY 3Y	13.5 " "		13.5 " "		13.5 " "		" " "
I_{OL1}		33 34 35	" " 0.4V "	5.0V " "	GND 5.0V GND	" " "	" " "	5.0V GND GND	" " "	" " "	" " "	GND GND 5.0V	5.0V " "	0.4V	" 0.4V " "	5.0V " "	1aY 2aY 3Y	0.51 " "		0.36 " "		0.64 " "		mA " "
I_{OL2}		36 37 38	" " 1.5V "	15.0V " "	GND 15.0V GND	" " "	" " "	15.0V GND GND	" " "	" " "	" " "	GND GND 15.0V	15.0V " "	1.5V	" 1.5V " "	15.0V " "	1aY 2aY 3Y	3.4 " "		2.4 " "		4.2 " "		" " "
I_{OH1}		39 40 41	" " 4.6V "	" " "	5.0V GND 5.0V	" " "	" " "	GND 5.0V 5.0V	" " "	" " "	" " "	5.0V 5.0V GND	" " "	4.6V	" 4.6V " "	5.0V " "	1aY 2aY 3Y	-0.51 " "		-0.36 " "		-0.64 " "		mA " "
I_{OH2}		42 43 44	" " 13.5V "	15.0V " "	15.0V GND 15.0V	" " "	" " "	GND 15.0V 15.0V	" " "	" " "	" " "	15.0V 15.0V GND	15.0V " "	13.5V	" 13.5V " "	15.0V " "	1aY 2aY 3Y	-3.4 " "		-2.4 " "		-4.2 " "		" " "

See footnotes at end of device type 51.

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TABLE III. Group A inspection for device type 51 – Continued.

Symbol	MIL-STD-883 method	Cases A, C,D,T,X,Y	Terminal conditions 1/														Measured terminal	Test limits						Unit
		Symbol	2aY	2aS	2A	2bS	2bY	1A	V_{SS}	1bY	3bS	3A	3aS	3Y	1aY	V_{DD}		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14		Min	Max	Min	Max	Min	Max	
I_{IH1} 5/	3010	45	(5) 2/	18.0V	18.0V	GND	(1) 2/	18.0V	GND	(13) 2/	GND	18.0V	18.0V		(8) 2/	18.0V	All inputs together		6.0					nA
I_{IH2} 5/		46	"	"	GND	"	"	18.0V	"	"	"	GND	"		"	"	1A		1.0		45				"
		47	"	"	18.0V	"	"	GND	"	"	"	GND	"		"	"	2A		"		"				"
		48	"	"	GND	"	"	"	"	"	"	18.0V	"		"	"	3A		"		"				"
I_{IL1} 5/	3009	49	"	"	GND	"	"	"	"	"	"	GND	"		"	"	All inputs together		-6.0						"
I_{IL2} 5/		50	"	"	18.0V	"	"	"	"	"	"	18.0V	"		"	"	1A		-1.0		-45				"
		51	"	"	GND	"	"	18.0V	"	"	"	18.0V	"		"	"	2A		"		"				"
		52	"	"	18.0V	"	"	18.0V	"	"	"	GND	"		"	"	3A		"		"				"
																		Subgroup 4 T_C = +25°C						
																		Min	Max					
C_I	3012	53	(5) 2/	GND		GND	(1) 2/	6/	GND	(13) 2/	GND		GND		(8) 2/	GND	1A		12					pF
		54	"	"	6/	"	"		"	"	"		"		"	"	2A		"						"
		55	"	"		"	"		"	"	"	6/	"		"	"	3A		"						"
																		Subgroup 9 T_C = +25°C		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C		
																		Min	Max	Min	Max	Min	Max	
t_{PHL}	3003 (Fig. 4)	56	(5) 2/	5.0V	GND	GND	(1) 2/	IN	GND	(13) 2/	GND	GND	5.0V		(8) 2/	5.0V	1A to 1aY	6	110	9	155	6	110	ns
		57	"	"	IN	"	"	GND	"	"	"	GND	"		"	"	2A to 2aY	"	"	"	"	"	"	"
		58	"	"	GND	"	"	GND	"	"	"	IN	"		"	"	3A to 3Y	"	"	"	"	"	"	"
t_{PLH}		59	"	"	GND	"	"	IN	"	"	"	GND	"		"	"	1A to 1aY	"	"	"	"	"	"	"
		60	"	"	IN	"	"	GND	"	"	"	GND	"		"	"	2A to 2aY	"	"	"	"	"	"	"
		61	"	"	GND	"	"	GND	"	"	"	IN	"		"	"	3A to 3Y	"	"	"	"	"	"	"
t_{THL}	3004 (Fig. 4)	62	"	"	GND	"	"	IN	"	"	"	GND	"		"	"	1aY	"	200	"	280	"	200	"
		63	"	"	IN	"	"	GND	"	"	"	GND	"		"	"	2aY	"	"	"	"	"	"	"
		64	"	"	GND	"	"	GND	"	"	"	IN	"		"	"	3Y	"	"	"	"	"	"	"
t_{TLH}		65	"	"	GND	"	"	IN	"	"	"	GND	"		"	"	1aY	"	"	"	"	"	"	"
		66	"	"	IN	"	"	GND	"	"	"	GND	"		"	"	2aY	"	"	"	"	"	"	"
		67	"	"	GND	"	"	GND	"	"	"	IN	"		"	"	3Y	"	"	"	"	"	"	"

1/ Pins not designated may be "high" level logic, "low" level logic, or open. Exceptions are as follows: For $V_{IC(POS)}$ tests, the V_{SS} terminal shall be open; for $V_{IC(NEG)}$ tests, the V_{DD} terminal shall be open; for I_{SS} tests, the outputs shall be open.

2/ Terminals in parentheses are connected together as indicated by the included number.

3/ I_{SS} measurements shall be performed in sequence.

4/ The V_{IL} and V_{IH} input test requirements are acceptable if the output state list is met, attributes data required.

5/ The device manufacturer may, at his option, measure I_{IL} and I_{IH} at +25°C for each individual input or measure all inputs together.

6/ See 4.4.1c.

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TABLE III. Group A inspection for device type 52.

Symbol	MIL-STD-883 method	Cases E,F,N,Z	Terminal conditions 1/																Measured terminal	Test limits						Unit
		Symbol	B4	A3	B3	A2	B2	A1	B1	V_{SS}	Ka	D1	D2	D3	D4	Kb	A4	V_{DD}		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Min	Max	Min	Max	Min	Max	
$V_{IC(POS)}$		1						1mA	1mA									GND	B1		1.5					V
		2																"	A1		"					"
		3					1mA											"	B2		"					"
		4				1mA												"	A2		"					"
		5			1mA													"	B3		"					"
		6		1mA														"	A3		"					"
		7	1mA															"	B4		"					"
		8															1mA	"	A4		"					"
		9									1mA							"	Ka		"					"
		10														1mA		"	Kb		"					"
$V_{IC(NEG)}$		11						-1mA	-1mA	GND									B1		-6.0					"
		12								"									A1		"					"
		13					-1mA			"									B2		"					"
		14				-1mA				"									A2		"					"
		15			-1mA					"									B3		"					"
		16		-1mA						"									A3		"					"
		17	-1mA							"									B4		"					"
		18								"							-1mA		A4		"					"
		19								"	-1mA								Ka		"					"
		20								"						-1mA			Kb		"					"
I_{SS} 2/	3005	21	GND	18.0V	GND	18.0V	GND	18.0V	GND	"	18.0V					18.0V	18.0V	18.0V	V_{SS}		-75		-750			nA
		22	GND	GND	GND	GND	GND	GND	GND	"	"					"	GND	"	"		"		"			"
		23	18.0V	GND	18.0V	GND	18.0V	GND	18.0V	"	"					"	GND	"	"		"		"			"
		24	18.0V	18.0V	18.0V	18.0V	18.0V	18.0V	18.0V	"	GND					GND	18.0V	"	"		"		"			"
V_{OH5}	3006	25	15.0V	GND	GND	GND	GND	GND	GND	"	15.0V					15.0V	GND	15.0V	D4	14.95		14.95		14.95			V
		26	GND	15.0V	GND	"	"	"	"	"	"					"	"	"	D3	"		"		"			"
		27	"	GND	15.0V	"	"	"	"	"	"					"	"	"	D3	"		"		"			"
		28	"	"	GND	15.0V	"	"	"	"	"					"	"	"	D2	"		"		"			"
		29	"	"	"	GND	15.0V	"	"	"	"					"	"	"	D2	"		"		"			"
		30	"	"	"	"	GND	15.0V	"	"	"					"	"	"	D1	"		"		"			"
		31	"	"	"	"	"	GND	15.0V	"	"					"	"	"	D1	"		"		"			"
		32	"	"	"	"	"	"	GND	"	"					"	15.0V	"	D4	"		"		"			"
V_{OL6}	3007	33	15.0V	15.0V	15.0V	15.0V	15.0V	15.0V	15.0V	"	GND					GND	"	"	D4		0.05		0.05		0.05		"
		34	"	"	"	"	"	"	"	"	"					"	"	"	D3		"		"		"		"
		35	"	"	"	"	"	"	"	"	"					"	"	"	D2		"		"		"		"
		36	"	"	"	"	"	"	"	"	"					"	"	"	D1		"		"		"		"
		37	GND	GND	GND	GND	GND	GND	GND	"	15.0V					15.0V	GND	"	D4		"		"		"		"
		38	"	"	"	"	"	"	"	"	"					"	"	"	D3		"		"		"		"
		39	"	"	"	"	"	"	"	"	"					"	"	"	D2		"		"		"		"
		40	"	"	"	"	"	"	"	"	"					"	"	"	D1		"		"		"		"
V_{IH1} 3/		41	3.5V	3.5V	3.5V	3.5V	3.5V	3.5V	3.5V	"	1.5V					1.5V	3.5V	5.0V	D4		0.5		0.5		0.5		"
		42	"	"	"	"	"	"	"	"	"					"	"	"	D3		"		"		"		"
		43	"	"	"	"	"	"	"	"	"					"	"	"	D2		"		"		"		"
		44	"	"	"	"	"	"	"	"	"					"	"	"	D1		"		"		"		"
		45	1.5V	1.5V	1.5V	1.5V	1.5V	1.5V	1.5V	"	5.0V					5.0V	1.5V	"	D4		"		"		"		"
		46	"	"	"	"	"	"	"	"	"					"	"	"	D3		"		"		"		"
		47	"	"	"	"	"	"	"	"	"					"	"	"	D2		"		"		"		"
		48	"	"	"	"	"	"	"	"	"					"	"	"	D1		"		"		"		"
V_{IH2} 3/		49	7.0V	7.0V	7.0V	7.0V	7.0V	7.0V	7.0V	"	3.0V					3.0V	7.0V	10.0V	D4		1.0		1.0		1.0		"
		50	"	"	"	"	"	"	"	"	"					"	"	"	D3		"		"		"		"
		51	"	"	"	"	"	"	"	"	"					"	"	"	D2		"		"		"		"
		52	"	"	"	"	"	"	"	"	"					"	"	"	D1		"		"		"		"
		53	3.0V	3.0V	3.0V	3.0V	3.0V	3.0V	3.0V	"	10.0V					10.0V	3.0V	"	D4		"		"		"		"
		54	"	"	"	"	"	"	"	"	"					"	"	"	D3		"		"		"		"
		55	"	"	"	"	"	"	"	"	"					"	"	"	D2		"		"		"		"
		56	"	"	"	"	"	"	"	"	"					"	"	"	D1		"		"		"		"

See footnotes at end of device type 52.

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TABLE III. Group A inspection for device type 52 – Continued.

Symbol	MIL-STD-883 method	Cases E,F,N,Z	Terminal conditions 1/																Measured terminal	Test limits						Unit
		Symbol	B4	A3	B3	A2	B2	A1	B1	V_{SS}	Ka	D1	D2	D3	D4	Kb	A4	V_{DD}		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Min	Max	Min	Max	Min	Max	
V_{IH3} 3/		57	11.0V	11.0V	11.0V	11.0V	11.0V	11.0V	11.0V	GND	4.0V					4.0V	11.0V	15.0V	D4		1.5		1.5		1.5	V
		58	"	"	"	"	"	"	"	"	"					"	"	"	D3		"		"		"	"
		59	"	"	"	"	"	"	"	"	"					"	"	"	D2		"		"		"	"
		60	"	"	"	"	"	"	"	"	"					"	"	"	D1		"		"		"	"
		61	4.0V	4.0V	4.0V	4.0V	4.0V	4.0V	4.0V	"	15.0V					15.0V	4.0V	"	D4		"		"		"	"
		62	"	"	"	"	"	"	"	"	"					"	"	"	D3		"		"		"	"
		63	"	"	"	"	"	"	"	"	"					"	"	"	D2		"		"		"	"
		64	"	"	"	"	"	"	"	"	"					"	"	"	D1		"		"		"	"
V_{IL1} 3/		65	3.5V	GND	GND	GND	GND	GND	GND	"	5.0V					5.0V	1.5V	5.0V	D4	4.5		4.5		4.5		"
		66	GND	3.5V	1.5V	"	"	"	"	"	"					"	GND	"	D3	"		"		"		"
		67	"	1.5V	3.5V	"	"	"	"	"	"					"	"	"	D3	"		"		"		"
		68	"	GND	GND	3.5V	1.5V	"	"	"	"					"	"	"	D2	"		"		"		"
		69	"	"	"	1.5V	3.5V	"	"	"	"					"	"	"	D2	"		"		"		"
		70	"	"	"	GND	GND	3.5V	1.5V	"	"					"	"	"	D1	"		"		"		"
		71	"	"	"	"	"	1.5V	3.5V	"	"					"	"	"	D1	"		"		"		"
		72	1.5V	"	"	"	"	GND	GND	"	"					"	3.5V	"	D4	"		"		"		"
V_{IL2} 3/		73	7.0V	"	"	"	"	"	"	"	10.0V					10.0V	3.0V	10.0V	D4	9.0		9.0		9.0		"
		74	GND	7.0V	3.0V	"	"	"	"	"	"					"	GND	"	D3	"		"		"		"
		75	"	3.0V	7.0V	"	"	"	"	"	"					"	"	"	D3	"		"		"		"
		76	"	GND	GND	7.0V	3.0V	"	"	"	"					"	"	"	D2	"		"		"		"
		77	"	"	"	3.0V	7.0V	"	"	"	"					"	"	"	D2	"		"		"		"
		78	"	"	"	GND	GND	7.0V	3.0V	"	"					"	"	"	D1	"		"		"		"
		79	"	"	"	"	"	3.0V	7.0V	"	"					"	"	"	D1	"		"		"		"
		80	3.0V	"	"	"	"	GND	GND	"	"					"	7.0V	"	D4	"		"		"		"
V_{IL3} 3/		81	11.0V	"	"	"	"	"	"	"	15.0V					15.0V	4.0V	15.0V	D4	13.5		13.5		13.5		"
		82	GND	11.0V	4.0V	"	"	"	"	"	"					"	GND	"	D3	"		"		"		"
		83	"	4.0V	11.0V	"	"	"	"	"	"					"	"	"	D3	"		"		"		"
		84	"	GND	GND	11.0V	4.0V	"	"	"	"					"	"	"	D2	"		"		"		"
		85	"	"	"	4.0V	11.0V	"	"	"	"					"	"	"	D2	"		"		"		"
		86	"	"	"	GND	GND	11.0V	4.0V	"	"					"	"	"	D1	"		"		"		"
		87	"	"	"	"	"	4.0V	11.0V	"	"					"	"	"	D1	"		"		"		"
		88	4.0V	"	"	"	"	GND	GND	"	"					"	11.0V	"	D4	"		"		"		"
I_{OL1}		89	5.0V	5.0V	5.0V	5.0V	5.0V	5.0V	5.0V	"	GND			0.4V	0.4V	GND	5.0V	5.0V	D4	0.51		0.36		0.64		mA
		90	"	"	"	"	"	"	"	"	"		0.4V			"	"	"	D3	"		"		"		"
		91	"	"	"	"	"	"	"	"	"					"	"	"	D2	"		"		"		"
		92	"	"	"	"	"	"	"	"	"	0.4V				"	"	"	D1	"		"		"		"
		93	GND	GND	GND	GND	GND	GND	GND	"	5.0V				0.4V	5.0V	GND	"	D4	"		"		"		"
		94	"	"	"	"	"	"	"	"	"			0.4V		"	"	"	D3	"		"		"		"
		95	"	"	"	"	"	"	"	"	"		0.4V			"	"	"	D2	"		"		"		"
		96	"	"	"	"	"	"	"	"	"	0.4V				"	"	"	D1	"		"		"		"
I_{OL2}		97	15.0V	15.0V	15.0V	15.0V	15.0V	15.0V	15.0V	"	GND				1.5V	GND	15.0V	15.0V	D4	3.4		2.4		4.2		"
		98	"	"	"	"	"	"	"	"	"			1.5V		"	"	"	D3	"		"		"		"
		99	"	"	"	"	"	"	"	"	"		1.5V			"	"	"	D2	"		"		"		"
		100	"	"	"	"	"	"	"	"	"	1.5V				"	"	"	D1	"		"		"		"
		101	GND	GND	GND	GND	GND	GND	GND	"	15.0V				1.5V	15.0V	GND	"	D4	"		"		"		"
		102	"	"	"	"	"	"	"	"	"			1.5V		"	"	"	D3	"		"		"		"
		103	"	"	"	"	"	"	"	"	"		1.5V			"	"	"	D2	"		"		"		"
		104	"	"	"	"	"	"	"	"	"	1.5V				"	"	"	D1	"		"		"		"
I_{OH1}		105	5.0V	"	"	"	"	"	"	"	5.0V				4.6V	5.0V	GND	5.0V	D4	-0.51		-0.36		-0.64		"
		106	GND	5.0V	"	"	"	"	"	"	"			4.6V	4.6V	"	"	"	D3	"		"		"		"
		107	"	GND	5.0V	"	"	"	"	"	"			4.6V		"	"	"	D3	"		"		"		"
		108	"	"	GND	5.0V	"	"	"	"	"		4.6V			"	"	"	D2	"		"		"		"
		109	"	"	"	GND	5.0V	"	"	"	"		4.6V			"	"	"	D2	"		"		"		"
		110	"	"	"	"	GND	5.0V	"	"	"	4.6V				"	"	"	D1	"		"		"		"
		111	"	"	"	"	"	GND	5.0V	"	"	4.6V				"	"	"	D1	"		"		"		"
		112	"	"	"	"	"	GND	GND	"	"				4.6V	"	5.0V	"	D4	"		"		"		"

See footnotes at end of device type 52.

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TABLE III. Group A inspection for device type 52 – Continued.

Symbol	MIL-STD-883 method	Cases E,F,N,Z	Terminal conditions 1/																Measured terminal	Test limits						Unit
		Symbol	B4	A3	B3	A2	B2	A1	B1	V_{SS}	Ka	D1	D2	D3	D4	Kb	A4	V_{DD}		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Min	Max	Min	Max	Min	Max	
I_{OH2}		113	15.0V	GND	GND	GND	GND	GND	GND	GND	15.0V				13.5V	15.0V	GND	15.0V	D4	-3.4		-2.4		-4.2		mA
		114	GND	15.0V	GND	"	"	"	"	"	"			13.5V		"	"	"	D3	"		"		"		"
		115	"	GND	15.0V	"	"	"	"	"	"			13.5V		"	"	"	D3	"		"		"		"
		116	"	"	GND	15.0V	"	"	"	"	"		13.5V			"	"	"	D2	"		"		"		"
		117	"	"	"	GND	15.0V	"	"	"	"		13.5V			"	"	"	D2	"		"		"		"
		118	"	"	"	"	GND	15.0V	"	"	"	13.5V				"	"	"	D1	"		"		"		"
		119	"	"	"	"	"	GND	15.0V	"	"	13.5V				"	"	"	D1	"		"		"		"
		120	"	"	"	"	"	GND	GND	"	"				13.5V	"	15.0V	"	D4	"		"		"		"
I_{IH1} 4/	3010	121	18.0V	18.0V	18.0V	18.0V	18.0V	18.0V	18.0V	"	18.0V					18.0V	18.0V	18.0V	All inputs together		1000					nA
I_{IH2} 4/		122	18.0V	GND	GND	GND	GND	GND	GND	"	GND					GND	GND	"	B4		100		100			"
		123	GND	18.0V	GND	"	"	"	"	"	"					"	"	"	A3		"		"			"
		124	"	GND	18.0V	"	"	"	"	"	"					"	"	"	B3		"		"			"
		125	"	"	GND	18.0V	"	"	"	"	"					"	"	"	A2		"		"			"
		126	"	"	"	GND	18.0V	"	"	"	"					"	"	"	B2		"		"			"
		127	"	"	"	"	GND	18.0V	"	"	"					"	"	"	A1		"		"			"
		128	"	"	"	"	"	GND	18.0V	"	"					"	"	"	B1		"		"			"
		129	"	"	"	"	"	"	GND	"	18.0V					"	"	"	Ka		"		"			"
		130	"	"	"	"	"	"	"	"	GND					18.0V	"	"	Kb		"		"			"
		131	"	"	"	"	"	"	"	"	"					GND	18.0V	"	A4		"		"			"
I_{IL1} 4/	3009	132	"	"	"	"	"	"	"	"	"					GND	GND	"	All inputs together		-1000					"
I_{IL2} 4/		133	"	18.0V	18.0V	18.0V	18.0V	18.0V	18.0V	"	18.0V					18.0V	18.0V	"	B4		-100		-100			"
		134	18.0V	GND	18.0V	"	"	"	"	"	"					"	"	"	A3		"		"			"
		135	"	18.0V	GND	"	"	"	"	"	"					"	"	"	B3		"		"			"
		136	"	"	18.0V	GND	"	"	"	"	"					"	"	"	A2		"		"			"
		137	"	"	"	18.0V	GND	"	"	"	"					"	"	"	B2		"		"			"
		138	"	"	"	"	18.0V	GND	"	"	"					"	"	"	A1		"		"			"
		139	"	"	"	"	"	18.0V	GND	"	"					"	"	"	B1		"		"			"
		140	"	"	"	"	"	"	18.0V	"	GND					"	"	"	Ka		"		"			"
		141	"	"	"	"	"	"	"	"	18.0V					GND	"	"	Kb		"		"			"
		142	"	"	"	"	"	"	"	"	18.0V					18.0V	GND	"	A4		"		"			"
																				Subgroup 4 T_C = +25°C						
																				Min	Max					
C_I	3012	143	5/							GND								GND	B4		12.0					pF
		144		5/						"								"	A3		"					"
		145			5/					"								"	B3		"					"
		146				5/				"								"	A2		"					"
		147					5/			"								"	B2		"					"
		148						5/		"								"	A1		"					"
		149							5/	"								"	B1		"					"
		150								"							5/	"	A4		"					"
		151								"	5/							"	Ka		"					"
		152								"						5/		"	Kb		"					"
																				Subgroup 9 T_C = +25°C		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C		
																				Min	Max	Min	Max	Min	Max	
t_{PHL}	3003 (Fig. 4)	153	IN	GND	GND	GND	GND	GND	GND	GND	5.0V				OUT	5.0V	GND	5.0V	D4	13	300	18	420	13	300	ns
		154	GND	"	IN	"	GND	"	"	"	"				"	"	"	"	D3	"	"	"	"	"	"	"
		155	"	"	GND	"	IN	"	"	"	"		OUT		"	"	"	"	D2	"	"	"	"	"	"	"
		156	"	"	"	"	GND	"	IN	"	"	OUT		OUT	"	"	"	"	D1	"	"	"	"	"	"	"
		157	"	"	"	"	GND	"	GND	"	"				OUT	"	IN	"	D4	"	"	"	"	"	"	"

See footnotes at end of device type 52.

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TABLE III. Group A inspection for device type 52 – Continued.

Symbol	MIL-STD-883 method	Cases E,F,N,Z	Terminal conditions 1/																Measured terminal	Test limits						Unit
		Symbol	B4	A3	B3	A2	B2	A1	B1	V_{SS}	Ka	D1	D2	D3	D4	Kb	A4	V_{DD}		Subgroup 9 T_C = +25°C		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C		
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Min	Max	Min	Max	Min	Max	
t_{PHL}	3003 (Fig. 4)	158	GND	IN	GND	GND	GND	GND	GND	GND	5.0V		OUT	OUT		5.0V	GND	5.0V	D3	13	300	18	420	13	300	ns
		159	"	GND	"	IN	"	GND	"	"	"		"	"		"	"	"	D2	"	"	"	"	"	"	"
		160	"	"	"	GND	"	IN	"	"	"	OUT	"	"		"	"	"	D1	"	"	"	"	"	"	"
		161	"	"	"	"	"	GND	"	"	IN		"	"	OUT	"	5.0V	"	D4	"	"	"	"	"	"	"
		162	"	5.0V	"	"	"	"	"	"	"		"	OUT		"	GND	"	D3	"	"	"	"	"	"	"
		163	"	GND	"	5.0V	"	"	"	"	"		OUT			"	"	"	D2	"	"	"	"	"	"	"
		164	"	"	"	GND	"	5.0V	"	"	"	OUT	"	"		"	"	"	D1	"	"	"	"	"	"	"
		165	5.0V	"	"	"	"	GND	"	"	5.0V		"	"	OUT	IN	"	"	D4	"	"	"	"	"	"	"
		166	GND	"	5.0V	"	"	"	"	"	"		"	OUT		"	"	"	D3	"	"	"	"	"	"	"
		167	"	"	GND	"	5.0V	"	"	"	"		OUT			"	"	"	D2	"	"	"	"	"	"	"
		168	"	"	"	"	GND	"	5.0V	"	"	OUT				"	"	"	D1	"	"	"	"	"	"	"
t_{PLH}		169	IN	"	"	"	"	"	GND	"	"				OUT	5.0V	"	"	D4	"	"	"	"	"	"	"
		170	GND	"	IN	"	"	"	"	"	"		OUT	OUT		"	"	"	D3	"	"	"	"	"	"	"
		171	"	"	GND	"	IN	"	"	"	"					"	"	"	D2	"	"	"	"	"	"	"
		172	"	"	"	"	GND	"	IN	"	"	OUT			OUT	"	"	"	D1	"	"	"	"	"	"	"
		173	"	"	"	"	"	"	GND	"	"					"	IN	"	D4	"	"	"	"	"	"	"
		174	"	IN	"	"	"	"	"	"	"			OUT		"	GND	"	D3	"	"	"	"	"	"	"
		175	"	GND	"	IN	"	"	"	"	"		OUT			"	"	"	D2	"	"	"	"	"	"	"
		176	"	"	"	GND	"	IN	"	"	"	OUT				"	"	"	D1	"	"	"	"	"	"	"
		177	"	5.0V	"	"	"	GND	"	"	IN		"	"	OUT	"	5.0V	"	D4	"	"	"	"	"	"	"
		178	"	GND	"	5.0V	"	"	"	"	"		"	OUT		"	GND	"	D3	"	"	"	"	"	"	"
		179	"	"	"	GND	"	5.0V	"	"	"		OUT			"	"	"	D2	"	"	"	"	"	"	"
		180	"	"	"	"	"	GND	"	"	"	OUT				"	"	"	D1	"	"	"	"	"	"	"
		181	5.0V	"	"	"	"	"	"	"	5.0V		"	"	OUT	IN	"	"	D4	"	"	"	"	"	"	"
		182	GND	"	5.0V	"	"	"	"	"	"		"	OUT		"	"	"	D3	"	"	"	"	"	"	"
		183	"	"	GND	"	5.0V	"	"	"	"		OUT			"	"	"	D2	"	"	"	"	"	"	"
		184	"	"	"	"	GND	"	5.0V	"	"	OUT				"	"	"	D1	"	"	"	"	"	"	"
t_{THL}	3004 (Fig. 4)	185	IN	"	"	"	"	"	GND	"	"				OUT	5.0V	"	"	D4	10	"	14	"	10	"	"
		186	GND	"	IN	"	"	"	"	"	"		OUT	OUT		"	"	"	D3	"	"	"	"	"	"	"
		187	"	"	GND	"	IN	"	"	"	"					"	"	"	D2	"	"	"	"	"	"	"
		188	"	"	"	"	GND	"	IN	"	"	OUT			OUT	"	"	"	D1	"	"	"	"	"	"	"
		189	"	"	"	"	"	"	GND	"	"					"	IN	"	D4	"	"	"	"	"	"	"
		190	"	IN	"	"	"	"	"	"	"			OUT		"	GND	"	D3	"	"	"	"	"	"	"
		191	"	GND	"	IN	"	"	"	"	"		OUT			"	"	"	D2	"	"	"	"	"	"	"
		192	"	"	"	GND	"	IN	"	"	"	OUT				"	"	"	D1	"	"	"	"	"	"	"
		193	"	"	"	"	"	GND	"	"	IN		"	"	OUT	"	5.0V	"	D4	"	"	"	"	"	"	"
		194	"	5.0V	"	"	"	"	"	"	"		"	OUT		"	GND	"	D3	"	"	"	"	"	"	"
		195	"	GND	"	5.0V	"	"	"	"	"		OUT			"	"	"	D2	"	"	"	"	"	"	"
		196	"	"	"	GND	"	5.0V	"	"	"	OUT				"	"	"	D1	"	"	"	"	"	"	"
		197	5.0V	"	"	"	"	GND	"	"	5.0V		"	"	OUT	IN	"	"	D4	"	"	"	"	"	"	"
		198	GND	"	5.0V	"	"	"	"	"	"		"	OUT		"	"	"	D3	"	"	"	"	"	"	"
		199	"	"	GND	"	5.0V	"	"	"	"		OUT			"	"	"	D2	"	"	"	"	"	"	"
		200	"	"	"	"	GND	"	5.0V	"	"	OUT				"	"	"	D1	"	"	"	"	"	"	"
t_{TLH}		201	IN	"	"	"	"	"	GND	"	"				OUT	5.0V	"	"	D4	"	"	"	"	"	"	"
		202	GND	"	IN	"	"	"	"	"	"		OUT	OUT		"	"	"	D3	"	"	"	"	"	"	"
		203	"	"	GND	"	IN	"	"	"	"					"	"	"	D2	"	"	"	"	"	"	"
		204	"	"	"	"	GND	"	IN	"	"	OUT			OUT	"	"	"	D1	"	"	"	"	"	"	"
		205	"	"	"	"	"	"	GND	"	"					"	IN	"	D4	"	"	"	"	"	"	"
		206	"	IN	"	"	"	"	"	"	"			OUT		"	GND	"	D3	"	"	"	"	"	"	"
		207	"	GND	"	IN	"	"	"	"	"		OUT			"	"	"	D2	"	"	"	"	"	"	"
		208	"	"	"	GND	"	IN	"	"	"	OUT				"	"	"	D1	"	"	"	"	"	"	"
		209	"	"	"	"	"	GND	"	"	IN		"	"	OUT	"	5.0V	"	D4	"	"	"	"	"	"	"
		210	"	5.0V	"	"	"	"	"	"	"		"	OUT		"	GND	"	D3	"	"	"	"	"	"	"
		211	"	GND	"	5.0V	"	"	"	"	"		OUT			"	"	"	D2	"	"	"	"	"	"	"
		212	"	"	"	GND	"	5.0V	"	"	"	OUT				"	"	"	D1	"	"	"	"	"	"	"
		213	5.0V	"	"	"	"	GND	"	"	5.0V		"	"	OUT	IN	"	"	D4	"	"	"	"	"	"	"
		214	GND	"	5.0V	"	"	"	"	"	"		"	OUT		"	"	"	D3	"	"	"	"	"	"	"
		215	"	"	GND	"	5.0V	"	"	"	"		OUT			"	"	"	D2	"	"	"	"	"	"	"
		216	"	"	"	"	GND	"	5.0V	"	"	OUT				"	"	"	D1	"	"	"	"	"	"	"

1/ Pins not designated may be "high" level logic, "low" level logic, or open. Exceptions are as follows: For $V_{IC(POS)}$ tests, the V_{SS} terminal shall be open; for $V_{IC(NEG)}$ tests, the V_{DD} terminal shall be open; for I_{SS} tests, the outputs shall be open.
2/ I_{SS} measurements shall be performed in sequence.
3/ The V_{IL} and V_{IH} input test requirements are acceptable if the output state limit is met, attributes data required.
4/ The device manufacturer may, at his option, measure I_{IL} and I_{IH} at +25°C for each individual input or measure all inputs together.
5/ See 4.4.1c.

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TABLE III. Group A inspection for device type 53.

Symbol	MIL-STD-883 method	Cases A, C,D,T,X,Y	Terminal conditions 1/														Measured terminal	Test limits						Unit
		Symbol	1A	1B	1Y	2Y	2A	2B	V_{SS}	3A	3B	3Y	4Y	4A	4B	V_{DD}		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14		Min	Max	Min	Max	Min	Max	
$V_{IC(POS)}$		1	1mA													GND	1A		1.5					V
		2		1mA												"	1B		"					"
		3					1mA									"	2A		"					"
		4						1mA								"	2B		"					"
		5								1mA						"	3A		"					"
		6									1mA					"	3B		"					"
		7												1mA		"	4A		"					"
		8													1mA	"	4B		"					"
$V_{IC(NEG)}$		9	-1mA						GND								1A		-6					"
		10		-1mA					"								1B		"					"
		11					-1mA		"								2A		"					"
		12						-1mA	"								2B		"					"
		13							"	-1mA							3A		"					"
		14							"		-1mA						3B		"					"
		15							"					-1mA			4A		"					"
		16							"						-1mA		4B		"					"
I_{SS} 2/	3005	17	GND	GND			GND	GND	"	GND	GND			GND	GND	18.0V	V_{SS}		-25		-750			nA
		18	GND	18.0V			GND	18.0V	"	GND	18.0V			GND	18.0V	"	"		"		"			"
		19	18.0V	GND			18.0V	GND	"	18.0V	GND			18.0V	GND	"	"		"		"			"
		20	18.0V	18.0V			18.0V	18.0V	"	18.0V	18.0V			18.0V	18.0V	"	"		"		"			"
V_{OH5}	3006	21	15.0V	GND			GND	GND	"	GND	GND			GND	GND	15.0V	1Y	14.95		14.95		14.95		V
		22	GND	15.0V			GND	"	"	"	"			"	"	"	1Y	"		"		"		"
		23	"	GND			15.0V	"	"	"	"			"	"	"	2Y	"		"		"		"
		24	"	"			GND	15.0V	"	"	"			"	"	"	2Y	"		"		"		"
		25	"	"			"	GND	"	15.0V	"			"	"	"	3Y	"		"		"		"
		26	"	"			"	"	"	GND	15.0V			"	"	"	3Y	"		"		"		"
		27	"	"			"	"	"	"	GND			15.0V	"	"	4Y	"		"		"		"
		28	"	"			"	"	"	"	"			GND	15.0V	"	4Y	"		"		"		"
V_{OL6}	3007	29	15.0V	15.0V			"	"	"	"	"			"	GND	"	1Y		0.05		0.05		0.05		"
		30	GND	GND			"	"	"	"	"			"	"	"	1Y		"		"		"		"
		31	"	"			15.0V	15.0V	"	"	"			"	"	"	2Y		"		"		"		"
		32	"	"			GND	GND	"	"	"			"	"	"	2Y		"		"		"		"
		33	"	"			"	"	"	15.0V	15.0V			"	"	"	3Y		"		"		"		"
		34	"	"			"	"	"	GND	GND			"	"	"	3Y		"		"		"		"
		35	"	"			"	"	"	"	"			15.0V	15.0V	"	4Y		"		"		"		"
		36	"	"			"	"	"	"	"			GND	GND	"	4Y		"		"		"		"
V_{IH1} 3/		37	3.5V	1.5V			"	"	"	"	"			"	"	5.0V	1Y	4.5		4.5		4.5		"
		38	1.5V	3.5V			"	"	"	"	"			"	"	"	1Y	"		"		"		"
		39	GND	GND			3.5V	1.5V	"	"	"			"	"	"	2Y	"		"		"		"
		40	"	"			1.5V	3.5V	"	"	"			"	"	"	2Y	"		"		"		"
		41	"	"			GND	GND	"	3.5V	1.5V			"	"	"	3Y	"		"		"		"
		42	"	"			"	"	"	1.5V	3.5V			"	"	"	3Y	"		"		"		"
		43	"	"			"	"	"	GND	GND			3.5V	1.5V	"	4Y	"		"		"		"
		44	"	"			"	"	"	"	"			1.5V	3.5V	"	4Y	"		"		"		"
V_{IH2} 3/		45	7.0V	3.0V			"	"	"	"	"			GND	GND	10.0V	1Y	9.0		9.0		9.0		"
		46	3.0V	7.0V			"	"	"	"	"			"	"	"	1Y	"		"		"		"
		47	GND	GND			7.0V	3.0V	"	"	"			"	"	"	2Y	"		"		"		"
		48	"	"			3.0V	7.0V	"	"	"			"	"	"	2Y	"		"		"		"
		49	"	"			GND	GND	"	7.0V	3.0V			"	"	"	3Y	"		"		"		"
		50	"	"			"	"	"	3.0V	7.0V			"	"	"	3Y	"		"		"		"
		51	"	"			"	"	"	GND	GND			7.0V	3.0V	"	4Y	"		"		"		"
		52	"	"			"	"	"	GND	GND			3.0V	7.0V	"	4Y	"		"		"		"

See footnotes at the end of device type 53.

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TABLE III. Group A inspection for device type 53 – Continued.

Symbol	MIL-STD-883 method	Cases A,C,D,T,X,Y	Terminal conditions 1/														Measured terminal	Test limits						Unit
		Symbol	1A	1B	1Y	2Y	2A	2B	V_{SS}	3A	3B	3Y	4Y	4A	4B	V_{DD}		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14		Min	Max	Min	Max	Min	Max	
V_{IH3} 3/		53	11.0V	4.0V			GND	GND	GND	GND	GND			GND	GND	15.0V	1Y	13.5		13.5		13.5		V
		54	4.0V	11.0V			GND	GND	"	"	"			"	"	"	1Y	"		"		"		"
		55	GND	GND			11.0V	4.0V	"	"	"			"	"	"	2Y	"		"		"		"
		56	"	"			4.0V	11.0V	"	"	"			"	"	"	2Y	"		"		"		"
		57	"	"			GND	GND	"	11.0V	4.0V			"	"	"	3Y	"		"		"		"
		58	"	"			"	"	"	4.0V	11.0V			"	"	"	3Y	"		"		"		"
		59	"	"			"	"	"	GND	GND			11.0V	4.0V	"	4Y	"		"		"		"
		60	"	"			"	"	"	"	"			4.0V	11.0V	"	4Y	"		"		"		"
V_{IL1} 3/		61	1.5V	"			"	"	"	"	"			GND	GND	5.0V	1Y		0.5		0.5		0.5	"
		62	GND	"			1.5V	"	"	"	"			"	"	"	2Y		"		"		"	"
		63	"	"			GND	"	"	1.5V	"			"	"	"	3Y		"		"		"	"
		64	"	"			"	"	"	GND	"			1.5V	"	"	4Y		"		"		"	"
V_{IL2} 3/		65	3.0V	"			"	"	"	"	"			GND	"	10.0V	1Y		1.0		1.0		1.0	"
		66	GND	"			3.0V	"	"	"	"			"	"	"	2Y		"		"		"	"
		67	"	"			GND	"	"	3.0V	"			"	"	"	3Y		"		"		"	"
		68	"	"			"	"	"	GND	"			3.0V	"	"	4Y		"		"		"	"
V_{IL3} 3/		69	4.0V	"			"	"	"	"	"			GND	"	15.0V	1Y		1.5		1.5		1.5	"
		70	GND	"			4.0V	"	"	"	"			"	"	"	2Y		"		"		"	"
		71	"	"			GND	"	"	4.0V	"			"	"	"	3Y		"		"		"	"
		72	"	"			"	"	"	GND	"			4.0V	"	"	4Y		"		"		"	"
I_{OL1}		73	"	"	0.4V		"	"	"	"	"			GND	"	5.0V	1Y	0.51		0.36		0.64		mA
		74	"	"		0.4V	"	"	"	"	"			"	"	"	2Y	"		"		"		"
		75	"	"			"	"	"	"	"	0.4V		"	"	"	3Y	"		"		"		"
		76	"	"			"	"	"	"	"		0.4V	"	"	"	4Y	"		"		"		"
I_{OL2}		77	"	"	1.5V		"	"	"	"	"			GND	"	15.0V	1Y	3.4		2.4		4.2		"
		78	"	"		1.5V	"	"	"	"	"			"	"	"	2Y	"		"		"		"
		79	"	"			"	"	"	"	"	1.5V		"	"	"	3Y	"		"		"		"
		80	"	"			"	"	"	"	"		1.5V	"	"	"	4Y	"		"		"		"
I_{OH1}		81	5.0V	"	4.6V		5.0V	"	"	5.0V	"			5.0V	"	5.0V	1Y	-0.51		-0.36		-0.64		"
		82	"	"		4.6V	"	"	"	"	"			"	"	"	2Y	"		"		"		"
		83	"	"			"	"	"	"	"	4.6V		"	"	"	3Y	"		"		"		"
		84	"	"			"	"	"	"	"		4.6V	"	"	"	4Y	"		"		"		"
I_{OH2}		85	15.0V	"	13.5V		15.0V	"	"	15.0V	"			15.0V	"	15.0V	1Y	-3.4		-2.4		-4.2		"
		86	"	"		13.5V	"	"	"	"	"			"	"	"	2Y	"		"		"		"
		87	"	"			"	"	"	"	"	13.5V		"	"	"	3Y	"		"		"		"
		88	"	"			"	"	"	"	"		13.5V	"	"	"	4Y	"		"		"		"
I_{IH1} 4/	3010	89	18.0V	18.0V			18.0V	18.0V	"	18.0V	18.0V			18.0V	18.0V	18.0V	All inputs together		800.0					nA
I_{IH2}		90	18.0V	GND			GND	GND	"	GND	GND			GND	GND	"	1A		100.0		100.0			"
		91	GND	18.0V			GND	"	"	"	"			"	"	"	1B		"		"			"
		92	"	GND			18.0V	"	"	"	"			"	"	"	2A		"		"			"
		93	"	"			GND	18.0V	"	"	"			"	"	"	2B		"		"			"
		94	"	"			"	GND	"	18.0V	"			"	"	"	3A		"		"			"
		95	"	"			"	"	"	GND	18.0V			"	"	"	3B		"		"			"
		96	"	"			"	"	"	"	GND			18.0V	"	"	4A		"		"			"
		97	"	"			"	"	"	"	"			GND	18.0V	"	4B		"		"			"
I_{IL1} 4/	3009	98	"	"			"	"	"	"	"			GND	GND	"	All inputs together		-800.0					"

See footnotes at end of device type 53.

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TABLE III. Group A inspection for device type 53 – Continued.

Symbol	MIL-STD-883 method	Cases A, C,D,T,X,Y	Terminal conditions 1/														Measured terminal	Test limits						Unit
		Symbol	1A	1B	1Y	2Y	2A	2B	V_{SS}	3A	3B	3Y	4Y	4A	4B	V_{DD}		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14		Min	Max	Min	Max	Min	Max	
I_{IL2}	3009	99	GND	18.0V			18.0V	18.0V	GND	18.0V	18.0V			18.0V	18.0V	18.0V	1A		-100.0		-100.0			nA
		100	18.0V	GND			18.0V	"	"	"	"			"	"	"	1B		"		"			"
		101	"	18.0V			GND	"	"	"	"			"	"	"	2A		"		"			"
		102	"	"			18.0V	GND	"	"	"			"	"	"	2B		"		"			"
		103	"	"			"	18.0V	"	GND	"			"	"	"	3A		"		"			"
		104	"	"			"	"	"	18.0V	GND			"	"	"	3B		"		"			"
		105	"	"			"	"	"	"	18.0V			GND	"	"	4A		"		"			"
		106	"	"			"	"	"	"	18.0V			18.0V	GND	"	4B		"		"			"
																		Subgroup 4 T_C = +25°C						
																		Min	Max					
C_I	3012	107	5/						GND							GND	1A		12					pF
		108		5/					"							"	1B		"					"
		109					5/		"							"	2A		"					"
		110						5/	"							"	2B		"					"
		111							"	5/						"	3A		"					"
		112							"		5/					"	3B		"					"
		113							"					5/		"	4A		"					"
		114							"						5/	"	4B		"					"
																		Subgroup 9 T_C = +25°C		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C		
																		Min	Max	Min	Max	Min	Max	
t_{PHL}	3003 (Fig. 6)	115	IN	GND	OUT		GND	GND	GND	GND	GND			GND	GND	5.0V	1A to 1Y	10	250	14	350	10	250	ns
		116	GND	IN	OUT		GND	"	"	"	"			"	"	"	1B to 1Y	"	"	"	"	"	"	"
		117	"	GND		OUT	IN	"	"	"	"			"	"	"	2A to 2Y	"	"	"	"	"	"	"
		118	"	"		OUT	GND	"	"	"	"			"	"	"	2B to 2Y	"	"	"	"	"	"	"
		119	"	"			"	IN	"	"	"	OUT		"	"	"	3A to 3Y	"	"	"	"	"	"	"
		120	"	"			"	GND	"	IN	"	OUT		"	"	"	3B to 3Y	"	"	"	"	"	"	"
		121	"	"			"	"	"	GND	IN		OUT	IN	"	"	4A to 4Y	"	"	"	"	"	"	"
		122	"	"			"	"	"	"	GND		OUT	GND	IN	"	4B to 4Y	"	"	"	"	"	"	"
t_{PLH}		123	IN	"	OUT		"	"	"	"	"			"	GND	"	1A to 1Y	"	"	"	"	"	"	"
		124	GND	IN	OUT		"	"	"	"	"			"	"	"	1B to 1Y	"	"	"	"	"	"	"
		125	"	GND		OUT	IN	"	"	"	"			"	"	"	2A to 2Y	"	"	"	"	"	"	"
		126	"	"		OUT	GND	"	"	"	"			"	"	"	2B to 2Y	"	"	"	"	"	"	"
		127	"	"			"	IN	"	"	"	OUT		"	"	"	3A to 3Y	"	"	"	"	"	"	"
		128	"	"			"	GND	"	IN	"	OUT		"	"	"	3B to 3Y	"	"	"	"	"	"	"
		129	"	"			"	"	"	GND	IN		OUT	IN	"	"	4A to 4Y	"	"	"	"	"	"	"
		130	"	"			"	"	"	"	GND		OUT	GND	IN	"	4B to 4Y	"	"	"	"	"	"	"
t_{TLH}	3004 (Fig. 6)	131	IN	"	OUT		"	"	"	"	"			"	GND	"	1Y	10	200	14	280	10	200	"
		132	GND	IN	OUT		"	"	"	"	"			"	"	"	1Y	"	"	"	"	"	"	"
		133	"	GND		OUT	IN	"	"	"	"			"	"	"	2Y	"	"	"	"	"	"	"
		134	"	"		OUT	GND	"	"	"	"			"	"	"	2Y	"	"	"	"	"	"	"
		135	"	"			"	IN	"	"	"	OUT		"	"	"	3Y	"	"	"	"	"	"	"
		136	"	"			"	GND	"	IN	"	OUT		"	"	"	3Y	"	"	"	"	"	"	"
		137	"	"			"	"	"	GND	IN		OUT	IN	"	"	4Y	"	"	"	"	"	"	"
		138	"	"			"	"	"	"	GND		OUT	GND	IN	"	4Y	"	"	"	"	"	"	"
t_{TLH}		139	IN	"	OUT		"	"	"	"	"			"	GND	"	1Y	"	"	"	"	"	"	"
		140	GND	IN	OUT		"	"	"	"	"			"	"	"	1Y	"	"	"	"	"	"	"
		141	"	GND		OUT	IN	"	"	"	"			"	"	"	2Y	"	"	"	"	"	"	"
		142	"	"		OUT	GND	"	"	"	"			"	"	"	2Y	"	"	"	"	"	"	"
		143	"	"			"	IN	"	"	"	OUT		"	"	"	3Y	"	"	"	"	"	"	"
		144	"	"			"	GND	"	IN	"	OUT		"	"	"	3Y	"	"	"	"	"	"	"
		145	"	"			"	"	"	GND	IN		OUT	IN	"	"	4Y	"	"	"	"	"	"	"
		146	"	"			"	"	"	"	GND		OUT	GND	IN	"	4Y	"	"	"	"	"	"	"

1/ Pins not designated may be "high" level logic, "low" level logic, or open. Exceptions are as follows: For $V_{IC(POS)}$ tests, the V_{SS} terminal shall be open; for $V_{IC(NEG)}$ tests, the V_{DD} terminal shall be open; for I_{SS} tests, the outputs shall be open.
2/ I_{SS} measurements shall be performed in sequence.
3/ The V_{IL} and V_{IH} input test requirements are acceptable if the output state limit is met, attributes data required.
4/ The device manufacturer may, at his option, measure I_{IL} and I_{IH} at +25°C for each individual input or measure all inputs together.
5/ See 4.4.1c.

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TABLE III. Group A inspection for device type 54.

Symbol	MIL-STD-883 method	Cases E,F,Z	Terminal conditions 1/															Measured terminal	Test limits						Unit
		Symbol	J	Kd	H	G	F	E	Kb	V_{SS}	Kc	Ka	D	C	B	A	EX	V_{DD}		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Min	Max	Min	Max	Min	Max	
$V_{IC(POS)}$		1		1mA														GND	Kd		1.5					V
		2			1mA													"	H		"					"
		3				1mA												"	G		"					"
		4					1mA											"	F		"					"
		5						1mA										"	E		"					"
		6							1mA									"	Kb		"					"
		7									1mA							"	Kc		"					"
		8										1mA						"	Ka		"					"
		9											1mA					"	D		"					"
		10												1mA				"	C		"					"
		11													1mA			"	B		"					"
		12														1mA		"	A		"					"
		13															1mA	"	EX		"					"
$V_{IC(NEG)}$		14		-1mA						GND									Kd		-6.0					"
		15			-1mA					"									H		"					"
		16				-1mA				"									G		"					"
		17					-1mA			"									F		"					"
		18						-1mA		"									E		"					"
		19							-1mA	"									Kb		"					"
		20								"	-1mA								Kc		"					"
		21								"		-1mA							Ka		"					"
		22								"			-1mA						D		"					"
		23								"				-1mA					C		"					"
		24								"					-1mA				B		"					"
		25								"						-1mA			A		"					"
		26								"							-1mA		EX		"					"
I_{SS} 2/	3005	27		GND	GND	GND	GND	GND	GND	"	GND	GND	GND	GND	GND	GND	GND	18.0V	V_{SS}		-1.9		-3			µA
		28		18.0V	GND	GND	GND	GND	18.0V	"	18.0V	18.0V	GND	GND	GND	GND	GND	"	"		"		"			"
		29		GND	18.0V	18.0V	18.0V	18.0V	GND	"	GND	GND	18.0V	18.0V	18.0V	18.0V	18.0V	"	"		"		"			"
		30		18.0V	18.0V	18.0V	18.0V	18.0V	18.0V	"	18.0V	18.0V	18.0V	18.0V	18.0V	18.0V	18.0V	"	"		"		"			"
V_{OH5}	3006	31		15.0V	GND	GND	GND	GND	GND	"	GND	GND	GND	GND	GND	GND	GND	15.0V	J	14.95		14.95		14.95		V
		32		"	GND	GND	GND	GND	"	"	15.0V	15.0V	GND	GND	GND	GND	"	"	"	14.95		14.95		14.95		"
V_{OL6}	3007	33		"	15.0V	15.0V	15.0V	15.0V	"	"	GND	GND	15.0V	15.0V	15.0V	15.0V	"	"	"		0.05		0.05		0.05	"
		34		"	15.0V	15.0V	15.0V	15.0V	"	"	15.0V	15.0V	15.0V	15.0V	15.0V	15.0V	"	"	"		0.05		0.05		0.05	"
V_{IH1}		35		3.5V	3.5V	1.5V	1.5V	1.5V	1.5V	"	1.5V	1.5V	1.5V	1.5V	1.5V	1.5V	"	5.0V	"	0.5		0.5		0.5		"
		36		"	1.5V	3.5V	1.5V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
		37		"	"	1.5V	3.5V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
		38		"	"	"	1.5V	3.5V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
		39		"	"	"	"	1.5V	"	"	"	"	3.5V	"	"	"	"	"	"	"	"	"	"	"	"	"	"
		40		"	"	"	"	"	"	"	"	"	1.5V	3.5V	"	"	"	"	"	"	"	"	"	"	"	"	"
		41		"	"	"	"	"	"	"	"	"	"	1.5V	3.5V	"	"	"	"	"	"	"	"	"	"	"	"
		42		"	"	"	"	"	"	"	"	"	"	"	1.5V	1.5V	3.5V	"	"	"	"	"	"	"	"	"	"
		43		"	3.5V	3.5V	3.5V	3.5V	"	"	3.5V	3.5V	3.5V	3.5V	3.5V	3.5V	"	"	"	"		"		"		"
V_{IH2}		44		7.0V	7.0V	3.0V	3.0V	3.0V	3.0V	"	3.0V	3.0V	3.0V	3.0V	3.0V	3.0V	"	10.0V	"	1.0		1.0		1.0		"
		45		"	3.0V	7.0V	3.0V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
		46		"	"	3.0V	7.0V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
		47		"	"	"	3.0V	7.0V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
		48		"	"	"	"	3.0V	"	"	"	"	7.0V	"	"	"	"	"	"	"	"	"	"	"	"	"	"
		49		"	"	"	"	"	"	"	"	"	3.0V	7.0V	"	"	"	"	"	"	"	"	"	"	"	"	"
		50		"	"	"	"	"	"	"	"	"	"	3.0V	7.0V	"	"	"	"	"	"	"	"	"	"	"	"
		51		"	"	"	"	"	"	"	"	"	"	3.0V	3.0V	7.0V	"	"	"	"	"	"	"	"	"	"	"
		52		"	7.0V	7.0V	7.0V	7.0V	"	"	7.0V	7.0V	7.0V	7.0V	7.0V	7.0V	"	"	"	"		"		"		"

See footnotes at end of device type 54.

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TABLE III. Group A inspection for device type 54 – Continued.

Symbol	MIL-STD-883 method	Cases E,F,Z	Terminal conditions 1/																Measured terminal	Test limits						Unit
		Symbol	J	Kd	H	G	F	E	Kb	V_{SS}	Kc	Ka	D	C	B	A	EX	V_{DD}		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Min	Max	Min	Max	Min	Max	
V_{IH3}		53		11.0V	11.0V	4.0V	4.0V	4.0V	4.0V	GND	4.0V	4.0V	4.0V	4.0V	4.0V	4.0V	GND	15.0V	J	1.5		1.5		1.5		V
		54		"	4.0V	11.0V	4.0V	"	"	"	"	"	"	"	"	"	"	"	"	"		"		"		"	
		55		"	"	4.0V	11.0V	"	"	"	"	"	"	"	"	"	"	"	"	"		"		"		"	
		56		"	"	"	4.0V	11.0V	"	"	"	"	"	"	"	"	"	"	"	"		"		"		"	
		57		"	"	"	"	4.0V	"	"	"	"	11.0V	"	"	"	"	"	"	"		"		"		"	
		58		"	"	"	"	"	"	"	"	"	4.0V	11.0V	"	"	"	"	"	"		"		"		"	
		59		"	"	"	"	"	"	"	"	"	"	4.0V	11.0V	"	"	"	"	"		"		"		"	
		60		"	"	"	"	"	"	"	"	"	"	4.0V	4.0V	11.0V	"	"	"	"		"		"		"	
		61		"	11.0V	11.0V	11.0V	11.0V	"	"	11.0V	11.0V	11.0V	11.0V	11.0V	11.0V	"	"	"	"		"		"		"	
V_{IL1}		62		3.5V	1.5V	1.5V	1.5V	1.5V	1.5V	"	1.5V	1.5V	1.5V	1.5V	1.5V	1.5V	"	5.0V	"		4.5		4.5		4.5	"
		63		"	1.5V	3.5V	3.5V	3.5V	"	"	3.5V	3.5V	3.5V	3.5V	3.5V	3.5V	"	"	"		"		"		"	"
		64		"	3.5V	1.5V	3.5V	"	"	"	"	"	"	"	"	"	"	"	"		"		"		"	"
		65		"	"	3.5V	1.5V	"	"	"	"	"	"	"	"	"	"	"	"		"		"		"	"
		66		"	"	"	3.5V	1.5V	"	"	"	"	"	"	"	"	"	"	"		"		"		"	"
		67		"	"	"	"	3.5V	"	"	"	"	1.5V	"	"	"	"	"	"		"		"		"	"
		68		"	"	"	"	"	"	"	"	"	3.5V	1.5V	"	"	"	"	"		"		"		"	"
		69		"	"	"	"	"	"	"	"	"	"	3.5V	1.5V	"	"	"	"		"		"		"	"
		70		"	"	"	"	"	"	"	"	"	"	3.5V	3.5V	1.5V	"	"	"		"		"		"	"
V_{IL2}		71		7.0V	3.0V	3.0V	3.0V	3.0V	3.0V	"	3.0V	3.0V	3.0V	3.0V	3.0V	3.0V	"	10.0V	"		9.0		9.0		9.0	"
		72		"	3.0V	7.0V	7.0V	7.0V	"	"	7.0V	7.0V	7.0V	7.0V	7.0V	7.0V	"	"	"		"		"		"	"
		73		"	7.0V	3.0V	7.0V	"	"	"	"	"	"	"	"	"	"	"	"		"		"		"	"
		74		"	"	7.0V	3.0V	"	"	"	"	"	"	"	"	"	"	"	"		"		"		"	"
		75		"	"	"	7.0V	3.0V	"	"	"	"	"	"	"	"	"	"	"		"		"		"	"
		76		"	"	"	"	7.0V	"	"	"	"	3.0V	"	"	"	"	"	"		"		"		"	"
		77		"	"	"	"	"	"	"	"	"	7.0V	3.0V	"	"	"	"	"		"		"		"	"
		78		"	"	"	"	"	"	"	"	"	"	7.0V	3.0V	"	"	"	"		"		"		"	"
		79		"	"	"	"	"	"	"	"	"	"	7.0V	7.0V	3.0V	"	"	"		"		"		"	"
V_{IL3}		80		11.0V	4.0V	4.0V	4.0V	4.0V	4.0V	"	4.0V	4.0V	4.0V	4.0V	4.0V	4.0V	"	15.0V	"		13.5		13.5		13.5	"
		81		"	4.0V	11.0V	11.0V	11.0V	"	"	11.0V	11.0V	11.0V	11.0V	11.0V	11.0V	"	"	"		"		"		"	"
		82		"	11.0V	4.0V	11.0V	"	"	"	"	"	"	"	"	"	"	"	"		"		"		"	"
		83		"	"	11.0V	4.0V	"	"	"	"	"	"	"	"	"	"	"	"		"		"		"	"
		84		"	"	"	11.0V	4.0V	"	"	"	"	"	"	"	"	"	"	"		"		"		"	"
		85		"	"	"	"	11.0V	"	"	"	"	4.0V	"	"	"	"	"	"		"		"		"	"
		86		"	"	"	"	"	"	"	"	"	11.0V	4.0V	"	"	"	"	"		"		"		"	"
		87		"	"	"	"	"	"	"	"	"	"	11.0V	4.0V	"	"	"	"		"		"		"	"
		88		"	"	"	"	"	"	"	"	"	"	11.0V	11.0V	4.0V	"	"	"		"		"		"	"
I_{OL1}		89	0.4V	5.0V	5.0V	GND	GND	GND	GND	"	GND	GND	GND	GND	GND	GND	"	5.0V	"	0.51		0.36		0.64		mA
		90	"	"	GND	5.0V	GND	"	"	"	"	"	"	"	"	"	"	"	"	"		"		"		"	
		91	"	"	"	GND	5.0V	"	"	"	"	"	"	"	"	"	"	"	"	"		"		"		"	
		92	"	"	"	"	GND	5.0V	"	"	"	"	"	"	"	"	"	"	"	"		"		"		"	
		93	"	"	"	"	"	GND	"	"	"	"	5.0V	"	"	"	"	"	"	"		"		"		"	
		94	"	"	"	"	"	"	"	"	"	"	GND	5.0V	"	"	"	"	"	"		"		"		"	
		95	"	"	"	"	"	"	"	"	"	"	"	GND	5.0V	"	"	"	"	"		"		"		"	
		96	"	"	"	"	"	"	"	"	"	"	"	GND	GND	5.0V	"	"	"	"		"		"		"	
		97	"	"	5.0V	5.0V	5.0V	5.0V	"	"	5.0V	5.0V	5.0V	5.0V	5.0V	5.0V	"	"	"	"		"		"		"	
I_{OL2}		98	1.5V	15.0V	15.0V	GND	GND	GND	"	"	GND	GND	GND	GND	GND	GND	"	15.0V	"	3.4		2.4		4.2		"
		99	"	"	GND	15.0V	GND	"	"	"	"	"	"	"	"	"	"	"	"	"		"		"		"	
		100	"	"	"	GND	15.0V	"	"	"	"	"	"	"	"	"	"	"	"	"		"		"		"	
		101	"	"	"	"	GND	15.0V	"	"	"	"	"	"	"	"	"	"	"	"		"		"		"	
		102	"	"	"	"	"	GND	"	"	"	"	15.0V	"	"	"	"	"	"	"		"		"		"	
		103	"	"	"	"	"	"	"	"	"	"	GND	15.0V	"	"	"	"	"	"		"		"		"	
		104	"	"	"	"	"	"	"	"	"	"	"	GND	15.0V	"	"	"	"	"		"		"		"	
		105	"	"	"	"	"	"	"	"	"	"	"	GND	GND	15.0V	"	"	"	"		"		"		"	
		106	"	"	15.0V	15.0V	15.0V	15.0V	"	"	15.0V	15.0V	15.0V	15.0V	15.0V	15.0V	"	"	"	"		"		"		"	

See footnotes at end of device type 54.

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TABLE III. Group A inspection for device type 54 – Continued.

Symbol	MIL-STD-883 method	Cases E,F,Z Symbol	Terminal conditions 1/ J	Kd	H	G	F	E	Kb	V_{SS}	Kc	Ka	D	C	B	A	EX	V_{DD}	Measured terminal	Test limits Subgroup 1 T_C = +25°C Min	Max	Subgroup 2 T_C = +125°C Min	Max	Subgroup 3 T_C = -55°C Min	Max	Unit
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
I_{OH1}		107	4.6V	5.0V	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	5.0V	J	-0.51		-0.36		-0.64		mA
		108	"	"	GND	5.0V	5.0V	5.0V	"	"	5.0V	5.0V	5.0V	5.0V	5.0V	5.0V	"	"	"	"		"		"		"
		109	"	"	5.0V	GND	5.0V	"	"	"	"	"	"	"	"	"	"	"	"	"		"		"		"
		110	"	"	"	5.0V	GND	"	"	"	"	"	"	"	"	"	"	"	"	"		"		"		"
		111	"	"	"	"	5.0V	GND	"	"	"	"	"	"	"	"	"	"	"	"		"		"		"
		112	"	"	"	"	"	5.0V	"	"	"	"	GND	"	"	"	"	"	"	"		"		"		"
		113	"	"	"	"	"	"	"	"	"	"	5.0V	GND	"	"	"	"	"	"		"		"		"
		114	"	"	"	"	"	"	"	"	"	"	"	5.0V	GND	"	"	"	"	"		"		"		"
		115	"	"	"	"	"	"	"	"	"	"	"	5.0V	5.0V	GND	"	"	"	"		"		"		"
I_{OH2}		116	13.5V	15.0V	GND	GND	GND	GND	"	"	GND	GND	GND	GND	GND	GND	"	15.0V	"	-3.4		-2.4		-4.2		"
		117	"	"	GND	15.0V	15.0V	15.0V	"	"	15.0V	15.0V	15.0V	15.0V	15.0V	15.0V	"	"	"	"		"		"		"
		118	"	"	15.0V	GND	15.0V	"	"	"	"	"	"	"	"	"	"	"	"	"		"		"		"
		119	"	"	"	15.0V	GND	"	"	"	"	"	"	"	"	"	"	"	"	"		"		"		"
		120	"	"	"	"	15.0V	GND	"	"	"	"	"	"	"	"	"	"	"	"		"		"		"
		121	"	"	"	"	"	15.0V	"	"	"	"	GND	"	"	"	"	"	"	"		"		"		"
		122	"	"	"	"	"	"	"	"	"	"	15.0V	GND	"	"	"	"	"	"		"		"		"
		123	"	"	"	"	"	"	"	"	"	"	"	15.0V	GND	"	"	"	"	"		"		"		"
		124	"	"	"	"	"	"	"	"	"	"	"	15.0V	15.0V	GND	"	"	"	"		"		"		"
I_{IH1} 3/	3010	125		18.0V	18.0V	18.0V	18.0V	18.0V	18.0V	"	18.0V	18.0V	18.0V	18.0V	18.0V	18.0V	18.0V	18.0V	All inputs together		13					nA
I_{IH2}		126		18.0V	GND	GND	GND	GND	GND	"	GND	GND	GND	GND	GND	GND	GND	"	Kd		1		45			"
		127		GND	18.0V	GND	"	"	"	"	"	"	"	"	"	"	"	"	H		"		"			"
		128		"	GND	18.0V	"	"	"	"	"	"	"	"	"	"	"	"	G		"		"			"
		129		"	"	GND	18.0V	"	"	"	"	"	"	"	"	"	"	"	F		"		"			"
		130		"	"	"	GND	18.0V	"	"	"	"	"	"	"	"	"	"	E		"		"			"
		131		"	"	"	"	GND	18.0V	"	"	"	"	"	"	"	"	"	Kb		"		"			"
		132		"	"	"	"	"	GND	"	18.0V	"	"	"	"	"	"	"	Kc		"		"			"
		133		"	"	"	"	"	"	"	GND	18.0V	"	"	"	"	"	"	Ka		"		"			"
		134		"	"	"	"	"	"	"	"	GND	18.0V	"	"	"	"	"	D		"		"			"
		135		"	"	"	"	"	"	"	"	"	GND	18.0V	"	"	"	"	C		"		"			"
		136		"	"	"	"	"	"	"	"	"	"	GND	18.0V	"	"	"	B		"		"			"
		137		"	"	"	"	"	"	"	"	"	"	"	GND	18.0V	"	"	A		"		"			"
		138		"	"	"	"	"	"	"	"	"	"	"	"	GND	18.0V	"	EX		"		"			"
I_{IL1} 3/	3009	139		"	"	"	"	"	"	"	"	"	"	"	"	"	"	GND	"	All inputs together		-13					"
I_{IL2}		140		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	Kd		-1.0		-45			"
		141		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	H		"		"			"
		142		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	G		"		"			"
		143		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	F		"		"			"
		144		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	E		"		"			"
		145		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	Kb		"		"			"
		146		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	Kc		"		"			"
		147		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	Ka		"		"			"
		148		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	D		"		"			"
		149		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	C		"		"			"
		150		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B		"		"			"
		151		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A		"		"			"
		152		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	EX		"		"			"
I_{OC1}		153	18.0V	"	18.0V	18.0V	18.0V	18.0V	18.0V	"	18.0V	18.0V	18.0V	18.0V	18.0V	18.0V	"	"	J		25		90		25	"
I_{OC2}		154	GND	"	18.0V	18.0V	18.0V	18.0V	18.0V	"	18.0V	18.0V	18.0V	18.0V	18.0V	18.0V	"	"	J		-25		-90		-25	"

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TABLE III. Group A inspection for device type 54 – Continued.

Symbol	MIL-STD-883 method	Cases E,F,Z Symbol	Terminal conditions 1/ J	Kd	H	G	F	E	Kb	V_{SS}	Kc	Ka	D	C	B	A	EX	V_{DD}	Measured terminal	Test limits Subgroup 4 T_C = +25°C Min	Max	Unit
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16				
C_I	3012	155		4/						GND								GND	Kd		12	pF
		156			4/					"								"	H		"	"
		157				4/				"								"	G		"	"
		158					4/			"								"	F		"	"
		159						4/		"								"	E		"	"
		160							4/	"								"	Kb		"	"
		161								"	4/							"	Kc		"	"
		162								"		4/						"	Ka		"	"
		163								"			4/					"	D		"	"
		164								"				4/				"	C		"	"
		165								"					4/			"	B		"	"
		166								"						4/		"	A		"	"
		167								"							4/	"	EX		"	"
																				Subgroup 9 T_C = +25°C Min	Max	Subgroup 10 T_C = +125°C Min	Max	Subgroup 11 T_C = -55°C Min	Max	
t_{PHL1}	3003 (Fig. 5)	168	OUT	5.0V	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND GND	GND IN	IN GND	GND	5.0V	A to J	34	685	41	820	30	600	ns
		169	"	"	"	"	"	"	"	"	"	"	"	IN GND	IN GND	"	"	"	B to J	"	"	"	"	"	"	"
		170	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	C to J	"	"	"	"	"	"	"
		171	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	D to J	"	"	"	"	"	"	"
		172	"	"	"	"	"	IN GND	"	"	"	"	"	"	"	"	"	"	E to J	"	"	"	"	"	"	"
		173	"	"	"	"	IN GND	"	"	"	"	"	"	"	"	"	"	"	F to J	"	"	"	"	"	"	"
		174	"	"	"	IN GND	"	"	"	"	"	"	"	"	"	"	"	"	G to J	"	"	"	"	"	"	"
		175	"	"	IN	"	"	"	"	"	"	"	"	"	"	"	"	"	H to J	"	"	"	"	"	"	"
t_{PLH1}		176	"	"	GND	"	"	"	"	"	"	"	"	"	"	"	IN GND	"	A to J	27	535	32	640	24	475	"
		177	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B to J	"	"	"	"	"	"	"
		178	"	"	"	"	"	"	"	"	"	"	"	"	"	IN GND	"	"	C to J	"	"	"	"	"	"	"
		179	"	"	"	"	"	"	"	"	"	"	"	"	IN GND	"	"	"	D to J	"	"	"	"	"	"	"
		180	"	"	"	"	"	IN GND	"	"	"	"	"	"	"	"	"	"	E to J	"	"	"	"	"	"	"
		181	"	"	"	"	IN GND	"	"	"	"	"	"	"	"	"	"	"	F to J	"	"	"	"	"	"	"
		182	"	"	"	IN GND	"	"	"	"	"	"	"	"	"	"	"	"	G to J	"	"	"	"	"	"	"
		183	"	"	IN	"	"	"	"	"	"	"	"	"	"	"	"	"	H to J	"	"	"	"	"	"	"
t_{PHL2}	3003 (Fig. 6)	184	"	"	GND	"	"	"	"	"	5.0V	"	"	"	"	IN GND	"	"	A to J	32	635	38	755	28	555	"
		185	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B to J	"	"	"	"	"	"	"
		186	"	"	"	"	"	"	"	"	"	"	"	"	"	IN GND	"	"	C to J	"	"	"	"	"	"	"
		187	"	"	"	"	"	"	"	"	"	"	"	"	IN GND	"	"	"	D to J	"	"	"	"	"	"	"
		188	"	"	"	"	"	IN GND	"	"	"	"	"	"	"	"	"	"	E to J	"	"	"	"	"	"	"
		189	"	"	"	"	IN GND	"	"	"	"	"	"	"	"	"	"	"	F to J	"	"	"	"	"	"	"
		190	"	"	"	IN GND	"	"	"	"	"	"	"	"	"	"	"	"	G to J	"	"	"	"	"	"	"
		191	"	"	IN	"	"	"	"	"	"	"	"	"	"	"	"	"	H to J	"	"	"	"	"	"	"
t_{PLH2}		192	"	"	GND	"	"	"	"	"	5.0V	"	"	"	"	IN GND	"	"	A to J	31	620	"	750	"	550	"
		193	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B to J	"	"	"	"	"	"	"
		194	"	"	"	"	"	"	"	"	"	"	"	"	"	IN GND	"	"	C to J	"	"	"	"	"	"	"
		195	"	"	"	"	"	"	"	"	"	"	"	"	IN GND	"	"	"	D to J	"	"	"	"	"	"	"
		196	"	"	"	"	"	IN GND	"	"	"	"	"	"	"	"	"	"	E to J	"	"	"	"	"	"	"
		197	"	"	"	"	IN GND	"	"	"	"	"	"	"	"	"	"	"	F to J	"	"	"	"	"	"	"
		198	"	"	"	IN GND	"	"	"	"	"	"	"	"	"	"	"	"	G to J	"	"	"	"	"	"	"
		199	"	"	IN	"	"	"	"	"	"	"	"	"	"	"	"	"	H to J	"	"	"	"	"	"	"
t_{PHL3}	3003 (Fig. 7)	200	"	"	5.0V	5.0V	5.0V	5.0V	"	"	GND	5.0V	5.0V	5.0V	5.0V	IN 5.0V	IN 5.0V	"	A to J	38	760	45	905	33	665	"
		201	"	"	"	"	"	"	"	"	"	"	"	"	IN 5.0V	"	"	"	B to J	"	"	"	"	"	"	"
		202	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	C to J	"	"	"	"	"	"	"
		203	"	"	"	"	"	"	"	"	"	"	"	"	IN 5.0V	"	"	"	D to J	"	"	"	"	"	"	"
		204	"	"	"	"	"	IN 5.0V	"	"	"	"	"	"	"	"	"	"	E to J	"	"	"	"	"	"	"
		205	"	"	"	"	IN 5.0V	"	"	"	"	"	"	"	"	"	"	"	F to J	"	"	"	"	"	"	"
		206	"	"	"	IN 5.0V	"	"	"	"	"	"	"	"	"	"	"	"	G to J	"	"	"	"	"	"	"
		207	"	"	IN	"	"	"	"	"	"	"	"	"	"	"	"	"	H to J	"	"	"	"	"	"	"

See footnotes at end of device type 54.

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TABLE III. Group A inspection for device type 54 – Continued.

Symbol	MIL-STD-883 method	Cases E,F,Z	Terminal conditions 1/																Measured terminal	Test limits						Unit
		Symbol	J	Kd	H	G	F	E	Kb	V_{SS}	Kc	Ka	D	C	B	A	EX	V_{DD}		Subgroup 9 T_C = +25°C		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C		
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Min	Max	Min	Max	Min	Max	
t_{PLH3}	3003 (Fig. 7)	208	OUT	5.0V	5.0V	5.0V	5.0V	5.0V	GND	GND	GND	5.0V	5.0V	5.0V	5.0V	IN	GND	5.0V	A to J	32	650	32	655	25	490	ns
		209	"	"	"	"	"	"	"	"	"	"	"	5.0V	IN	5.0V	"	"	B to J	"	"	"	"	"	"	"
		210	"	"	"	"	"	"	"	"	"	"	"	IN	5.0V	"	"	"	C to J	"	"	"	"	"	"	"
		211	"	"	"	"	"	"	"	"	"	"	IN	5.0V	"	"	"	"	D to J	"	"	"	"	"	"	"
		212	"	"	"	"	"	IN	"	"	"	"	5.0V	"	"	"	"	"	E to J	"	"	"	"	"	"	"
		213	"	"	"	"	IN	5.0V	"	"	"	"	"	"	"	"	"	"	F to J	"	"	"	"	"	"	"
		214	"	"	"	IN	5.0V	"	"	"	"	"	"	"	"	"	"	"	G to J	"	"	"	"	"	"	"
		215	"	"	IN	5.0V	"	"	"	"	"	"	"	"	"	"	"	"	H to J	"	"	"	"	"	"	"
t_{PHL4}	3003 (Fig. 8)	216	"	"	5.0V	"	"	"	"	"	5.0V	"	"	"	"	IN	"	"	A to J	"	"	33	670	29	570	"
		217	"	"	"	"	"	"	"	"	"	"	"	"	IN	5.0V	"	"	B to J	"	"	"	"	"	"	"
		218	"	"	"	"	"	"	"	"	"	"	"	IN	5.0V	"	"	"	C to J	"	"	"	"	"	"	"
		219	"	"	"	"	"	"	"	"	"	"	IN	5.0V	"	"	"	"	D to J	"	"	"	"	"	"	"
		220	"	"	"	"	"	IN	"	"	"	"	5.0V	"	"	"	"	"	E to J	"	"	"	"	"	"	"
		221	"	"	"	"	IN	5.0V	"	"	"	"	"	"	"	"	"	"	F to J	"	"	"	"	"	"	"
		222	"	"	"	IN	5.0V	"	"	"	"	"	"	"	"	"	"	"	G to J	"	"	"	"	"	"	"
		223	"	"	IN	5.0V	"	"	"	"	"	"	"	"	"	"	"	"	H to J	"	"	"	"	"	"	"
t_{PLH4}		224	"	"	5.0V	"	"	"	"	"	"	"	"	"	"	IN	"	"	A to J	35	695	42	830	31	610	"
		225	"	"	"	"	"	"	"	"	"	"	"	"	IN	5.0V	"	"	B to J	"	"	"	"	"	"	"
		226	"	"	"	"	"	"	"	"	"	"	"	IN	5.0V	"	"	"	C to J	"	"	"	"	"	"	"
		227	"	"	"	"	"	"	"	"	"	"	IN	5.0V	"	"	"	"	D to J	"	"	"	"	"	"	"
		228	"	"	"	"	"	IN	"	"	"	"	5.0V	"	"	"	"	"	E to J	"	"	"	"	"	"	"
		229	"	"	"	"	IN	5.0V	"	"	"	"	"	"	"	"	"	"	F to J	"	"	"	"	"	"	"
		230	"	"	"	IN	5.0V	"	"	"	"	"	"	"	"	"	"	"	G to J	"	"	"	"	"	"	"
		231	"	"	IN	5.0V	"	"	"	"	"	"	"	"	"	"	"	"	H to J	"	"	"	"	"	"	"
t_{PHL5}	3003 (Fig. 9)	232	"	"	GND	GND	GND	"	5.0V	"	GND	GND	GND	GND	GND	IN	"	"	A to J	33	665	39	775	29	575	"
		233	"	"	"	"	"	"	"	"	"	"	"	GND	IN	GND	"	"	B to J	"	"	"	"	"	"	"
		234	"	"	"	"	"	"	"	"	"	"	"	IN	GND	"	"	"	C to J	"	"	"	"	"	"	"
		235	"	"	"	"	"	"	"	"	"	"	IN	GND	"	"	"	"	D to J	"	"	"	"	"	"	"
		236	"	"	"	"	"	IN	"	"	"	"	GND	"	"	5.0V	"	"	E to J	"	"	"	"	"	"	"
		237	"	"	"	"	IN	GND	"	"	"	"	"	"	"	"	"	"	F to J	"	"	"	"	"	"	"
		238	"	"	"	IN	GND	"	"	"	"	"	"	"	"	"	"	"	G to J	"	"	"	"	"	"	"
		239	"	"	IN	GND	"	"	"	"	"	"	"	"	"	"	"	"	H to J	"	"	"	"	"	"	"
t_{PHL5}		240	"	"	GND	"	"	5.0V	"	"	"	"	"	"	"	IN	"	"	A to J	27	540	32	645	24	485	"
		241	"	"	"	"	"	"	"	"	"	"	"	"	IN	GND	"	"	B to J	"	"	"	"	"	"	"
		242	"	"	"	"	"	"	"	"	"	"	"	IN	GND	"	"	"	C to J	"	"	"	"	"	"	"
		243	"	"	"	"	"	"	"	"	"	"	IN	GND	"	"	"	"	D to J	"	"	"	"	"	"	"
		244	"	"	"	"	"	IN	"	"	"	"	GND	"	"	5.0V	"	"	E to J	"	"	"	"	"	"	"
		245	"	"	"	"	IN	GND	"	"	"	"	"	"	"	"	"	"	F to J	"	"	"	"	"	"	"
		246	"	"	"	IN	GND	"	"	"	"	"	"	"	"	"	"	"	G to J	"	"	"	"	"	"	"
		247	"	"	IN	GND	"	"	"	"	"	"	"	"	"	"	"	"	H to J	"	"	"	"	"	"	"
t_{PHL6}	3003 (Fig. 10)	248	"	"	GND	"	"	5.0V	"	"	5.0V	"	"	"	"	IN	"	"	A to J	32	640	38	755	27	540	"
		249	"	"	"	"	"	"	"	"	"	"	"	"	IN	GND	"	"	B to J	"	"	"	"	"	"	"
		250	"	"	"	"	"	"	"	"	"	"	"	IN	GND	"	"	"	C to J	"	"	"	"	"	"	"
		251	"	"	"	"	"	"	"	"	"	"	IN	GND	"	"	"	"	D to J	"	"	"	"	"	"	"
		252	"	"	"	"	"	IN	"	"	"	"	GND	"	"	5.0V	"	"	E to J	"	"	"	"	"	"	"
		253	"	"	"	"	IN	GND	"	"	"	"	"	"	"	"	"	"	F to J	"	"	"	"	"	"	"
		254	"	"	"	IN	GND	"	"	"	"	"	"	"	"	"	"	"	G to J	"	"	"	"	"	"	"
		255	"	"	IN	GND	"	"	"	"	"	"	"	"	"	"	"	"	H to J	"	"	"	"	"	"	"
t_{PLH6}		256	"	"	GND	"	"	5.0V	"	"	"	"	"	"	"	IN	"	"	A to J	30	585	35	700	26	520	"
		257	"	"	"	"	"	"	"	"	"	"	"	"	IN	GND	"	"	B to J	"	"	"	"	"	"	"
		258	"	"	"	"	"	"	"	"	"	"	"	IN	GND	"	"	"	C to J	"	"	"	"	"	"	"
		259	"	"	"	"	"	"	"	"	"	"	IN	GND	"	"	"	"	D to J	"	"	"	"	"	"	"
		260	"	"	"	"	"	IN	"	"	"	"	GND	"	"	5.0V	"	"	E to J	"	"	"	"	"	"	"
		261	"	"	"	"	IN	GND	"	"	"	"	"	"	"	"	"	"	F to J	"	"	"	"	"	"	"
		262	"	"	"	IN	GND	"	"	"	"	"	"	"	"	"	"	"	G to J	"	"	"	"	"	"	"
		263	"	"	IN	GND	"	"	"	"	"	"	"	"	"	"	"	"	H to J	"	"	"	"	"	"	"

See footnotes at end of device type 54.

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TABLE III. Group A inspection for device type 54 – Continued.

Symbol	MIL-STD-883 method	Cases E,F,Z	Terminal conditions 1/																Measured terminal	Test limits						Unit
		Symbol	J	Kd	H	G	F	E	Kb	V_{SS}	Kc	Ka	D	C	B	A	EX	V_{DD}		Subgroup 9 T_C = +25°C		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C		
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Min	Max	Min	Max	Min	Max	
t_{PHL7}	3003 (Fig. 11)	264	OUT	5.0V	GND	GND	GND	GND	5.0V	GND	GND	5.0V	5.0V	5.0V	5.0V	IN	GND	5.0V	A to J	34	675	40	790	30	590	ns
		265	"	"	"	"	"	"	"	"	"	"	"	5.0V	IN	5.0V	"	"	B to J	"	"	"	"	"	"	"
		266	"	"	"	"	"	"	"	"	"	"	"	IN	5.0V	"	"	"	C to J	"	"	"	"	"	"	"
		267	"	"	"	"	"	"	"	"	"	"	IN	5.0V	5.0V	"	"	"	D to J	"	"	"	"	"	"	"
		268	"	"	5.0V	5.0V	5.0V	IN	"	"	"	"	GND	GND	GND	GND	"	"	E to J	"	"	"	"	"	"	"
		269	"	"	"	5.0V	IN	5.0V	"	"	"	"	"	"	"	"	"	"	F to J	"	"	"	"	"	"	"
		270	"	"	"	IN	5.0V	"	"	"	"	"	"	"	"	"	"	"	G to J	"	"	"	"	"	"	"
		271	"	"	IN	5.0V	5.0V	"	"	"	"	"	"	"	"	"	"	"	H to J	"	"	"	"	"	"	"
t_{PLH7}		272	"	"	GND	GND	GND	GND	"	"	"	"	5.0V	5.0V	5.0V	IN	"	"	A to J	31	"	36	730	27	550	"
		273	"	"	"	"	"	"	"	"	"	"	"	5.0V	IN	5.0V	"	"	B to J	"	"	"	"	"	"	"
		274	"	"	"	"	"	"	"	"	"	"	"	IN	5.0V	"	"	"	C to J	"	"	"	"	"	"	"
		275	"	"	"	"	"	"	"	"	"	"	IN	5.0V	5.0V	"	"	"	D to J	"	"	"	"	"	"	"
		276	"	"	5.0V	5.0V	5.0V	IN	"	"	"	"	GND	GND	GND	GND	"	"	E to J	"	"	"	"	"	"	"
		277	"	"	"	5.0V	IN	5.0V	"	"	"	"	"	"	"	"	"	"	F to J	"	"	"	"	"	"	"
		278	"	"	"	IN	5.0V	"	"	"	"	"	"	"	"	"	"	"	G to J	"	"	"	"	"	"	"
		279	"	"	IN	5.0V	5.0V	"	"	"	"	"	"	"	"	"	"	"	H to J	"	"	"	"	"	"	"
t_{PHL8}	3003 (Fig. 12)	280	"	"	GND	GND	GND	GND	"	"	5.0V	"	5.0V	5.0V	5.0V	IN	"	"	A to J	36	715	42	840	31	630	"
		281	"	"	"	"	"	"	"	"	"	"	"	5.0V	IN	5.0V	"	"	B to J	"	"	"	"	"	"	"
		282	"	"	"	"	"	"	"	"	"	"	"	IN	5.0V	"	"	"	C to J	"	"	"	"	"	"	"
		283	"	"	"	"	"	"	"	"	"	"	IN	5.0V	5.0V	"	"	"	D to J	"	"	"	"	"	"	"
		284	"	"	5.0V	5.0V	5.0V	IN	"	"	"	"	GND	GND	GND	GND	"	"	E to J	"	"	"	"	"	"	"
		285	"	"	"	5.0V	IN	5.0V	"	"	"	"	"	"	"	"	"	"	F to J	"	"	"	"	"	"	"
		286	"	"	"	IN	5.0V	"	"	"	"	"	"	"	"	"	"	"	G to J	"	"	"	"	"	"	"
		287	"	"	IN	5.0V	5.0V	"	"	"	"	"	"	"	"	"	"	"	H to J	"	"	"	"	"	"	"
t_{PLH8}		288	"	"	GND	GND	GND	GND	"	"	"	"	5.0V	5.0V	5.0V	IN	"	"	A to J	30	600	36	720	27	535	"
		289	"	"	"	"	"	"	"	"	"	"	"	5.0V	IN	5.0V	"	"	B to J	"	"	"	"	"	"	"
		290	"	"	"	"	"	"	"	"	"	"	"	IN	5.0V	"	"	"	C to J	"	"	"	"	"	"	"
		291	"	"	"	"	"	"	"	"	"	"	IN	5.0V	5.0V	"	"	"	D to J	"	"	"	"	"	"	"
		292	"	"	5.0V	5.0V	5.0V	IN	"	"	"	"	GND	GND	GND	GND	"	"	E to J	"	"	"	"	"	"	"
		293	"	"	"	5.0V	IN	5.0V	"	"	"	"	"	"	"	"	"	"	F to J	"	"	"	"	"	"	"
		294	"	"	"	IN	5.0V	"	"	"	"	"	"	"	"	"	"	"	G to J	"	"	"	"	"	"	"
		295	"	"	IN	5.0V	5.0V	"	"	"	"	"	"	"	"	"	"	"	H to J	"	"	"	"	"	"	"
t_{PHL9}	3003 (Fig. 13)	296	"	"	GND	GND	GND	GND	GND	"	GND	IN	"	"	"	"	"	"	Ka to J	34	690	38	760	30	595	"
t_{PLH9}		297	"	"	"	"	"	"	GND	"	"	IN	"	"	"	"	"	"	Ka to J	31	625	37	730	28	555	"
t_{PHL10}	3003 (Fig. 14)	298	"	"	"	"	"	"	IN	"	"	GND	"	"	"	"	"	"	Kb to J	27	535	31	615	23	460	"
t_{PLH10}		299	"	"	"	"	"	"	IN	"	"	"	"	"	"	"	"	"	Kb to J	26	515	30	605	23	465	"
t_{PHL11}	3003 (Fig. 15)	300	"	"	"	"	"	"	GND	"	IN	"	"	"	"	"	"	"	Kc to J	21	420	24	475	18	360	"
t_{PLH11}		301	"	"	"	"	"	"	"	"	IN	"	"	"	"	"	"	"	Kc to J	19	380	24	445	17	340	"
t_{THL}	3004 (Fig. 5)	302	"	"	"	"	"	"	"	"	GND	"	"	"	"	IN	"	"	J	7	135	8	160	6	115	"
		303	"	"	"	"	"	"	"	"	"	"	"	"	IN	GND	"	"		"	"	"	"	"	"	"
		304	"	"	"	"	"	"	"	"	"	"	"	IN	GND	"	"	"		"	"	"	"	"	"	"
		305	"	"	"	"	"	"	"	"	"	"	IN	GND	"	"	"	"		"	"	"	"	"	"	"
		306	"	"	"	"	"	IN	"	"	"	"	GND	"	"	"	"	"		"	"	"	"	"	"	"
		307	"	"	"	"	IN	GND	"	"	"	"	"	"	"	"	"	"		"	"	"	"	"	"	"
		308	"	"	"	IN	GND	"	"	"	"	"	"	"	"	"	"	"		"	"	"	"	"	"	"
		309	"	"	IN	GND	GND	"	"	"	"	"	"	"	"	"	"	"		"	"	"	"	"	"	"

See footnotes at end of device type 54.

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TABLE III. Group A inspection for device type 54 – Continued.

Symbol	MIL-STD-883 method	Cases E,F,Z	Terminal conditions 1/																Measured terminal	Test limits						Unit
		Symbol	J	Kd	H	G	F	E	Kb	V_{SS}	Kc	Ka	D	C	B	A	EX	V_{DD}		Subgroup 9 T_C = +25°C		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C		
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Min	Max	Min	Max	Min	Max	
t_{TLH}	3004 (Fig. 5)	310	OUT	5.0V	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	IN	GND	5.0V	J	6	115	7	140	5	100	ns
		311	"	"	"	"	"	"	"	"	"	"	"	GND	IN	GND	"	"	"	"	"	"	"	"	"	"	"
		312	"	"	"	"	"	"	"	"	"	"	"	IN	GND	"	"	"	"	"	"	"	"	"	"	"	"
		313	"	"	"	"	"	"	"	"	"	"	IN	GND	"	"	"	"	"	"	"	"	"	"	"	"	"
		314	"	"	"	"	"	IN	"	"	"	"	GND	"	"	"	"	"	"	"	"	"	"	"	"	"	"
		315	"	"	"	"	IN	GND	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
		316	"	"	"	IN	GND	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
		317	"	"	IN	GND	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
t_{PLZ}	3003 (Fig. 16)	318	"	IN	GND	"	"	"	"	"	"	"	"	"	"	5.0V	"	"	Kd to J	18	350	25	490	18	350	"
t_{PHZ}		319	"	"	"	"	"	"	"	"	"	"	"	"	"	GND	"	"	"	18	350	25	490	18	350	"
t_{PZL}		320	"	"	"	"	"	"	"	"	"	"	"	"	"	5.0V	"	"	"	23	450	32	630	23	450	"
t_{PZH}		321	"	"	"	"	"	"	"	"	"	"	"	"	"	GND	"	"	"	23	450	32	630	23	450	"

1/ Pins not designated may be "high" level logic, "low" level logic, or open. Exceptions are as follows: For $V_{IC(POS)}$ tests, the V_{SS} terminal shall be open; for $V_{IC(NEG)}$ tests, the V_{DD} terminal shall be open; for I_{SS} tests, the outputs shall be open.

2/ I_{SS} measurements shall be performed in sequence.

3/ The device manufacturer may, at his option, measure I_{IL} and I_{IH} at +25°C for each individual input or measure all inputs together.

4/ See 4.4.1c.

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4.4.4 Group D inspection. Group D inspection shall be in accordance with table V of MIL-PRF-38535. End-point electrical parameters shall be as specified in table II herein.

4.4.5 Group E inspection. Group E inspection is required only for parts intended to be marked as radiation hardness assured (see 3.7 herein). RHA levels for device classes B and S shall be as specified in MIL-PRF-38535 and 4.5.4 herein.

4.5 Methods of inspection. Methods of inspection shall be specified and as follows:

4.5.1 Voltage and current. Unless otherwise specified, all voltages given are referenced to the microcircuit V_{SS} terminal. Currents given are conventional current and positive when flowing into the referenced terminal.

4.5.2 Burn-in and life test cool down procedures. When the burn-in and life tests are completed and prior to removal of bias voltages, the devices under test (DUT) shall be cooled to a temperature of 25°C ±3°C; then, electrical parameter end-point measurements shall be performed.

TABLE IV. Delta limits at 25°C.

Parameter 1/	Device types	
	01, 02, 03, 04	51, 52, 53, 54
I_{SS}	±10 nA	±10 nA
V_{OL1}	±0.04 V	2/
V_{OH1}	±0.08 V	2/
I_{OL1}	2/	±15%
I_{OH1}	2/	±15%

1/ Each of the above parameters shall be recorded before and after the required burn-in and life tests to determine delta (Δ).
2/ Not applicable.

4.5.3 Quiescent supply current (I_{SS} test). When performing quiescent supply current measurements (I_{SS}), the meter shall be placed so that all currents flow through the meter.

4.5.4 Radiation hardness assurance (RHA) testing. The RHA testing shall be performed in accordance with test procedures and sampling specified in MIL-PRF-38535 and herein.

a. Before irradiation, selected samples shall be assembled in qualified packages and pass the governing electrical parameters (group A subgroup 1 at 25°C) and also be subjected to the threshold-voltage test in table VII in order to calculate the delta threshold (ΔV_T) after irradiation.

b. The devices shall be subjected to a total radiation dose as specified in MIL-PRF-38535 for the radiation hardness assurance level being tested, and meet the end-point electrical parameters as defined in table V at 25°C, after exposure. The start and completion of the end-point electrical parameter measurements shall not exceed 2 hours following irradiation.

c. Threshold-voltage test circuit conditions shall be as specified in table VII and on figure 17. In situ and remote testing, the tests shall be performed with the devices biased in accordance with table VI and the bias may be interrupted for up to 1 minute to remove devices to the remote bias fixture.

d. After irradiation, the devices shall pass the truth table test as specified in subgroup 7 in table III or if subgroup 7 is not required, then an equivalent truth table test shall be performed.

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TABLE V. Radiation hardened end-point electrical parameters at 25°C.

Parameter	Test limits (All device types)	V_{DD}	
		Device types 01, 02, 03, 04	Device types 51, 52, 53, 54
V_{TN}	0.3 V min	10 V	10 V
V_{TP}	2.8 V max	10 V	10 V
ΔV_T	1.4 V max	10 V	10 V
I_{SS}	100 x max limit	15 V	18 V
t_{PLH}	1.35 x max limit	5 V	5 V
t_{PHL}	1.35 x max limit	5 V	5 V

TABLE VI. Bias during exposure to radiation.

Device type	Pin connections 1/		
	V_{DD} = 10 V dc (through a 30 kΩ to 60 kΩ resistor)	V_{SS} = GND	V_{DD} = 10 V dc
01, 51	2, 3, 6, 10, 11,	4, 7, 9	14
02, 52	1, 2, 3, 4, 5, 6, 7, 9, 14, 15	8	16
03, 53	1, 2, 5, 6, 8, 9, 12, 13	7	14
04, 54	2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, 15	8	16

1/ Pins not designated are open, or tied to 10 V dc through a 30 kΩ to 60 kΩ resistor.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

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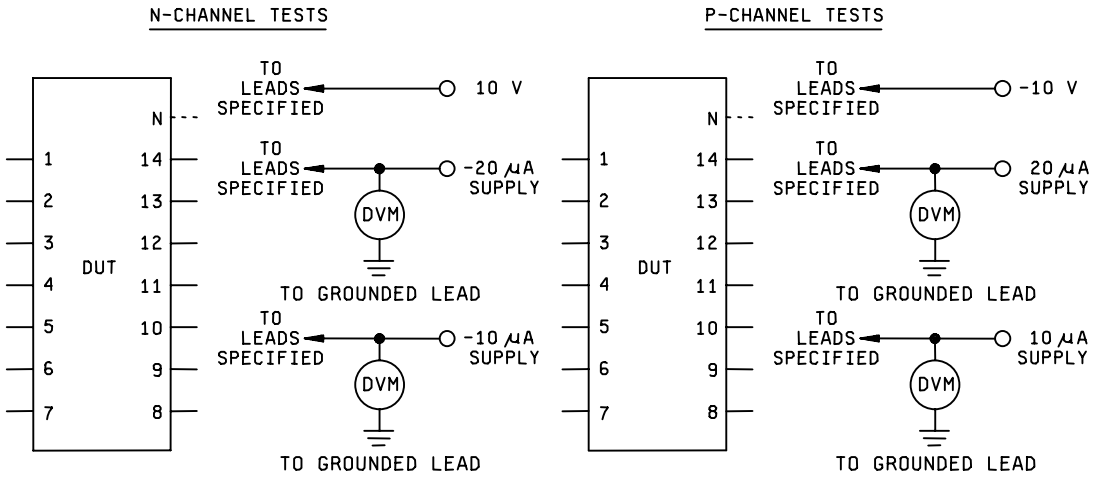


FIGURE 17. Threshold-voltage test circuit.

TABLE VII. Threshold-voltage test circuit conditions.

Device type	GND	10 V	V_{TN} measured at		GND	-10 V	V_{TP} measured at	
			-20 μA supply	-10 μA supply			20 μA supply	10 μA supply
01, 51	6	8, 14		7	6	7, 13		14
02, 52	9	14-16	1-8		9	1-8	14-16	
03, 53	8	14		1, 2, 5-7, 9, 12, 13	8	1, 2-5, 7, 9, 12, 13		14
04, 54	10	16	2-9, 11-15		10	2-9, 11-15	16	

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. PIN and compliance identifier, if applicable (see 1.2).

c. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

d. Requirements for certificate of compliance, if applicable.

e. Requirements for notification of change of product or process to contracting activity in addition to notification to the qualifying activity, if applicable.

f. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

g. Requirements for product assurance and radiation hardness assurance options.

h. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements should not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

i. Requirements for "JAN" marking.

j. Packaging requirements. (see 5.1)

6.3 Superseding information. The requirements of MIL-M-38510 have been superseded to take advantage of the available Qualified Manufacturer Listing (QML) system provided by MIL-PRF-38535. Previous references to MIL-M-38510 in this document have been replaced by appropriate references to MIL-PRF-38535. All technical requirements now consist of this specification and MIL-PRF-38535. The MIL-M-38510 specification sheet number and PIN have been retained to avoid adversely impacting existing government logistics systems and contractors parts lists.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List QML-38535 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DSCC-VQ, P.O. Box 3990, Columbus, Ohio 43218-3990.

6.5 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-PRF-38535, MIL-HDBK-1331, and as follows:

C_I ..... Input terminal-to-GND capacitance.
GND ..... Ground zero voltage potential.`
I_{OC} ..... Three-state output leakage current.
I_{SS} ..... Quiescent supply current.
T_A ..... Free air temperature.
V_{DD} ..... Positive supply voltage.
V_{SS} ..... Negative supply voltage.

6.6 Logistic support. Lead materials and finishes (see 3.4) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class S for National Aeronautics and Space Administration or class B for Department of Defense (see 1.2.2), lead material and finish A (see 3.4). Longer length leads and lead forming should not affect the part number.

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6.7 Data reporting. When specified in the purchase order or contract, a copy of the following data, as applicable, will be supplied.

a. Attributes data for all screening tests (see 4.2) and variables data for all static burn-in, dynamic burn-in, and steady-state life tests (see 3.6).

b. A copy of each radiograph.

c. The technology conformance inspection (TCI) data (see 4.4).

d. Parameter distribution data on parameters evaluated during burn-in (see 3.6).

e. Final electrical parameters data (see 4.2d).

f. RHA delta limits.

6.8 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges, post irradiation performance or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information should not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-PRF-38535.

Military device type	Generic-industry type
01	4007A
02	4019A
03	4030A
04	4048A
51	4007UB
52	4019B
53	4030B
54	4048B

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5962-2065)

Review activities:
Army - MI, SM
Navy - AS, CG, MC, SH, TD
Air Force – 03, 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST Online database at http://assist.daps.dla.mil.

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